Terrorism and Regional Development: Evidence from Peru's Internal War

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Abstract

How do regions recover from civil conflict? This paper follows Peru's 1981 census districts through 1993, 2007, and 2017 to study the 1980–2000 internal armed conflict, which claimed about 69,000 lives. In the highlands, the most affected districts lost about 21 percent of their population relative to other highland districts by 1993, a loss concentrated in Ayacucho, and they had recovered about half of that loss by 2007. The cohort of young adults at the war's onset and the small districts where violence reached a large share of residents did not recover. The literacy gap with other districts fell 61 percent between 1981 and 2017. Infrastructure recovered unevenly, as services delivered through public networks lagged while services built on site expanded faster. Night-time lights show that light covered less of the territory of the most affected districts. Recovery was demographically uneven, educationally convergent, and spatially concentrated.

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1 Introduction

How do regions recover from sustained internal conflict? Beyond the immediate casualties, war damages infrastructure, displaces populations, and disrupts schooling, but its long-run consequences are still poorly understood. Affected areas may eventually recover, or the conflict shock may produce a permanent divergence. When recovery occurs, it can come from return migration, from demographic replacement, or from rebuilding in situ, and recovery in one dimension (population, education, infrastructure) does not necessarily translate into recovery in others (economic activity, living standards).

I study these questions for Peru's internal armed conflict (1980–2000), one of the most destructive civil wars in Latin America. The conflict between the Maoist insurgency Sendero Luminoso (Shining Path), the smaller Túpac Amaru Revolutionary Movement (MRTA), and Peruvian state forces claimed approximately 69,000 lives and forced approximately 600,000 people to leave their homes, generating large-scale internal displacement concentrated in the rural highlands (Comisión de la Verdad y Reconciliación, 2003, Anexo 2 and Tomo VI). I follow Peruvian districts from the 1981 census, taken before the violence escalated, through the censuses of 1993, 2007, and 2017, with the 1972 census as an earlier pre-war observation, and I compare the districts that recorded more than 100 deaths and disappearances in the database of Peru's Truth and Reconciliation Commission (Comisión de la Verdad y Reconciliación, CVR) with all other districts. Because district boundaries changed after 1981, every census is aggregated to the territory of the 1981 districts.

Recovery was demographically uneven, educationally convergent, and spatially concentrated. Three findings support this description.

The first is a severe regional demographic shock followed by recovery. By 1993 the most affected highland districts had lost about a fifth of their population relative to other highland districts, which had grown at the same pace in the decade before the war. The loss was centered on Ayacucho, where Sendero Luminoso began its war. Within the department it struck districts regardless of how many victims each recorded, and population moved towards the regional capital, Huamanga. By 2007 these districts had recovered about half of that loss, and nationally the most affected districts regained their relative position, growing faster than the rest between 1993 and 2007. Recovery was uneven, however. The cohort that was entering adulthood when the war began remained underrepresented for at least two decades, and the small districts where violence reached a large share of residents, many of them with 100 or fewer victims, lost population and had not recovered it by 2017.

The second finding is long-run literacy convergence despite severe conflict exposure. Among residents aged 15 and over, the literacy gap between the most affected districts

and the rest fell by 61 percent between 1981 and 2017, as the expansion of schooling after the war reached places that had started far behind. Secondary completion among adults lagged, a mark of where these districts started.

The third finding is an uneven recovery of public services. Services delivered through networks lagged, and electricity coverage expanded about 10 percentage points less in the most affected districts by 2007, also outside Ayacucho, while services that households and communities could build on site, such as latrines, expanded faster. Night-time lights add a spatial dimension. Light grew in the towns of the most affected districts at a pace similar to elsewhere, but it reached less of their territory, and within Ayacucho the districts with more victims gained the least light in the last decade.

Recovery also shows suggestive differences by perpetrator. Districts where Sendero Luminoso committed most of the killings and districts where state agents did both lagged in electricity and caught up in literacy, while the districts where state agents were the main perpetrators followed a weaker population path.

The Peruvian context is well suited to studying long-run conflict effects, because the conflict had a clear beginning (1980) and end (approximately 2000) and the 1972 and 1981 censuses precede large-scale violence. The turning point of the conflict was the capture of Sendero Luminoso's leader, Abimael Guzmán, in September 1992.

The paper contributes three findings on how places recover from a long internal war. The first is that demographic recovery was real but uneven. Nationally, the districts with the most victims regained their relative population within about fifteen years, while the cohort that was young when the war began and the small districts where violence reached a large share of residents did not recover. The second is long-run educational convergence, which places the war within a broader process of catching up instead of making it the engine of that process. The third is an uneven recovery of basic services, with a lag in networked electricity and a faster expansion of on-site sanitation. These findings rest on a measurement step. District boundaries and codes changed repeatedly after 1981, and following places across four decades requires tracing every later district back to the district it came from, without which comparisons across censuses mix the evolution of places with the reorganization of the administrative map.

1.1 Related Literature

This paper is related to four bodies of work. The first is quantitative research on Peru's conflict that uses the CVR records. León (2012) shows that children exposed to political violence early in life accumulated less schooling as adults, Galdo (2013) finds lower adult

earnings among those exposed in early childhood, and Grimard and Laszlo (2014) document long-term effects on women's health. These studies follow cohorts of individuals wherever they live, while this paper follows places. The two perspectives are complementary, because a district can catch up on literacy through younger cohorts even when the cohorts exposed to violence carry permanent losses. Anthropological and political research on the war, including McClintock (1984), Degregori (2010), Stern (1998), and Theidon (2013), documents the local social and political processes that district aggregates cannot observe.

The second body of work asks whether places recover from large violent shocks. Japanese cities returned to their relative size within about fifteen years of the Allied bombing (Davis and Weinstein, 2002), the bombing of German cities had temporary effects on city growth in western Germany and persistent effects in eastern Germany (Brakman et al., 2004), and the most heavily bombed districts of Vietnam show no lasting differences in poverty or consumption a quarter of a century later (Miguel and Roland, 2011). However, those shocks were concentrated in time and came from outside, while Peru's war lasted two decades and was fought among the residents of the affected districts. It displaced about 600,000 people (Comisión de la Verdad y Reconciliación, 2003, Tomo VI), and displacement imposes large welfare losses on those who flee (Ibáñez and Vélez, 2008). My population results for the districts with the most victims are close to the bombing studies, while the smaller districts where violence reached a large share of residents and the networked public services recovered more slowly.

The third body of work studies the consequences of civil war more broadly (Blattman and Miguel, 2010; Justino, 2009; Verwimp et al., 2019). Cross-country evidence describes a conflict trap, in which countries emerging from civil war face low growth and a high risk of renewed fighting (Collier et al., 2003). Peru did not return to large-scale civil war after 2000, and its affected regions partly recovered population while the national economy grew. Within-country evidence therefore speaks to the recovery of regions inside a country that left the conflict behind, a different question from the recovery of countries. Sendero Luminoso relied systematically on terror against civilians (Comisión de la Verdad y Reconciliación, 2003), so this paper is also related to work on the economic costs of terrorism (Abadie and Gardeazabal, 2003), measured here on population and public services instead of output.

The fourth body of work uses night-time lights as a proxy for economic activity (Henderson et al., 2012; Chen and Nordhaus, 2011, 2019). Lights measure economic activity poorly in rural and sparsely populated areas (Gibson et al., 2020), which describes most of the districts studied here, and the treatment of districts with no detected light changes the estimates (Chen and Roth, 2024). For these reasons I report lights as complementary evidence on where activity spread, not as a measure of output.

The rest of the paper is organized as follows. Section 2 provides historical background and Section 3 describes the data and the empirical strategy. Sections 4, 5, and 6 report the results for population, human capital, and infrastructure, Section 7 compares districts by the main perpetrator of violence, and Section 8 presents the night-time lights analysis. Section 9 concludes.

2 Historical Background

Peru's internal armed conflict (1980–2000) was one of the deadliest insurgencies in Latin America, and the Truth and Reconciliation Commission estimates that it claimed approximately 69,000 lives (Comisión de la Verdad y Reconciliación, 2003, Anexo 2). The conflict set Maoist guerrillas against the Peruvian state, and rural indigenous communities suffered most of the violence from both sides. Appendix B provides a fuller account of the conflict's origins, the state response, and its end.

The violence was overwhelmingly rural and highland. Most victims were Quechua-speaking peasants, and the CVR found that the share of victims who spoke Quechua as their mother tongue far exceeded the share of Quechua speakers in the national population (Comisión de la Verdad y Reconciliación, 2003). The war was also local in a literal sense. Sendero Luminoso recruited in the same communities it attacked, the armed forces organized peasant patrols that fought alongside them, and some massacres were committed by and against neighbors. This feature distinguishes the Peruvian case from the external bombing campaigns studied in the literature on city recovery, and it bears on recovery, because a community that lost members to violence committed by other members may rebuild its population and institutions differently from one that suffered an external attack.

The conflict's geographic footprint was far from random. Violence concentrated heavily in the central and southern highlands (Ayacucho, Huancavelica, Apurímac, Junín, and parts of Cusco and Puno) before spreading to other regions (Comisión de la Verdad y Reconciliación, 2003). Several factors explain this pattern. First, Shining Path deliberately targeted areas with weak state capacity, where small guerrilla columns could operate with relative impunity (Weinstein, 2007). Second, the organization exploited local grievances, particularly disputes over land, labor, and abusive local authorities (Stern, 1998). Third, the mountainous terrain of the central highlands provided natural refuge from military operations.

Pre-war poverty and inequality were highest where the war would later be fought, as Appendix Figure A1 shows. The five poorest departments in 1981 (Cajamarca, Puno, Huancavelica, Ayacucho, and Cusco) were also among the six most unequal, and three of them lie in the highland belt that became the epicenter of the violence, while coastal departments

had both higher incomes and lower inequality. Chuschi in Ayacucho, where Shining Path launched its armed struggle by burning ballot boxes on May 17, 1980, was among the poorest and most unequal districts of the country. Across districts the association is weaker. Districts without documented victims had the lowest median inequality in 1981, and within departments the 1981 Gini coefficient predicts the number of victims but not membership in the very-high-conflict group. Appendix C.2 describes how inequality is measured and reports these comparisons.

Inequality in these departments was extreme by any standard. Department Gini coefficients of declared income ranged from 0.63 to 0.67 among the five poorest departments, against 0.39 in Tumbes, 0.42 in Ica, 0.46 in Lambayeque, and 0.48 in Lima. Much of this inequality reflects the coexistence of subsistence agriculture, where declared monetary income is often zero, with a small commercial and public sector in provincial towns. In Chuschi, mean declared income was 6 soles per capita and the Gini reached 0.89, largely because many residents declared no monetary income at all. Measured inequality in these districts therefore describes the distance between a monetized minority and a largely non-monetized peasant majority, the social divide that Sendero Luminoso's rhetoric targeted.

This correlation between inequality and violence, while not establishing causality, is consistent with theoretical accounts emphasizing grievance-based mobilization (Gurr, 1970). Shining Path's rhetoric explicitly targeted class enemies (landlords, merchants, local officials) and promised radical redistribution (Degregori, 2010). The correlation also reflects the overlap between inequality, indigenous identity, and regional underdevelopment. The districts that would suffer the most violence had a much larger share of native-language speakers than the rest of the country in 1981, and the southern highlands had long lagged behind the coast in growth and in the economic structure that export-led development created (Thorp and Bertram, 1978).

The conflict evolved through distinct phases (Comisión de la Verdad y Reconciliación, 2003). The initial period (1980–1982) saw Shining Path consolidate control in rural Ayacucho through a combination of political work and selective violence against perceived enemies. The militarization phase (1983–1988) began when the government declared emergency zones and deployed the armed forces, leading to massive human rights violations as both insurgents and counterinsurgents targeted civilian populations. In the following phase (1989–1992) violence spread to more districts and reached Lima, culminating in the Tarata bombing of 1992, and after Guzmán's capture it declined until 2000. Throughout, the conflict's epicenter remained the central highlands, though violence spread to the Upper Huallaga Valley (coca-growing region) and eventually metropolitan Lima.

These phases frame the census comparison. The 1981 census was taken a year after the

first attack, when violence was confined to a few districts of Ayacucho, the 1993 census after the peak of the war and Guzmán's capture, and the 2007 and 2017 censuses seven and seventeen years after the conflict ended.

3 Data and Empirical Strategy

3.1 Census Data and Fixed Geography

I use district-level data from Peru's national population censuses of 1981, 1993, 2007, and 2017, conducted by the National Institute of Statistics and Informatics (INEI). The 1981 census provides the baseline. Sendero Luminoso launched its armed struggle in May 1980, but violence remained localized and small in scale until the militarization of 1983, so the 1981 census precedes the overwhelming majority of documented violence.[1] The 1993 census captures conditions shortly after Guzmán's capture, while violence continued in parts of the country.[2] The 2007 and 2017 censuses measure medium- and long-run recovery. From these censuses I construct population, age structure, literacy, educational attainment, migration, mother tongue, and dwelling services.

Population is the main outcome because it summarizes the demographic consequences of the war, which operated through deaths, disappearances, flight, and return, and through the births that did not take place while families were displaced. The census counts the residents of each district on census day, so the population of a district reflects both its losses during the war and its capacity to retain or attract residents afterwards. Literacy and schooling describe the human capital of those residents, and the dwelling characteristics describe access to public services and the quality of housing. None of these outcomes measures income, which the census does not record after 1981, and night-time lights are used as a partial substitute.

District boundaries changed after 1981, and INEI renumbered many districts, so the same numeric code can designate different districts in different censuses. I therefore aggregate every census to fixed territorial units that correspond to the 1981 district boundaries, each formed by a 1981 district together with the districts later created from it. The resulting

[1] Throughout the paper, "pre-conflict baseline" is shorthand for this pre-escalation baseline.

[2] INEI estimated national omission in the 1993 census at 2.35 percent (531,543 persons) by re-enumerating a 1 percent random sample of dwellings, and omission among women ranged from 1.2 percent in Lima to 3.4 percent in Huancavelica (INEI, *Perfil Sociodemográfico* N. 7, 1994, as reported in INEI, "Información básica y metodología," https://www.inei.gob.pe/media/MenuRecursivo/publicaciones_digitales/Est/Lib0014/infbasme.htm). Rates for the emergency zones and for districts are not reported, so differential omission in the most affected districts cannot be ruled out. Its size can be bounded, however. Omission one point above the national rate, as in Huancavelica, would account for about 0.01 of the 0.24 log points of the highland loss reported in Section 4.1, and omission would have to exceed the national rate by about 20 points, nine times the national rate, to account for the whole loss.

panel contains 1,540 units, 42 of which are classified as very-high-conflict districts. The 1981 census database does not cover the departments of Apurímac and San Martín and covers Loreto only through the six districts of Contamana, enumerated in 1981 under Ucayali, so 1,371 units are observed in 1981 and all 1,540 in 1993, 2007, and 2017. Appendix C.1 documents the reconstruction, and an alternative concordance that assigns each new district only to its main district of origin, which yields 1,645 units, serves as a robustness check. Throughout the paper, "district" refers to these units.

Geographic characteristics complete the data. Altitude and district area come from INEI's cartographic database, distances to Lima and to the coast are measured from district centroids, and terrain ruggedness follows Nunn and Puga (2012). These variables enter the extended specifications interacted with year, because remote and mountainous districts may have followed different paths regardless of the war. Night-time lights, described in Section 8, are summed over the same fixed district polygons.

3.2 Violence Data

Conflict exposure is measured with the victim-level database of the CVR, which documented 23,149 deaths and disappearances during 1980–2000, compiled through testimonies, forensic investigations, and cross-referenced administrative records. For each victim it records the year and district of the events, the perpetrator, and the sources that name the victim. Sendero Luminoso is responsible for 54.3 percent of these victims, state agents (armed forces and police) for 31.4 percent, and the self-defense committees, paramilitary groups, and the MRTA together for 3.8 percent. Throughout the paper, "state" refers to state agents only. Records geocoded only to a province are allocated across its districts in proportion to district-coded victims, and records without location are excluded.

The CVR records are the most complete district-level account of the war, but they are not a census of victims. The Commission's statistical estimate of total deaths and disappearances is about three times the number of documented victims (Comisión de la Verdad y Reconciliación, 2003, Anexo 2), and under-documentation was likely more severe in remote rural areas, where testimonies were harder to collect. If under-documentation were classical, it would attenuate the estimates. If it were concentrated in the most isolated districts, some districts with very high violence would be misclassified into lower categories, which would bias the comparison towards zero for outcomes where those districts behave like the treated group. The robustness of the results to thresholds from 25 to 200 victims, and the regional character of the main population shock, make the conclusions less sensitive to the exact classification of individual districts.

The treatment group is formed by the 42 districts with more than 100 documented victims, which together account for 65 percent of all victims. Twenty-one of them are in Ayacucho, seven each in Junín and Huánuco, and the rest in Huancavelica, San Martín, Cusco, and Lima. The comparison group is every other district, including districts with fewer victims. Most of the violence in these districts had already occurred when the 1993 census was taken. In the districts observed in 1972, 92 percent of the victims of very-high-conflict districts (96 percent in the highlands) were recorded by 1992, and 31 of these 35 districts had already passed the threshold of 100 victims by then. A count of victims favors large districts, so I also measure exposure as victims per 1,000 inhabitants of 1981, which singles out smaller districts where violence touched a large share of residents. Figure 1 maps victims by district. Appendix Figure A11 shows that the distribution of victims is skewed in counts and per capita, so the thresholds cut through continuous distributions, and Appendix Figure A10 shows how violence spread from 104 districts in 1980–1982 to more than 500 districts in 1989–1992 before receding after Guzmán's capture.

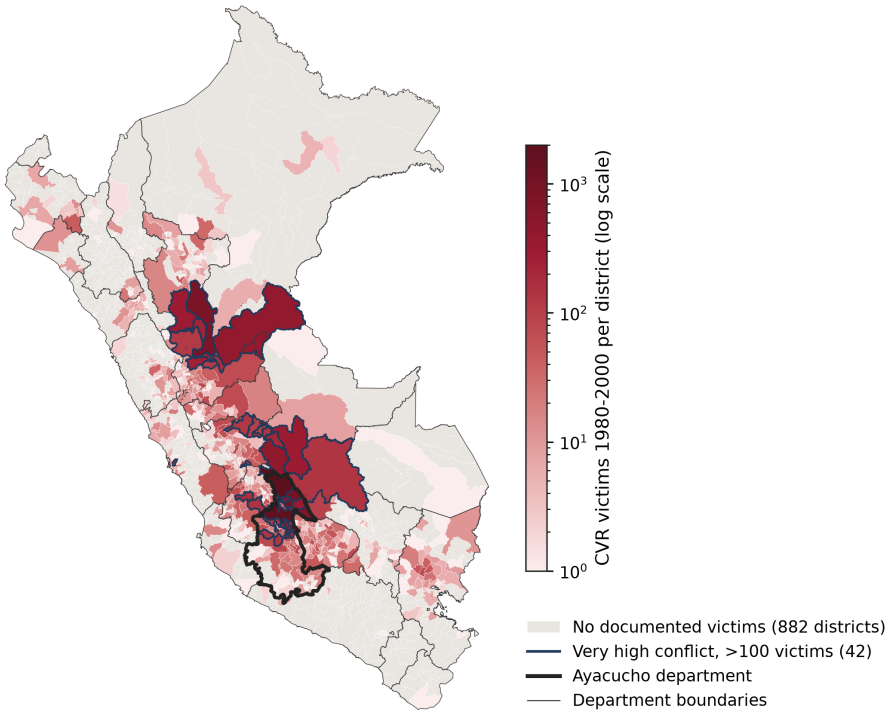


Figure 1: Geography of Violence, 1980–2000

Notes. CVR-documented victims (deaths and disappearances) per district in 1980–2000, log scale, on the 1,540 districts of the fixed 1981 geography. The 882 districts without documented victims are shown in gray. The 42 very-high-conflict districts (more than 100 victims) are outlined, and the thick line marks the department of Ayacucho.

3.3 Baseline Comparability

Table 1 compares very-high-conflict districts with districts without documented victims in 1981. The war was fought in different places from the rest of the country. The most affected districts were larger, younger, much more indigenous, less literate, and closer to Lima, the last because most of them lie in the central highlands. In occupational structure, agriculture, migration, altitude, and terrain, on the other hand, they resembled the districts that escaped the violence. These level differences are absorbed by district fixed effects, and the concern they raise is whether the two groups were on different trends, which the 1972 census allows me to examine for population.

The differences are large. Literacy among residents aged 15 and over was 12 percentage points lower in very-high-conflict districts, and the share of residents whose mother tongue was an indigenous language was 38 points higher. Their populations were about two years younger on average, with more children, and the average treated district had about 28,000 residents against about 6,500 in districts that escaped the violence, because the treatment group includes the cities of Ayacucho and Huancayo. The similarities are as informative as the differences. Very-high-conflict districts had the same occupational structure, the same shares in agriculture, manufacturing, and commerce, the same migration rates, and the same altitude and terrain as the districts without victims. The war therefore did not single out districts with a distinct economic base. It singled out districts that were poorer in human capital, more indigenous, and located in the central highlands. Appendix Table A1 extends the comparison to every intensity category. Literacy falls from 73 percent in districts without victims to 61 percent in very-high-conflict districts, and the indigenous-language share rises from 33 percent to more than 70 percent in high and very-high-conflict districts, so the most affected districts were the extreme of a gradient and not a separate group.

Table 1: Baseline Comparison of Very-High-Conflict and No-Conflict Districts, 1981

	No Conflict	Very High	Difference	SE	p-value
Total population	6,464.3	28,350.7	21,886.4	7,861.6	0.008
Mean age (years)	24.63	22.79	−1.84	0.37	0.000
Dependency ratio (%)	97.19	99.77	2.59	2.72	0.347
Sex ratio (M per 100 F)	102.58	100.73	−1.85	2.80	0.512
Children 0–14 (%)	43.26	45.22	1.96	0.47	0.000
Working age 15–64 (%)	51.04	50.40	−0.64	0.69	0.358
Literacy rate, ages 15+ (%)	73.16	60.76	−12.40	2.87	0.000
Higher education (%)	2.22	2.23	0.01	0.45	0.978
High-skill occupation (%)	3.42	3.55	0.13	0.56	0.815
Agriculture (%)	73.38	73.24	−0.14	3.92	0.971
Manufacturing (%)	5.02	4.43	−0.59	0.88	0.506
Commerce (%)	4.75	6.34	1.59	1.10	0.157
Indigenous language (%)	32.99	71.08	38.09	5.30	0.000
Migration rate (%)	26.16	26.60	0.44	3.92	0.910
Altitude (m)	2,273.1	2,363.3	90.2	197.1	0.650
Distance to Lima (km)	527.4	331.4	−196.0	15.6	0.000
Terrain ruggedness	279.5	267.2	−12.3	23.5	0.602

Notes. Welch two-sample t-tests with unequal variances on the 1981 baseline, comparing no-conflict districts (N = 827) with very-high-conflict districts (N = 39). Difference = Very High − No Conflict. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census.

3.4 Specification and Identification

The main specification is a difference-in-differences design with district and year fixed effects,

$$Y_{dt} = \alpha_d + \lambda_t + \sum_{t \neq 1981} \beta_t \left(\text{VeryHighConflict}_d \times \mathbf{1}[Year = t] \right) + \varepsilon_{dt} \tag{1}$$

where Y_{dt} is the outcome of district d in census year t, α_d and λ_t are district and year fixed effects, and $\text{VeryHighConflict}_d$ indicates more than 100 victims. Treatment is the cumulative violence of 1980–2000 and does not vary over time, so the coefficients β_t trace a dynamic difference-in-differences in calendar time relative to 1981. Where the 1972 census is available, adding a 1972 interaction tests whether the most affected districts were already growing differently before the war. Extended specifications add 1981 characteristics (literacy, indigenous share, population, and geography) or department-by-year fixed effects, which

allow regions to follow their own trends.

The identifying assumption is that, without the war, the most affected districts would have followed the trends of the comparison group. Violence was not randomly assigned, because Sendero Luminoso first targeted highland districts with indigenous populations, limited state presence, and historical grievances, and these characteristics may have shaped development on their own. The 1972 pre-trend test, the controls for 1981 characteristics, and the regional fixed effects address part of this concern, but selection on unobservables cannot be fully ruled out. The outcomes also describe the residents of each district at each census, not a fixed population, so the estimates measure what happened to places, including the effects of displacement and return. The 1981 census also records where residents lived in 1976, and Appendix Table A5 shows that very-high-conflict districts did not differ from the rest in the share of residents who lived in the same province five years earlier, and therefore in recent in-migration, or in the stock of residents born in another district.

Three further concerns deserve attention. The first is that very-high-conflict districts are few, so conventional inference may overstate precision. Randomization inference places the national 1993 deficit at the margin of what random assignment would produce, and the 2007 and 2017 coefficients well within that range. The second concerns the single pre-period of the main panel. For the highland loss, bounds that allow the post-war trend to deviate from the pre-war trend by up to twice the change estimated between 1972 and 1981 still exclude zero (Rambachan and Roth, 2023). The third concerns spillovers, which Section 4.3 examines. Appendix Table A37 reports these bounds together with a sensitivity analysis in the sense of Oster (2019), in which observed controls move the national 1993 coefficient away from zero.

Standard errors are clustered by district. Because violence was spatially concentrated, the main results are also examined with standard errors clustered by 1981 province (159 clusters) and with a wild cluster bootstrap by department (Roodman et al., 2019; Webb, 2023), and the highland population loss also with department clusters and Conley (1999) standard errors. The main tables report the province-cluster and wild bootstrap p-values.

4 Demographic Disruption and Recovery

Table 2 and Figure 2 report the population estimates. Nationally, very-high-conflict districts had grown 0.12 log points less than other districts by 1993 and then grew 0.17 log points more between 1993 and 2007, so that by 2007 and 2017 their coefficients are close to zero. The recovery between 1993 and 2007 holds with 1981-province clusters and with the department wild bootstrap, but the 1993 deficit does not, and the national comparison mixes regions

with different histories. In the jungle, several of the most affected districts were frontier areas of agricultural colonization that were already growing faster before the war, which is why the national 1972 coefficient is negative.

What recovery means here deserves precision. The coefficients measure the population of the most affected districts relative to other districts, against their 1981 difference. A coefficient close to zero in 2017 means that these districts regained their 1981 position relative to the rest of the country, not that they returned to the path they would have followed without the war. The data cannot rule out a lasting shortfall either, because the confidence interval of the 2017 coefficient is wide. The national evidence therefore supports relative recovery of population after 1993, and the highland evidence, discussed next, shows the shock that preceded it. Appendix Table A4 reports the full specification on the 1981 panel, with columns that change the comparison group, include every intensity category, add geographic controls, and exclude the department of Lima. The 1993 coefficient stays between -0.11 and -0.14 across these choices, and the recovery between 1993 and 2007 appears in all of them. Appendix Figure A12 shows the raw means behind the estimates.

Table 2: Population of Very-High-Conflict Districts, 1972–2017

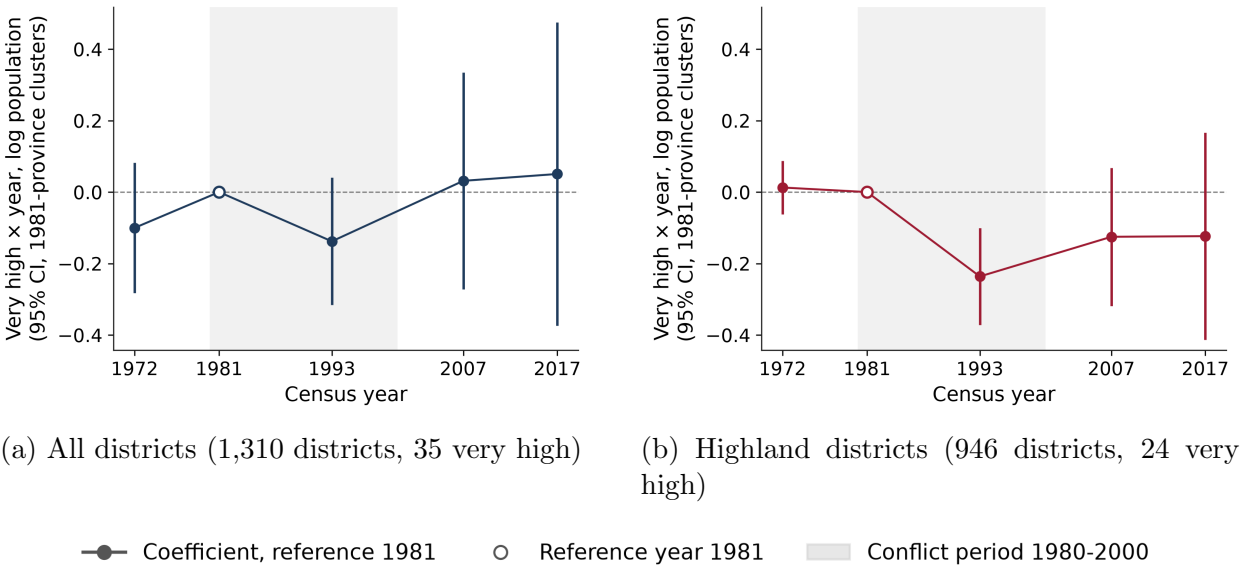
Exposure group	(1) All districts, 1981 reference	(2) All districts, with 1972	(3) Highland districts, with 1972	(4) All districts, with 1972, department × year FE	(5) All districts, with 1972
	Very high conflict, more than 100 victims				>5 victims per 1,000, ≤100 victims
Exposed × 1972		−0.100	0.013	−0.176**	0.049
		(0.092)	(0.038)	(0.083)	(0.031)
Exposed × 1993	−0.118	−0.138	−0.236***	0.019	−0.269***
	(0.092)	(0.090)	(0.069)	(0.068)	(0.062)
Exposed × 2007	0.049	0.031	−0.125	0.206	−0.166***
	(0.146)	(0.153)	(0.097)		(0.057)
Exposed × 2017	0.057	0.051	−0.123	0.357	−0.315***
	(0.198)	(0.215)	(0.146)		(0.071)
Change 1993 to 2007	0.167		0.111		
p, 1993, 1981-province clusters	0.203	0.128	<0.001	0.777	<0.001
p, 1993, wild bootstrap by department	0.798		0.084		0.032
p, 2017, wild bootstrap by department	0.803				0.005
p, change 1993 to 2007, 1981-province clusters	0.021		0.046		
p, change 1993 to 2007, wild bootstrap by department	0.073		0.138		
Observations	5,991	6,550	4,730	6,550	6,550
Districts	1,540	1,310	946	1,310	1,310
Exposed districts	42	35	24	35	89

Notes. Dependent variable log population. Each column reports exposure × census-year coefficients with district and year fixed effects and 1981 as the reference year. Column (1) uses all districts and the censuses of 1981 to 2017. Columns (2) to (4) use the districts observed in 1972 and 1981 and add the 1972 census, and Column (3) restricts them to highland districts. Column (4) replaces year fixed effects with department-by-year fixed effects. Standard errors clustered by 1981 province in parentheses, with stars from the same clustering. The change rows test the 2007 coefficient minus the 1993 coefficient. The wild cluster bootstrap uses Webb weights by department. The exposure group is the very-high-conflict districts in Columns (1) to (4). Column (5) reports, on the sample of Column (2), the coefficients of the districts with more than 5 CVR victims per 1,000 inhabitants of 1981 and 100 or fewer victims, estimated jointly with indicators for the very-high districts above and below 5 victims per 1,000, and its last row counts those small districts (Appendix Table A2). Empty cells are statistics not computed for that sample. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

4.1 The Highland Shock and Ayacucho

The highlands provide the cleaner comparison. There the most affected districts had grown like other highland districts between 1972 and 1981, and by 1993 they had lost 0.24 log points of population, about 21 percent, relative to them. This loss holds under every inference method, as Appendix Table A6 reports, and it holds when any single province or treated district is left out and grows with the threshold from 25 to 200 victims, as Appendix Table A3 shows. By 2007 the gap had narrowed and could no longer be distinguished from zero, although the confidence intervals of 2007 and 2017 are wide enough to include a lasting shortfall.

The highland estimate is the most informative result of the paper, because it combines a pre-war observation with a comparison group of districts that shared the geography, the economy, and the ethnic composition of the treated districts. A loss of a fifth of the population in twelve years is large. It exceeds the number of documented victims in these districts many times over, which suggests that displacement and births that did not take place account for more of it than deaths, although the census does not separate these sources. In these 24 districts the CVR documented about 9,200 victims, while a fifth of their 1981 population amounts to about 98,000 people. The CVR estimated that about 600,000 people left their homes during the war, most of them from the rural highlands (Comisión de la Verdad y Reconciliación, 2003, Tomo VI). The census loss of the highland districts is the local counterpart of that exodus.



(a) All districts (1,310 districts, 35 very high)

(b) Highland districts (946 districts, 24 very high)

Figure 2: Population of Very-High-Conflict Districts, 1972–2017

Notes. Coefficients on very-high-conflict districts (>100 CVR-documented victims) interacted with census-year indicators, with log population as the outcome, district and year fixed effects, and 1981 as the reference year (hollow marker). Panel (a) uses all districts observed in 1972 and 1981 (6,550 observations, 1,310 districts, 35 of them very high), and panel (b) the highland districts among them (4,730 observations, 946 districts, 24 very high). Nineteen of the 24 highland very-high districts are in Ayacucho. Bars show 95% confidence intervals with standard errors clustered by 1981 province. The shaded band marks the conflict period 1980–2000.

The shock was regional. With department-by-year fixed effects, which compare each district only with others in its own department, the 1993 coefficient is close to zero, as a shock that struck whole departments would produce, and within departments the most affected districts had been growing faster before the war, as Column (4) of Table 2 shows.

Nineteen of the 24 highland districts in the treatment group lie in Ayacucho, so the loss is above all a loss of Ayacucho. Within Ayacucho, population change between 1981 and 1993 is unrelated to district victim counts. Population shifted instead towards Huamanga, the province of the regional capital, which grew 0.35 log points more than the rest of the department by 1993 and kept growing faster afterwards. Compared with the other highland departments, Ayacucho lost about 13 percent of its highland population by 1993, without a differential trend before the war, and by 2007 the gap was no longer significant. The evidence therefore identifies a regional demographic shock in the epicenter of the war, followed by a reorganization of population within the region, and it does not separate the war from other factors specific to Ayacucho.

The concentration of population in Huamanga illustrates how the war reorganized the region. Huamanga was the seat of the regional government, of the university where Sendero Luminoso had begun, and of the military command of the emergency zone (Degregori, 2010; Comisión de la Verdad y Reconciliación, 2003). During the war it received families from the surrounding countryside who sought the relative security of the city, and after the war it kept growing as the administrative and commercial center of the department. Rural districts of Ayacucho lost population, Huamanga gained it, and the department as a whole lost population relative to the other highland departments. This pattern explains why the district-level victim count does not predict the population loss within Ayacucho. The loss was shared by districts with very different numbers of victims, because the threat of violence and the collapse of rural economies affected the whole countryside, not only the communities where killings took place.

4.2 Exposure Relative to Population

A count of victims measures how much violence a district suffered, not how much of its population it reached. Table A2 in the Appendix re-estimates the population path with exposure defined relative to the 1981 population and with victims recorded before the 1993 census. Restricting exposure to the victims of 1980–1992 leaves the results unchanged, because nearly all treated districts had passed the threshold by 1992.

The comparison across intensity categories points in the same direction. Nationally, districts with medium, high, and very high intensity all grew less than districts without victims by 1993, by similar amounts, so the national pattern does not rise with intensity. In the highlands, where the 1972 census provides a pre-period, the 1993 loss grows with the threshold, from about 0.14 log points above 25 victims to about 0.40 above 200. Violence of every intensity was associated with slower growth during the war, and the largest losses

appear where violence was most concentrated within the highland region.

The per-capita measures change the picture of recovery. Districts with more than 5 victims per 1,000 inhabitants lost about 0.26 log points of population by 1993 relative to other districts, without a differential trend between 1972 and 1981, and they were still about 0.25 log points behind in 2017. Controlling for the 1981 size of each district does not change this result. The persistence of this pooled estimate holds with the department wild bootstrap in the highlands but not in the national sample. When both definitions enter one regression, the 31 districts that exceed both thresholds follow the path of the main estimates, with a loss in 1993 and recovery by 2007. The 89 smaller districts that exceed the per-capita threshold with 100 or fewer victims lost about 0.27 log points by 1993 and remained 0.32 log points behind in 2017, a deficit that holds with the department wild bootstrap nationally and in the highlands and that Column (5) of Table 2 reports. Defining both thresholds with the victims recorded by 1992, before the 1993 census, gives 87 such districts and almost the same path, while the four large districts with more than 100 victims and a low rate per inhabitant grew faster than the rest before and after the war. Recovery therefore describes the districts with the most victims, and the small districts where violence touched a large share of residents did not recover their population. In the highlands, stricter per-capita thresholds select districts that were already shrinking before the war, so I keep the absolute threshold as the main definition and read the per-capita results as complementary evidence. They describe different places, not a contradiction of the main estimates, and they show that the population shock was more persistent where violence reached a large share of residents. Like the main definition, they rest on parallel trends, which the absence of a differential between 1972 and 1981 supports but cannot prove. Across intensity categories the national pattern is not monotonic, and I read comparisons across intensity levels as descriptive (Callaway et al., 2024).

Why would small places with intense violence fail to recover while large places with many victims did? The censuses cannot answer this question, but they narrow the possibilities. In a small district, a hundred victims can represent a large share of adult men, local authorities, and community leaders, and the CVR documents that many massacres destroyed or emptied entire communities (Comisión de la Verdad y Reconciliación, 2003). Families that left such places had fewer reasons to return, because the social and productive networks that sustained rural life had been broken. Larger districts, which often contain a provincial capital, had urban centers that kept functioning during the war and attracted migrants afterwards. The per-capita results are therefore consistent with recovery that depended on the survival of a local economic and institutional core, although the data do not observe that core directly.

4.3 Spatial Comparisons

Displacement could have moved population into neighboring districts and inflated the comparison group. Appendix Figure A2 maps the zones these designs use. Excluding the districts near the conflict core leaves the estimates almost unchanged, whether distance is measured between district centroids or between district boundaries, and no ring of nearby districts shows a significant population gain, also when districts with and without documented victims are separated within each ring. The losses that appear in wider rings, 75 to 100 km from the conflict core, look like the long-run depopulation of the highlands, whose share of the population of the districts observed in 1981 fell from 39 percent in 1981 to 30 percent in 2017, and not like a spillover that decays with distance. Comparing each very-high-conflict district only with districts of its own province tells the same story. Within provinces the 1993 coefficient is close to zero, so the loss was shared by the whole province, consistent with a regional shock. Appendix A.4 reports these designs. Excluding the districts within 25 to 100 km of a very-high-conflict district moves the 1993 coefficient only between -0.12 and -0.14, the ring design shows small coefficients within 50 km and losses beyond 75 km, and assigning the very-high label at random to other districts of the same provinces places the 1993 population deficit in the tail of the placebo distribution, as Appendix Tables A13, A14, A12, and A36 report.

These results matter for the interpretation of the main estimates. If refugees had moved to nearby comparison districts, the population loss of the most affected districts would overstate the effect of the war. The absence of gains near the conflict core and the shift of population towards Huamanga suggest that displaced families moved mostly to provincial capitals or beyond the neighboring districts.

4.4 Cohorts, Age Structure, and Migration

The composition of the population changed as well. Following birth cohorts across censuses, the cohort aged 15 to 29 in 1981 made up a smaller share of the population of very-high-conflict districts than elsewhere, by 2.1 percentage points in 1993, 3.4 in 2007, and 3.7 in 2017, as Figure 3 shows. The deficits of 1993 and 2007 hold at the 10 percent level with 1981-province clusters, while the 2017 estimate is less precise. Controlling for 1981 characteristics shrinks the deficit to between 1.2 and 1.4 points without changing its sign, and measured as a share of the adult population, which removes differences in fertility, it is larger. The cohort that was entering adulthood when the war began therefore remained underrepresented in the places where violence was most intense, although the census data cannot separate death from permanent departure or non-return.

The deficit of this cohort is the most direct trace of the war in the age structure. These were the men and women aged 27 to 41 in 1993, the age groups most exposed to recruitment by Sendero Luminoso, to suspicion and detention by the armed forces, and to flight. Their smaller share in 2007 and 2017, when they were in their forties to sixties, is consistent with limited return of those who left during the war, although cohort shares cannot show who left and who came back.

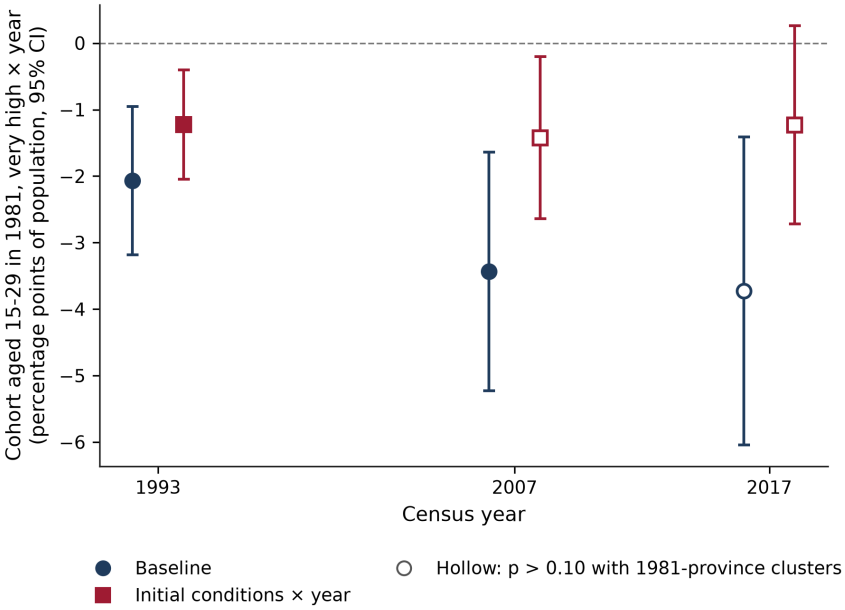


Figure 3: Cohort Aged 15 to 29 in 1981, Very-High-Conflict Districts

Notes. Coefficients on very-high-conflict districts interacted with census year for the population share of the cohort aged 15 to 29 in 1981 (percentage points), with district and year fixed effects, and 1981 as the reference year. Circles show the baseline specification (5,991 observations, 1,540 districts, 42 very high), and squares add 1981 initial conditions interacted with year (5,484 observations, 1,371 districts, 39 very high). Bars show 95% confidence intervals with standard errors clustered by district. Hollow markers indicate coefficients whose p-value with standard errors clustered by 1981 province exceeds 0.10.

The loss of a cohort did not become a permanent change in the age structure. Mean age and the share of the elderly show no robust differential, and the higher shares of children and the higher dependency ratio of 2007 had faded by 2017. Nor do the censuses show population turnover through in-migration. The share of residents born in another district did not rise differentially, and neither did the shares of indigenous-language or Quechua speakers. Appendix Tables A16 and A17 report these outcomes. Birthplace is recorded relative to the current district, so a resident born in another part of the same 1981 unit counts as born elsewhere, which raises measured migration in units that later split but affects

the comparison only through differences in splitting. The censuses also cannot distinguish the return of former residents from their permanent absence, and they do not record where the displaced settled.

The sex ratio tells a related story. The number of men per 100 women fell by about 3 in very-high-conflict districts by 1993 and by about 5 by 2017 relative to other districts. Men were the large majority of the documented victims (Comisión de la Verdad y Reconciliación, 2003) and were more likely to be recruited, detained, or forced to flee, so a lower male share in 1993 is what the war would produce. Its persistence through 2017 suggests that the demographic imprint of the war on these districts did not disappear with the return of peace.

5 Human Capital Convergence

Census definitions of literacy and schooling changed over time, so I compute all shares from district tabulations by age group. The main literacy measure covers residents aged 15 and over, a universe that is identical in the four censuses, and educational attainment is measured among residents aged 25 and over. Table 3 reports the estimates and Figure 4 separates two questions, whether the gap closed and whether the most affected districts gained more than districts that started from the same level.

Table 3: Effect of Conflict on Literacy and Educational Attainment

	Literacy 15+		Secondary+ 25+		Higher educ. 25+	
	(1)	(2)	(3)	(4)	(5)	(6)
	Baseline	Initial cond.	Baseline	Initial cond.	Baseline	Initial cond.
Very High × 1993	1.487	−1.987**	−2.354***	−1.341**	−0.838	−0.735
	(1.157)	(0.899)	(0.904)	(0.624)	(0.703)	(0.488)
Very High × 2007	5.508***	0.427	−6.283***	−1.232	−3.156**	−0.440
	(1.702)	(0.726)	(1.404)	(1.182)	(1.271)	(0.927)
Very High × 2017	6.615***	0.630	−5.495***	−1.615	−1.554	−0.730
	(1.876)	(0.674)	(1.432)	(1.155)	(1.333)	(0.947)
Observations	5,991	5,484	5,991	5,484	5,991	5,484
Districts	1,540	1,371	1,540	1,371	1,540	1,371
Treated districts (very high)	42	39	42	39	42	39
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Initial conditions × year	No	Yes	No	Yes	No	Yes

Notes. Outcomes are shares in percent of the age group, namely literacy among persons aged 15 and over, secondary complete or more and higher education (non-university or university, complete or incomplete) among persons aged 25 and over, with the same age universe in every census. All specifications include district and year fixed effects, with 1981 as the reference year. Standard errors clustered by district in parentheses. Initial conditions are the own 1981 level, 1981 literacy, indigenous share, log population and geography, each interacted with year, and these columns use the districts with a 1981 row. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

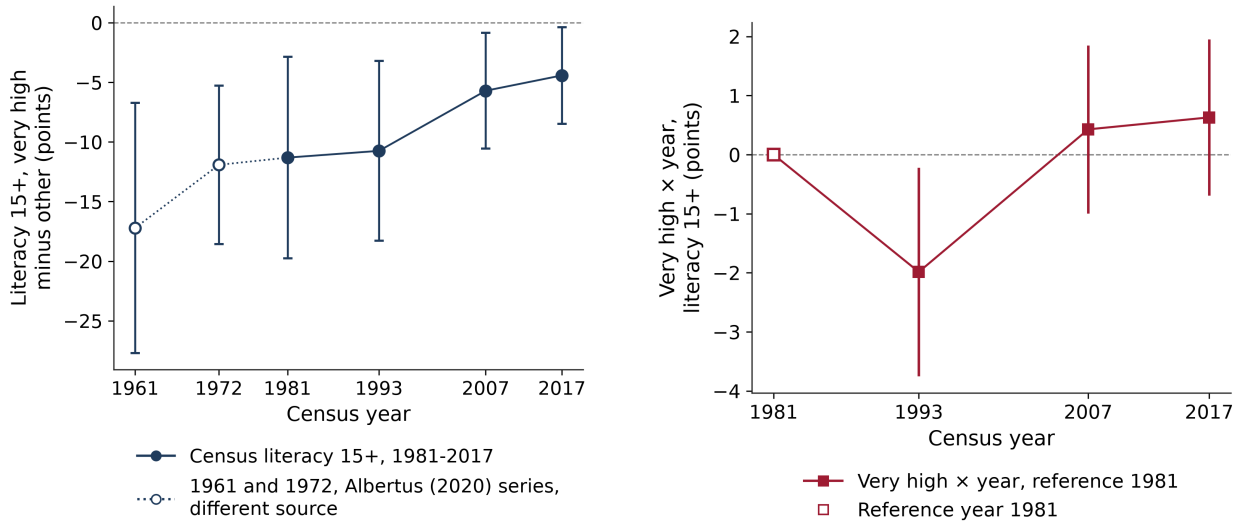
The gap closed. On a fixed sample of districts, the literacy gap between very-high-conflict districts and the rest was 11.3 percentage points in 1981, 10.8 in 1993, 5.7 in 2007, and 4.4 in 2017, a reduction of 61 percent that is robust to spatial inference. Places that lived through the most intense violence reduced most of their educational gap with the rest of the country, and the narrowing came after 1993. Earlier figures from a different source suggest that the gap had already been narrowing between 1961 and 1972, and the change of the conflict decade is similar to that of the decade before it.

The timing of the convergence is informative. The gap barely moved between 1981 and 1993, when teachers were among the targets of violence and families fled, and it narrowed rapidly after 1993. Unconditionally, literacy rose 5.5 percentage points more in very-high-conflict districts by 2007 and 6.6 points more by 2017. The same pattern appears among residents aged 15 to 24, the cohorts educated after the war, whose literacy rose 6.8 points

more by 2007. Much of the convergence therefore came from young people who went to school after the violence ended, in districts where older cohorts had very low literacy.

The convergence is what the low initial literacy of these districts predicts. Districts with low literacy in 1981 gained more everywhere, and conditional on initial conditions the most affected districts gained 2.0 points less than comparable districts between 1981 and 1993, a slowdown that disappears with regional fixed effects, and their gains by 2007 and 2017 cannot be distinguished from those of comparable districts. The war therefore did not prevent convergence, and the evidence does not show that it accelerated it. Within provinces, the literacy gains of the most affected districts were shared by their neighbors, which also started from low levels.

This distinction matters for the interpretation of post-war recovery. A finding that conflict-affected districts converged could be read as evidence that the war accelerated education, through reconstruction programs or through a stronger demand for schooling among displaced families. The conditional estimates do not support that reading. The most affected districts gained about as much as districts with the same initial literacy, indigenous share, and geography, and among young residents the conditional difference is small and vanishes by 2017. The expansion of schooling in rural Peru after 1990 reached the former conflict zones as it reached other poor highland districts, and the conditional estimates show no lasting differential in basic literacy once that expansion is taken into account, although they cannot rule out losses for particular cohorts.



(a) Descriptive gap, same 1,309 districts in every year

(b) Conditional on 1981 initial conditions × year

Figure 4: Literacy Convergence in Very-High-Conflict Districts

Notes. Literacy among residents aged 15 and over, in percentage points. Panel (a) is descriptive and shows the difference in mean literacy between very-high-conflict and other districts on the same 1,309 districts in every year, with 95% confidence intervals from standard errors clustered by 1981 province. The 1961 and 1972 points (hollow markers, dotted line) come from the replication data of Albertus (2020), a different source whose age universe is not documented. Panel (b) is conditional and shows the coefficients of Column (2) of Table 3, with 1981 initial conditions interacted with year and 1981 as the reference year (hollow marker), on 5,484 observations, 1,371 districts, and 39 very-high districts. Its bars show 95% confidence intervals with standard errors clustered by district.

Secondary schooling tells a less favorable story. The share of adults who completed secondary school rose between 5.5 and 6.3 percentage points less in very-high-conflict districts by 2007 and 2017, a deficit that holds with regional fixed effects and in the alternative unit definition. However, it shrinks to under 2 points and is no longer significant once the full set of 1981 characteristics is controlled for, so it reflects where these districts started more than what happened to them. Higher education shows a smaller deficit that is not robust.

The contrast between literacy and secondary completion has a natural reading. Literacy depends on primary schooling, which expanded across rural Peru during the 1990s and 2000s. Secondary completion among adults aged 25 and over reflects the schooling of cohorts educated during and before the war, and in the most affected districts these cohorts had less access to secondary schools to begin with. The deficit is therefore a mark of where these districts started, and the war may have deepened it for the cohorts in school during the violence, although the data cannot separate the two.

6 Infrastructure Recovery

Table 4 and Figure 5 report electricity and sanitation, and Appendix Table A10 covers six indicators drawn from the census housing modules. Each share is computed over dwellings with a valid answer.

Table 4: Electricity and Sanitation in Very-High-Conflict Districts

	(1) Baseline			(2)	(3)
	Coefficient	p, 1981 provinces	p, wild bootstrap	Initial cond. × year	Department × year FE
Electricity (% of dwellings)					
Very high × 1993	−3.723*	0.136	0.291	−5.536***	−4.503*
	(2.066)			(2.056)	(2.473)
Very high × 2007	−10.005***	0.003	0.051	−10.780***	−12.799***
	(2.655)			(2.699)	(3.079)
Very high × 2017	0.132	0.972	0.986	−2.406	−5.480**
	(2.626)			(1.970)	(2.473)
Observations	5,987			5,480	5,987
Sanitation, any facility (% of dwellings)					
Very high × 1993	−4.661	0.396	0.635	−2.438	2.526
	(3.609)			(2.650)	(3.237)
Very high × 2007	9.846***	0.018	0.147	9.632***	10.139***
	(3.526)			(2.606)	(3.795)
Very high × 2017	6.719**	0.073	0.465	5.401***	1.142
	(3.409)			(1.555)	(3.371)
Observations	5,987			5,480	5,987
Pit latrines (% of dwellings)					
Very high × 1993	−3.371	0.344	0.537	−1.292	1.601
	(2.137)			(2.094)	(2.096)
Very high × 2007	9.638***	0.027	0.481	8.978***	5.458
	(3.469)			(2.789)	(3.724)
Very high × 2017	9.501***	0.012	0.231	4.892	3.747
	(3.554)			(3.043)	(3.840)
Observations	5,987			5,480	5,987
Districts	1,539			1,370	1,539

Notes. Each cell is the very-high × year coefficient with 1981 as the reference year. Column (1) has district and year fixed effects, with standard errors clustered by district in parentheses and stars from district-clustered p-values, and also reports the p-values of the same coefficient with 1981-province clusters and with the department wild cluster bootstrap (Webb weights). Column (2) adds 1981 initial conditions interacted with year (own 1981 level where available, literacy, indigenous share, log population and geography) and uses the districts with a 1981 record. Column (3) adds department-by-year fixed effects on all districts. Sanitation counts dwellings with a public network, septic tank, or pit latrine. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Electricity lagged most clearly in 2007. Relative to 1981, the share of dwellings with

electricity rose about 10 percentage points less in very-high-conflict districts by 2007, a gap that holds with initial conditions, with regional fixed effects, in the alternative unit definition, and outside Ayacucho, and that sits at the margin of conventional significance with the department wild bootstrap. By 2017 the gap had closed in the baseline specification, although it remains with regional fixed effects and outside Ayacucho. The extension of the network to these districts was therefore delayed, and the delay was not confined to the epicenter of the war.

The size of the electricity gap is substantial. In 2007, about 42 percent of dwellings in very-high-conflict districts had electricity, against 56 percent in other districts, and the gap had widened since 1981, when the two groups were much closer. The gap appeared already in 1993, when the share of dwellings with electricity had risen about 4 points less in the most affected districts, so it began during the war, when electricity towers were frequent targets of sabotage (Comisión de la Verdad y Reconciliación, 2003). By 2017 the network reached most dwellings in these districts, 73 percent against 77 percent elsewhere, and the baseline gap had closed, but it remains when each district is compared with others in its own department and outside Ayacucho, which suggests that the most affected districts were still behind the other districts of their own regions.

Basic sanitation moved in the opposite direction. The share of dwellings with any sanitation facility, from a public sewer to a pit latrine, rose about 10 percentage points more in very-high-conflict districts by 2007 and about 7 points more by 2017. The 2007 estimate is stable across specifications and significant with province clusters but not with the department wild bootstrap, and pit latrines account for the increase. When both outcomes enter one regression, sanitation exceeds electricity by about 20 points in 2007 and by about 7 points in 2017, as Appendix Table A11 reports. The evidence favors divergent trajectories of the two services, although the strength of that conclusion depends on the inference method. Public sewerage, on the other hand, lags only in proportion to its lower coverage in 1981, and public water networks, wall materials, and floors show no robust differential.

The other housing indicators confirm that the divergence is specific to the two services. Public water connections show no robust differential, with a gain by 2017 that is significant only at the 10 percent level, and wall materials and dirt floors, which depend on household investment in the dwelling, show no difference between the most affected districts and the rest. The war did not leave these districts with worse housing, and the gaps that matter concern the services delivered through public networks.

Sanitation coverage was similar in the two groups in 1981 and higher in very-high-conflict districts from 2007 onwards. The increase came almost entirely from pit latrines, the most basic form of sanitation, which households can build with simple materials. Excluding septic

tanks, whose census definition changed over time, gives a smaller but similar estimate. Public sewer connections, which require the same kind of network investment as electricity, lagged in the most affected districts, but only in proportion to their lower coverage before the war.

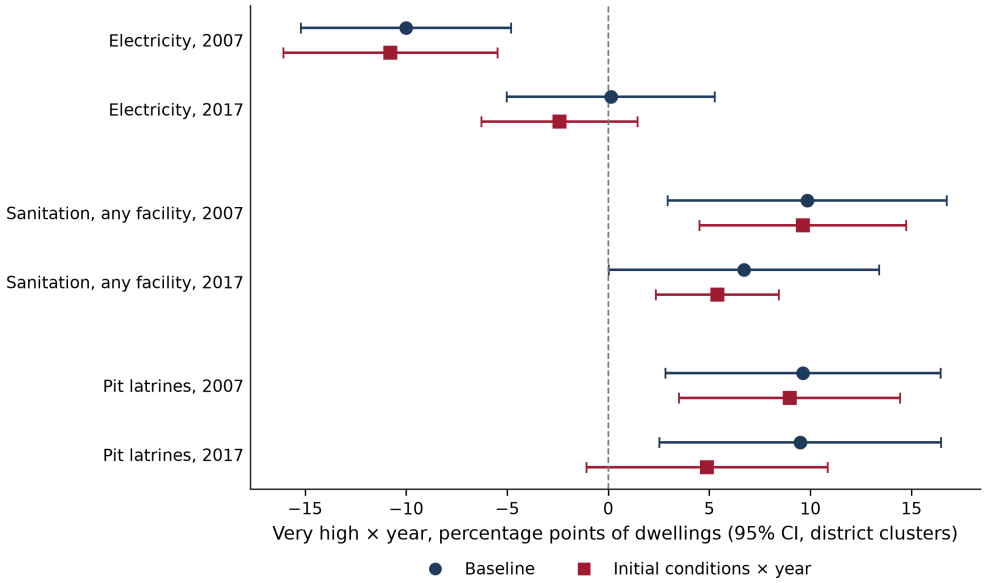


Figure 5: Electricity and Sanitation in Very-High-Conflict Districts, 2007 and 2017

Notes. Coefficients on very-high-conflict districts interacted with census year, in percentage points of dwellings, with district and year fixed effects and 1981 as the reference year. Circles show the baseline specification (5,987 observations, 1,539 districts), and squares add 1981 initial conditions interacted with year (5,480 observations, 1,370 districts). Bars show 95% confidence intervals with standard errors clustered by district. Sanitation counts dwellings with a public network, septic tank, or pit latrine.

The contrast is consistent with a difference in how the two services are delivered. Electricity reaches dwellings through networks that depend on utilities and public investment, which the conflict damaged or kept away, while pit latrines can be built by households and communities on site, often with modest program support. Peru's post-conflict programs, including the social investment fund FONCODES (Fondo de Cooperación para el Desarrollo Social), operated intensively in the poor highland areas where the conflict had concentrated, although program allocation is not observed directly. These channels are hypotheses that the census data cannot test.

7 Perpetrator Heterogeneity

The CVR attributed each documented death or disappearance to a perpetrator, so the most affected districts can be classified by the actor responsible for most of their victims. Of the

42 very-high-conflict districts, 24 are Sendero-dominant, 7 State-dominant, and 11 Mixed. Sendero-dominant districts cluster in the central highlands around Ayacucho, while State-dominant districts are more dispersed and appear along the jungle frontier and in areas of counterinsurgency operations, as Appendix Figure A5 shows. Table 5 estimates the three groups in one regression against the same comparison group and tests whether Sendero-dominant and State-dominant districts followed the same path.

The three groups differed before the war. Among all districts with documented victims, Sendero-dominant districts were smaller, with a mean population of about 7,900 against about 29,000 in State-dominant and 34,000 in Mixed districts. They were less literate, at about 66 percent among residents aged 15 and over against 71 and 74 percent, they were located about 500 meters higher, and they recorded more victims on average. Sendero Luminoso's violence therefore concentrated in small, poor, high-altitude communities, while state violence was more frequent in larger and somewhat more developed districts, often along the jungle frontier and in areas of counterinsurgency operations.

Table 5: Recovery by Dominant Perpetrator, Joint Model with a Common Comparison Group

	Sendero dominant	State dominant	Mixed	p, Sendero = State	
				District clusters	Province clusters
Log population					
Very high × 1993	−0.104	−0.232	−0.066	0.538	0.448
	(0.078)	(0.193)	(0.169)		
Very high × 2007	0.141	−0.224	0.030	0.128	0.106
	(0.105)	(0.217)	(0.187)		
Very high × 2017	0.130	−0.256	0.110	0.192	0.181
	(0.131)	(0.266)	(0.248)		
Joint p, three years				0.143	0.021
Literacy, ages 15+ (%)					
Very high × 1993	1.356	5.716	−1.984	0.103	0.116
	(1.550)	(2.197)	(1.357)		
Very high × 2007	7.374	9.466	−2.144	0.615	0.653
	(2.121)	(3.591)	(2.881)		
Very high × 2017	8.141	12.372	−1.439	0.374	0.416
	(2.247)	(4.215)	(3.403)		
Joint p, three years				0.227	0.010
Electricity (%)					
Very high × 1993	−6.301	−1.177	0.155	0.178	0.123
	(2.816)	(2.652)	(3.875)		
Very high × 2007	−8.429	−14.671	−10.601	0.313	0.077
	(3.978)	(4.833)	(3.266)		
Very high × 2017	1.595	3.708	−5.463	0.755	0.749
	(3.201)	(6.041)	(5.342)		
Joint p, three years				0.227	<0.001
Very-high districts	24	7	11		
Observations (log population)	5,991				

Notes. One regression per outcome with the very-high indicator interacted with the three dominant-perpetrator groups and census years, district and year fixed effects, reference year 1981. All districts below the very-high threshold form a common comparison group. Standard errors clustered by district in parentheses. Sendero (State) dominant means more than half of the documented victims were attributed to Sendero Luminoso (state agents), and Mixed covers the remaining very-high districts. The joint test covers the three census years. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district).

The two groups share the main patterns of the paper. Both caught up in literacy, by 8 points in Sendero-dominant districts and 12 points in State-dominant districts by 2017, and both lagged in electricity in 2007, by 8 and 15 points. They differ in population. Sendero-dominant districts recovered after 1993 and grew faster than the comparison group afterwards, while State-dominant districts remained between 0.22 and 0.26 log points behind through 2017. Year by year these differences are not statistically significant, except for electricity in 2007 at the 10 percent level with province clusters. Across the three years, the tests with province clusters reject equality for population, literacy, and electricity, while those with district clusters do not. Among all conflict districts, the share of victims attributed to Sendero is also related to the paths of population, literacy, and mean age.

The separate estimates by group tell a consistent story. When very-high-conflict districts are compared only with lower-intensity districts of their own group, Sendero-dominant districts grew faster than other Sendero-dominant districts after 1993, by about 0.20 log points by 2007 and 0.29 by 2017, and became younger. State-dominant very-high-conflict districts show no such gain. Among all conflict districts, a higher share of victims attributed to Sendero goes with slower population growth and faster literacy growth, while a higher state share goes with the opposite. The continuous shares use the variation among hundreds of districts with lower intensity, so they answer a different question from the comparison of very-high-conflict districts, and for population they point the other way, so the weaker population path of State-dominant districts is specific to those with very high violence. Appendix Tables A41, A42, A40, and A44 report these estimates.

These are suggestive differences by perpetrator, a weaker demographic recovery where state agents were the main perpetrators, and not differences the tests establish year by year. The CVR documents that Sendero targeted local authorities, teachers, and community leaders (Comisión de la Verdad y Reconciliación, 2003, Tomos II and VII), and state violence included massacres and disappearances in counterinsurgency operations (Comisión de la Verdad y Reconciliación, 2003), so the two kinds of violence may have affected trust, return, and local institutions in different ways. Appendix E reports the separate estimates by group and the continuous perpetrator shares.

8 Night-Time Lights

Census outcomes describe population and services, and satellite night-time lights add a view of where activity spread after the war. The main series is the harmonized annual product of Chiovelli et al. (2026), which places DMSP and VIIRS observations for 1992–2025 on a single scale and corrects for blooming and top-coding. Light is summed over the fixed district

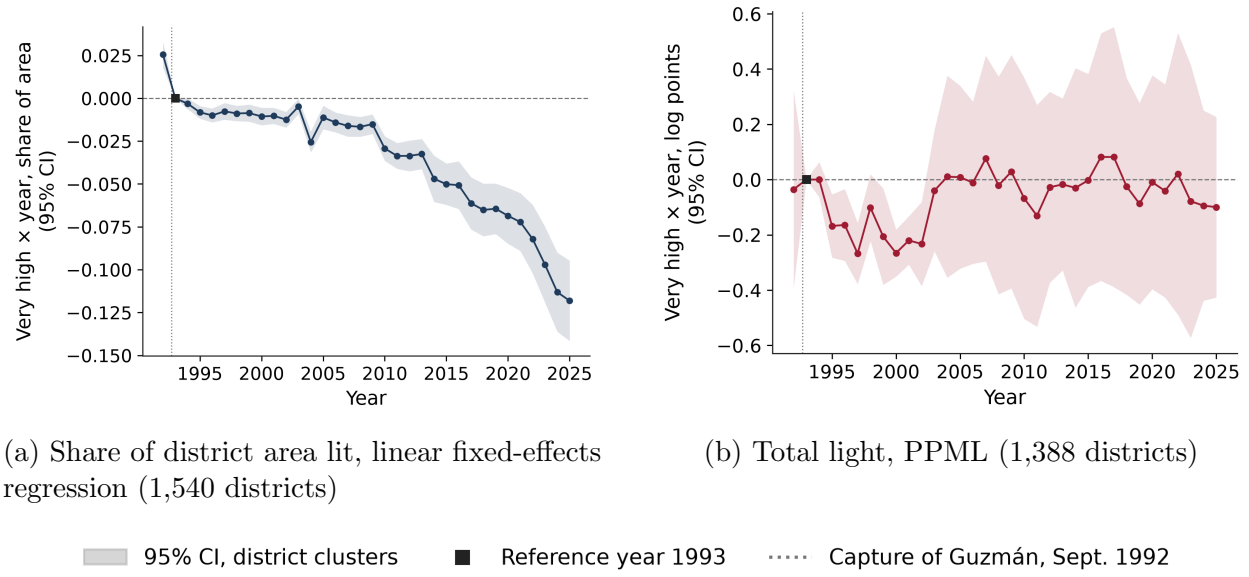
polygons, and I use total light and the share of each district's area that is lit. The satellite record begins in 1992, when the conflict was well advanced, so lights describe the period of recovery, with 1993 as the reference year, and cannot be compared with a pre-war baseline. Appendix G reports the full set of series, sensors, and specifications.

Total light in very-high-conflict districts did not grow at a detectably different rate from that of other districts. The Poisson estimates for 1993–2007 and 2007–2017 are small and far from significant, and the probability of any detected light shows no differential either. Panel (b) of Figure 6 shows the yearly path of total light, which fell below its 1993 level in the late 1990s and returned to it from 2004 onwards, with wide confidence bands. There is therefore no detectable differential convergence in total night-time luminosity, which does not mean that economic activity grew identically, because lights capture output poorly in rural districts.

The choice of series matters for this conclusion. Products that switch from the DMSP to the VIIRS sensor change scale at the switch, and dim districts gain detected light because VIIRS sees fainter sources. In the series of Chen et al. (2024) and Li et al. (2020) the most affected districts appear to gain total light after 2007, but the gain coincides with the change of sensor and disappears within each sensor period. Appendix Tables A52, A53, and A58 report the contrasts for every series, sensor period, and correction variant.

Light also spread over a smaller share of the territory of the most affected districts. In 1993, 3.2 percent of the area of the average very-high-conflict district was lit, against 8.0 percent elsewhere, and by 2017 the shares were 6.4 and 17.3 percent. Panel (a) of Figure 6 shows that the gap widened mainly after 2010. In proportional terms lit area roughly doubled in both groups, so the widening gap in percentage points reflects the lower starting coverage of the most affected districts more than a slower expansion. The sign of the gap is the same in every series, threshold, and correction variant. In absolute terms, then, light reached less of the territory of these districts, which is consistent with the concentration of population in Huamanga and with the delayed extension of the electricity network, although the data do not show that one explains the other.

Lit area captures something different from total light. Total light grows when existing towns become brighter, and lit area grows when light reaches new places, such as villages connected to the network. In the most affected districts light grew in the towns that already had it and reached new parts of the territory more slowly. In 2007, 4.1 percent of the area of the average very-high-conflict district was lit, against 10.5 percent elsewhere.



(a) Share of district area lit, linear fixed-effects regression (1,540 districts)

(b) Total light, PPML (1,388 districts)

Figure 6: Night-Time Lights in Very-High-Conflict Districts, 1992–2025

Notes. Coefficients on very-high-conflict districts interacted with year indicators, with district and year fixed effects and 1993 as the reference year, in the harmonized series of Chiovelli et al. (2026) (main correction variant, DMSP-like scale in digital numbers). Panel (a) shows the share of district area with positive light, estimated by linear regression with district and year fixed effects on 1,540 districts (52,360 district-year observations). Panel (b) shows total light estimated by Poisson pseudo-maximum-likelihood, in log points. PPML drops the districts never lit in 1992–2025, so panel (b) uses 1,388 districts (47,192 observations). Shaded bands are 95% confidence intervals with standard errors clustered by district.

Within Ayacucho, districts with more victims gained less light between 2007 and 2017, in total and per inhabitant, and the direction is the same in every series. Their lit area, on the other hand, grew no less than elsewhere in the department, so the shortfall came from the intensity of light in places that were already lit. The estimate rests on the seven provinces of the department and I read it as descriptive, but it suggests that within the region of the war, activity recovered least where violence had been most intense.

9 Conclusion

In this paper, I study what happened to the places that Peru's internal armed conflict devastated, following the districts of the 1981 census through 2017, with the 1972 census as an earlier benchmark. Recovery was demographically uneven, educationally convergent, and spatially concentrated.

The war produced a regional demographic shock. The highland districts with the most victims lost about a fifth of their population relative to their neighbors by 1993, a loss that

belongs to Ayacucho and that was accompanied by a shift of population towards the regional capital. Nationally, the districts with the most victims regained their relative population within about fifteen years, but the cohort that was young when the war began remained underrepresented, and small districts where violence touched a large share of residents did not recover. Literacy converged, as the low initial levels of these districts predict, so the war did not prevent the spread of basic education. Public services recovered unevenly, with a delayed extension of the electricity network and a faster expansion of on-site sanitation. Light reached less of the territory of the most affected districts. Recovery also shows suggestive differences by perpetrator, with a weaker demographic path where state agents committed most of the violence.

The timing of these changes shows how the pieces fit together. The population loss and the missing cohort appear between 1981 and 1993, while the war was being fought, and they are the largest changes of that decade. Literacy, sanitation, and electricity, on the other hand, changed mostly after 1993, once the violence had subsided. Literacy and on-site sanitation, which households, schools, and communities could expand with modest support, moved in favor of these districts, while electricity, which required the extension of a network, moved against them in 2007, a gap that had largely closed by 2017. The war therefore left its direct mark on who lived in these places, and the slower recovery of networked services suggests that its longer shadow fell on the capacity to deliver services that depend on large public investment.

For the debate on whether war traps places in poverty, the Peruvian evidence points to relative recovery of population in the main centers of violence within about fifteen years, and to slower recovery in small places, in the cohorts exposed as young adults, and in networked services. However, these findings concern districts and not people. Studies that follow individuals exposed to the violence find lasting losses in schooling and earnings (León, 2012; Galdo, 2013), and a district can recover its population and its literacy rate while those who lived through the war carry such losses. The data also cannot apportion the demographic changes between mortality, displacement, and return, and they do not observe reconstruction programs directly. Linking individual migration histories to places of origin, and the allocation of reconstruction spending to districts, would allow those mechanisms to be measured.

These findings have implications for post-conflict policy. Basic literacy converged as the general expansion of schooling reached the former conflict zones, and on-site sanitation expanded with modest support to households and communities. Networked services were slower, because they required public investment that the war had interrupted and that took years to resume. The places that did not recover their population were the small districts

where violence reached a large share of residents, which are also the places least visible in aggregate statistics. Reconstruction policies that target districts by the number of victims would miss many of them, and measures of exposure relative to population would identify them better.

References

Abadie, A. and Gardeazabal, J. (2003). The economic costs of conflict: A case study of the Basque Country. *American Economic Review*, 93(1):113–132.

Albertus, M. (2020). Land reform and civil conflict: Theory and evidence from Peru. *American Journal of Political Science*, 64(2):256–274.

Blattman, C. and Miguel, E. (2010). Civil war. *Journal of Economic Literature*, 48(1):3–57.

Brakman, S., Garretsen, H., and Schramm, M. (2004). The strategic bombing of German cities during World War II and its impact on city growth. *Journal of Economic Geography*, 4(2):201–218.

Burt, J.-M. (2007). *Political Violence and the Authoritarian State in Peru: Silencing Civil Society*. Palgrave Macmillan, New York.

Callaway, B., Goodman-Bacon, A., and Sant'Anna, P. H. C. (2024). Difference-in-differences with a continuous treatment. NBER Working Paper 32117, National Bureau of Economic Research.

Chen, J. and Roth, J. (2024). Logs with zeros? some problems and solutions. *The Quarterly Journal of Economics*, 139(2):891–936.

Chen, X. and Nordhaus, W. D. (2011). Using luminosity data as a proxy for economic statistics. *Proceedings of the National Academy of Sciences*, 108(21):8589–8594.

Chen, X. and Nordhaus, W. D. (2019). VIIRS nighttime lights in the estimation of cross-sectional and time-series GDP. *Remote Sensing*, 11(9):1057.

Chen, X., Wang, Z., Zhang, F., Shen, G., and Chen, Q. (2024). A global annual simulated VIIRS nighttime light dataset from 1992 to 2023. *Scientific Data*, 11(1):1380.

Chiovelli, G., Michalopoulos, S., Papaioannou, E., and Regan, T. (2026). Illuminating the global south. *The Economic Journal*. Advance article.

Collier, P., Elliott, V. L., Hegre, H., Hoeffler, A., Reynal-Querol, M., and Sambanis, N. (2003). *Breaking the Conflict Trap: Civil War and Development Policy*. World Bank and Oxford University Press, Washington, DC.

Comisión de la Verdad y Reconciliación (2003). *Informe Final*. CVR, Lima.

Conley, T. G. (1999). GMM estimation with cross sectional dependence. *Journal of Econometrics*, 92(1):1–45.

Davis, D. R. and Weinstein, D. E. (2002). Bones, bombs, and break points: The geography of economic activity. *American Economic Review*, 92(5):1269–1289.

Degregori, C. I. (2010). *El Surgimiento de Sendero Luminoso. Ayacucho 1969–1979. Del Movimiento por la Gratuidad de la Enseñanza al Inicio de la Lucha Armada*. Instituto de Estudios Peruanos, Lima, 3rd edition. First published 1990.

Galdo, J. (2013). The long-run labor-market consequences of civil war: Evidence from the Shining Path in Peru. *Economic Development and Cultural Change*, 61(4):789–823.

Gibson, J., Olivia, S., and Boe-Gibson, G. (2020). Night lights in economics: Sources and uses. *Journal of Economic Surveys*, 34(5):955–980.

Grimard, F. and Laszlo, S. (2014). Long-term effects of civil conflict on women's health outcomes in Peru. *World Development*, 54:139–155.

Gurr, T. R. (1970). *Why Men Rebel*. Princeton University Press, Princeton, NJ.

Henderson, J. V., Storeygard, A., and Weil, D. N. (2012). Measuring economic growth from outer space. *American Economic Review*, 102(2):994–1028.

Ibáñez, A. M. and Vélez, C. E. (2008). Civil conflict and forced migration: The micro determinants and welfare losses of displacement in Colombia. *World Development*, 36(4):659–676.

Instituto Nacional de Estadística e Informática (2020). Perú: Proyecciones de población por departamento, provincia y distrito, 2018–2020. capítulo 2: División política y administrativa al 31 de diciembre del 2019. Technical report, INEI, Lima.

Justino, P. (2009). Poverty and violent conflict: A micro-level perspective on the causes and duration of warfare. *Journal of Peace Research*, 46(3):315–333.

León, G. (2012). Civil conflict and human capital accumulation: The long-term effects of political violence in Perú. *Journal of Human Resources*, 47(4):991–1022.

Li, X., Zhou, Y., Zhao, M., and Zhao, X. (2020). A harmonized global nighttime light dataset 1992–2018. *Scientific Data*, 7(1):168.

Mauceri, P. (1995). State reform, coalitions, and the neoliberal autogolpe in peru. *Latin American Research Review*, 30(1):7–37.

McClintock, C. (1984). Why peasants rebel: The case of peru's sendero luminoso. *World Politics*, 37(1):48–84.

Miguel, E. and Roland, G. (2011). The long-run impact of bombing Vietnam. *Journal of Development Economics*, 96(1):1–15.

Nunn, N. and Puga, D. (2012). Ruggedness: The blessing of bad geography in Africa. *Review of Economics and Statistics*, 94(1):20–36.

Oster, E. (2019). Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business & Economic Statistics*, 37(2):187–204.

Palmer, D. S. (1995). The revolutionary terrorism of peru's shining path. In Crenshaw, M., editor, *Terrorism in Context*, pages 249–308. Pennsylvania State University Press, University Park, PA.

Rambachan, A. and Roth, J. (2023). A more credible approach to parallel trends. *Review of Economic Studies*, 90(5):2555–2591.

Roodman, D., Nielsen, M. Ø., MacKinnon, J. G., and Webb, M. D. (2019). Fast and wild: Bootstrap inference in Stata using boottest. *The Stata Journal*, 19(1):4–60.

Santos Silva, J. and Tenreyro, S. (2006). The log of gravity. *The Review of Economics and Statistics*, 88(4):641–658.

Starn, O. (1998). Villagers at arms: War and counterrevolution in the central-south Andes. In Stern, S. J., editor, *Shining and Other Paths: War and Society in Peru, 1980–1995*, pages 224–258. Duke University Press, Durham, NC.

Stern, S. J., editor (1998). *Shining and Other Paths: War and Society in Peru, 1980–1995*. Duke University Press, Durham, NC.

Theidon, K. (2013). *Intimate Enemies: Violence and Reconciliation in Peru*. University of Pennsylvania Press, Philadelphia.

Thorp, R. and Bertram, G. (1978). *Peru 1890–1977: Growth and Policy in an Open Economy*. Macmillan, London.

Verwimp, P., Justino, P., and Brück, T. (2019). The microeconomics of violent conflict. *Journal of Development Economics*, 141:102297.

Webb, M. D. (2023). Reworking wild bootstrap-based inference for clustered errors. *Canadian Journal of Economics*, 56(3):839–858.

Weinstein, J. M. (2007). *Inside Rebellion: The Politics of Insurgent Violence*. Cambridge University Press, Cambridge.

A Additional Tables and Figures

This appendix collects the tables and figures that the main text cites in support of its results, grouped by topic in the order in which the text discusses them.

A.1 Background and Sample

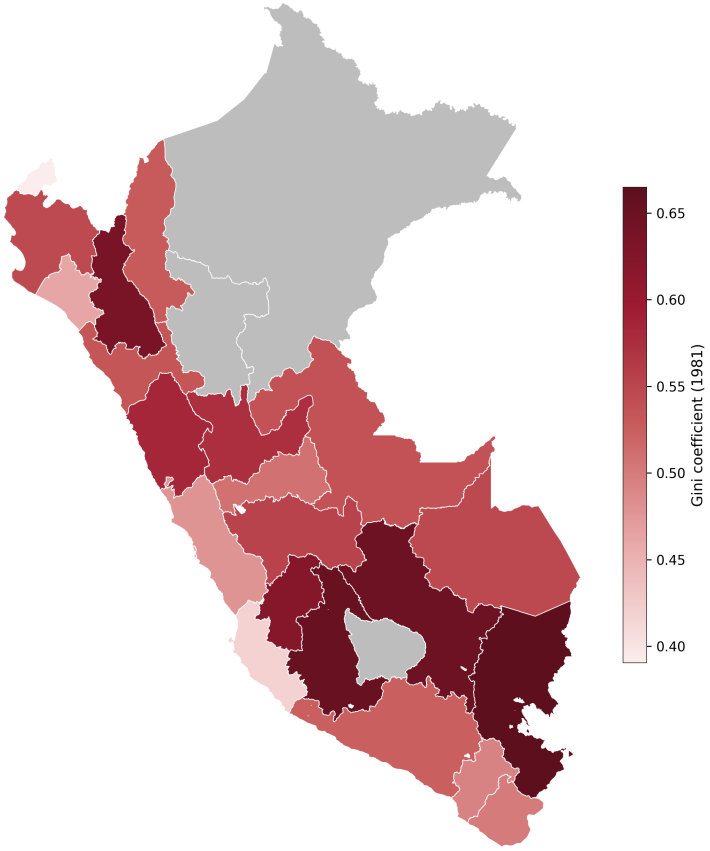


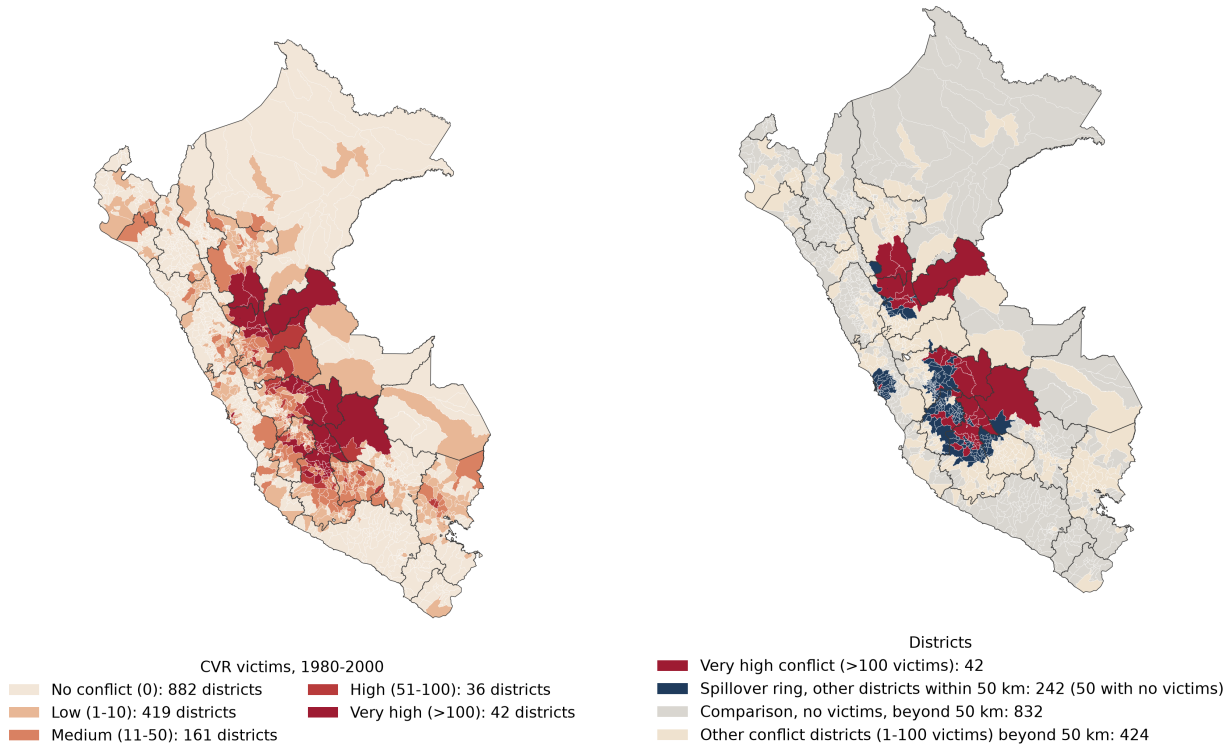
Figure A1: Pre-Conflict Income Inequality by Department, Peru 1981

Notes. Department-level Gini coefficients from grouped income data in the 1981 census. Gray areas (Loreto, San Martín, Apurímac) are not covered in the 1981 census income tabulations used here, except the six districts of Ucayali province (Contamana), today in Loreto, which were tabulated under Ucayali in 1981. Appendix C.2 describes the Gini calculation.

Table A1: Summary Statistics: 1981 Pre-Conflict Baseline by Conflict Intensity

	No Conflict		Low		Medium		High		Very High		VH minus	Normalized
	mean	sd	mean	sd	mean	sd	mean	sd	mean	sd	no conflict	difference
Total population	6,464.28	(13,670.74)	15,994.16	(39,792.53)	28,397.88	(90,496.06)	10,953.79	(15,023.39)	28,350.72	(49,005.84)	21,886.43	0.608
Mean age (years)	24.63	(2.98)	24.48	(2.80)	24.03	(2.50)	23.79	(2.14)	22.79	(2.20)	−1.84	−0.702
Dependency ratio (%)	97.19	(15.43)	97.21	(16.27)	100.47	(17.14)	105.68	(13.27)	99.77	(16.67)	2.59	0.161
Sex ratio (M per 100 F)	102.58	(15.51)	97.69	(12.62)	97.49	(11.94)	95.50	(11.94)	100.73	(17.13)	−1.85	−0.113
Literacy rate, ages 15+ (%)	73.16	(16.27)	72.09	(15.74)	65.43	(17.40)	62.08	(14.73)	60.76	(17.56)	−12.40	−0.732
Higher education (%)	2.22	(3.28)	3.17	(4.37)	2.43	(3.73)	2.52	(3.27)	2.23	(2.73)	0.01	0.004
High-skill occupation (%)	3.42	(3.95)	4.78	(5.30)	3.86	(4.57)	3.65	(3.72)	3.55	(3.38)	0.13	0.036
Agriculture (%)	73.38	(22.28)	64.70	(28.64)	72.05	(26.50)	74.81	(22.92)	73.24	(23.98)	−0.14	−0.006
Manufacturing (%)	5.02	(7.15)	5.81	(6.47)	4.24	(5.17)	4.64	(7.85)	4.43	(5.24)	−0.59	−0.094
Commerce (%)	4.75	(5.04)	6.92	(6.61)	5.69	(6.25)	6.07	(6.35)	6.34	(6.79)	1.59	0.266
Indigenous language (%)	32.99	(39.02)	46.81	(38.20)	60.28	(35.52)	74.03	(30.12)	71.08	(32.02)	38.09	1.067
Migration rate (%)	26.16	(20.97)	28.12	(23.01)	23.35	(22.17)	19.47	(17.59)	26.60	(24.04)	0.44	0.020
Altitude (m)	2,273.14	(1,287.04)	2,800.26	(1,166.94)	2,887.06	(1,115.04)	3,049.45	(722.05)	2,363.33	(1,199.01)	90.19	0.073
Distance to Lima (km)	527.38	(256.62)	364.81	(239.09)	322.89	(196.53)	321.42	(143.85)	331.38	(80.04)	−195.99	−1.031
Terrain ruggedness	279.54	(173.83)	249.74	(165.09)	237.38	(150.64)	210.77	(114.74)	267.20	(141.67)	−12.34	−0.078
Observations	827		351		121		33		39			

Notes. Sample restricted to the 1981 census baseline. Conflict intensity is based on total CVR-documented victims 1980–2000, with No conflict (0 victims), Low (1–10), Medium (11–50), High (51–100), and Very high (>100). Standard deviations in parentheses. The difference column subtracts the no-conflict mean from the very-high mean, and the normalized difference divides it by the square root of the mean of the two group variances. Indigenous language includes Quechua, Aymara, and other native languages. Dependency ratio = (population 0–14 + 65+) / (population 15–64) × 100. Literacy is the share literate among persons aged 15 and over. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.



(a) Violence intensity by district

(b) Spillover zones

Figure A2: Geographic Distribution of Conflict and Potential Spillover Zones

Notes. Panel (a) shows conflict intensity based on CVR-documented victims (deaths and disappearances) on the 1981 district units. The categories are No conflict (0 victims, 882 districts), Low (1–10, 419), Medium (11–50, 161), High (51–100, 36), and Very high (>100, 42). Panel (b) shows the spillover design, with treatment = the 42 very-high-conflict districts, spillover = the 242 other districts within 50 km of a treatment district (192 of them with 1–100 victims), comparison = the 832 districts without victims beyond 50 km, and the remaining 424 conflict districts beyond 50 km in a muted tone.

A.2 Population

Inference with few treated districts. Because the very-high-conflict group comprises only 42 districts, cluster-robust standard errors could overstate precision. Randomization inference, randomly reassigning the very-high-conflict label across districts (holding the number of treated districts fixed) over 500 draws, yields p-values of 0.050 (1993), 0.592 (2007), and 0.586 (2017) (Table A30, Panel A). The 1993 deficit is borderline relative to random assignment of the label across the whole country, and the spatial methods that compare it with geographically close districts do not reject zero.

Pre-conflict comparability. The 1981 census's retrospective migration questions, which record each resident's district of birth and place of residence in 1976, allow comparing *pre-*

conflict demographic dynamics. Appendix Table A5 shows that very-high-conflict districts are statistically indistinguishable from the rest in the share of residents living in the same province in 1976 and in 1976–1981 in-migration from other provinces, with or without department fixed effects and geographic controls (differences between −4.2 and 4.2 percentage points, $p > 0.34$), and in the stock of residents born in another district. Migration dynamics in the five years before the conflict's escalation show no differential trajectory.

Table A2 re-estimates the population path on the districts observed in 1972 and 1981 with alternative measures of exposure. Counting only the victims recorded by 1992, before the 1993 census, leaves the estimates close to those of the main definition. Thresholds in victims per 1,000 inhabitants of 1981 give larger losses in 1993 that persist through 2017, without a differential trend between 1972 and 1981 at the threshold of 5 per 1,000, and adding log 1981 population interacted with year leaves them unchanged. Panel B shows that the persistent loss comes from the smaller districts that exceed the per-capita threshold with 100 or fewer victims. In the highlands the thresholds of 10 and 20 per 1,000 show positive 1972 coefficients, so they select districts that were already shrinking before the war.

Table A2: Alternative Definitions of Conflict Exposure, Log Population

	All districts					Highland districts				
	1972	1993	2007	2017	Exposed	1972	1993	2007	2017	Exposed
Panel A. One exposure measure per regression										
>100 victims (main definition)	−0.100	−0.138	0.031	0.051	35	0.013	−0.236***	−0.125	−0.123	24
	(0.092)	(0.090)	(0.153)	(0.215)		(0.038)	(0.069)	(0.097)	(0.146)	
>100 victims by 1992	−0.088	−0.168*	−0.006	−0.012	31	0.027	−0.257***	−0.152	−0.168	22
	(0.096)	(0.090)	(0.147)	(0.205)		(0.036)	(0.071)	(0.098)	(0.146)	
>5 per 1,000	0.022	−0.261***	−0.139*	−0.254***	120	0.017	−0.262***	−0.132**	−0.213***	105
	(0.039)	(0.059)	(0.070)	(0.093)		(0.027)	(0.055)	(0.052)	(0.062)	
>10 per 1,000	0.036	−0.274***	−0.166*	−0.254**	71	0.055***	−0.301***	−0.213***	−0.270***	59
	(0.045)	(0.071)	(0.095)	(0.125)		(0.021)	(0.062)	(0.048)	(0.065)	
>20 per 1,000	0.018	−0.324***	−0.172	−0.272	33	0.087***	−0.308***	−0.245***	−0.303***	27
	(0.080)	(0.071)	(0.135)	(0.182)		(0.023)	(0.065)	(0.062)	(0.090)	
log(1 + victims)	−0.011	−0.025*	−0.004	−0.022		−0.011*	−0.029**	−0.007	−0.018	
	(0.009)	(0.013)	(0.017)	(0.023)		(0.006)	(0.013)	(0.014)	(0.019)	
log(1 + victims per 1,000)	0.021	−0.082***	−0.048*	−0.105***		0.012	−0.079***	−0.038*	−0.075***	
	(0.013)	(0.020)	(0.027)	(0.035)		(0.008)	(0.019)	(0.020)	(0.023)	
>5 per 1,000, log 1981 pop. × year	0.011	−0.256***	−0.134*	−0.242***	120	0.020	−0.263***	−0.132**	−0.215***	105
	(0.033)	(0.056)	(0.069)	(0.088)		(0.023)	(0.053)	(0.052)	(0.061)	
Panel B. Absolute and per-capita groups estimated jointly										
>100 victims and >5 per 1,000	−0.059	−0.232***	−0.053	−0.067	31	0.022	−0.272***	−0.147	−0.165	23
>100 victims, ≤5 per 1,000	−0.388**	0.426***	0.587***	0.771***	4	−0.168***	0.043***	0.111***	0.362***	1
>5 per 1,000, ≤100 victims	0.049	−0.269***	−0.166***	−0.315***	89	0.015	−0.259***	−0.128***	−0.226***	82
>100 and >5 per 1,000, victims by 1992	−0.043	−0.267***	−0.101	−0.137	28	0.029	−0.280***	−0.162	−0.186	22
>5 per 1,000, ≤100 victims, by 1992	0.047	−0.268***	−0.148**	−0.282***	87	0.021	−0.271***	−0.127***	−0.215***	79
Panel C. p-values, wild cluster bootstrap by department										
>100 victims (main definition)		0.778	0.867	0.854			0.089	0.774	0.765	
>5 per 1,000		0.063	0.337	0.156			0.002	0.108	0.038	
>100 victims and >5 per 1,000 (Panel B)		0.277	0.802	0.826			0.085	0.762	0.739	
>5 per 1,000, ≤100 victims (Panel B)		0.032	0.136	0.005			0.005	0.116	0.032	
>5 per 1,000, ≤100 victims, by 1992 (Panel B)		0.098	0.231	0.015			0.004	0.115	0.045	
Districts					1,310					946

Notes. Each row in Panel A is a separate regression of log population on the exposure measure interacted with census-year indicators, with district and year fixed effects and 1981 as the reference year, on the districts observed in 1972 and 1981. Victims are CVR-documented deaths and disappearances, and victims per 1,000 use the 1981 census population. The 1972 column is the pre-conflict change between 1972 and 1981, and Exposed counts the districts in the exposure group. The row with log 1981 population adds that variable interacted with year. Panel B enters three mutually exclusive groups in one regression, with all other districts as the reference. Its last two rows come from a second regression in which both thresholds use only the victims recorded by 1992, before the 1993 census, and its third group (more than 100 victims with 5 or fewer per 1,000, three districts nationally and none in the highlands) is not shown. Standard errors clustered by 1981 province in parentheses in Panel A, and stars from 1981-province-clustered p-values in Panels A and B. Panel C reports p-values of the wild cluster bootstrap by department (Webb weights, 9,999 replications, 23 departments for all districts and 18 for the highlands) for the corresponding coefficients. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Table A3 reports the highland loss for thresholds between 25 and 200 victims, for the highland districts inside and outside Ayacucho, and the range of the 1993 coefficient when one department, one 1981 province, or one treated district is left out.

Table A3: Highland Population Loss by Treatment Threshold and Leave-One-Out Samples

	1972	1993	2007	2017	Exposed districts
Panel A. Threshold of documented victims					
More than 25 victims	−0.055	−0.139	−0.059	−0.092	86
	(0.033)	(0.061)	(0.067)	(0.081)	
More than 50 victims	−0.068	−0.178	−0.124	−0.133	56
	(0.056)	(0.060)	(0.077)	(0.101)	
More than 75 victims	−0.049	−0.159	−0.047	−0.015	33
	(0.045)	(0.077)	(0.113)	(0.164)	
More than 100 victims (main definition)	0.013	−0.236	−0.125	−0.123	24
	(0.038)	(0.069)	(0.097)	(0.146)	
More than 150 victims	0.016	−0.302	−0.153	−0.173	15
	(0.041)	(0.079)	(0.109)	(0.173)	
More than 200 victims	−0.026	−0.399	−0.170	−0.232	11
	(0.039)	(0.090)	(0.112)	(0.157)	
Panel B. Ayacucho					
Highland districts outside Ayacucho	−0.095	−0.105	0.184	0.337	5
	(0.025)	(0.232)	(0.124)	(0.175)	
Districts of Ayacucho only	0.006	−0.034	−0.038	−0.006	19
	(0.048)	(0.070)	(0.066)	(0.102)	
Panel C. Range of the 1993 coefficient when one unit is left out					
One department (4 samples)		−0.260 to −0.105			
One 1981 province (9 samples)		−0.283 to −0.180			
One treated district (24 samples)		−0.263 to −0.200			

Notes. Highland districts observed in 1972 and 1981. Each row is a separate regression of log population on the exposure indicator interacted with census-year indicators, with district and year fixed effects and 1981 as the reference year. Standard errors clustered by 1981 province in parentheses, without stars. Panel C reports the smallest and largest 1993 coefficient across the regressions that leave out one department, one 1981 province, or one treated district at a time. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A4: Effect of Conflict on Log Population

	(1) VH vs all other	(2) VH and High vs ≤50 victims	(3) All categories vs no victims	(4) VH vs all other, geography × year	(5) VH vs all other, excluding Lima department
Very High × 1993	-0.118*	-0.122*	-0.131*	-0.111*	-0.140**
	(0.070)	(0.070)	(0.071)	(0.065)	(0.068)
Very High × 2007	0.049	0.044	0.037	0.048	0.030
	(0.088)	(0.088)	(0.088)	(0.075)	(0.086)
Very High × 2017	0.057	0.051	0.028	0.050	0.037
	(0.111)	(0.111)	(0.112)	(0.090)	(0.109)
High × 1993		-0.139	-0.149*		
		(0.085)	(0.085)		
High × 2007		-0.166*	-0.173*		
		(0.093)	(0.094)		
High × 2017		-0.257**	-0.279**		
		(0.124)	(0.125)		
Medium × 1993			-0.107***		
			(0.040)		
Medium × 2007			-0.088**		
			(0.040)		
Medium × 2017			-0.179***		
			(0.049)		
Low × 1993			0.001		
			(0.023)		
Low × 2007			0.001		
			(0.031)		
Low × 2017			-0.018		
			(0.039)		
Observations	5991	5991	5991	5991	5479
Adjusted R^2	0.950	0.950	0.950	0.957	0.948
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Geography × year	No	No	No	Yes	No

Notes. Standard errors clustered by district in parentheses. All columns include district and year fixed effects, with 1981 (pre-conflict) as the reference year. Column (1) compares very-high-conflict districts with all other districts. Column (2) adds High × year terms, so the comparison group is the districts with 50 or fewer victims. Column (3) adds High, Medium and Low × year terms, so the comparison group is the districts without documented victims. Column (4) adds altitude and distance to Lima interacted with year. Column (5) excludes every district of the department of Lima. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Table A5: Pre-Conflict Migration Balance, 1976–1981

	Same province 1976 (%)			Born other district (%)	
	(1)	(2)	(3)	(4)	(5)
	Raw	Dept FE	+Geo	Dept FE	+Geo
Very High Conflict	-3.856	-4.200	-2.786	7.626	-0.611
	(4.211)	(4.345)	(3.017)	(8.918)	(3.587)
Observations	1371	1371	1371	1371	1371
Adjusted R^2	0.010	0.102	0.180	0.243	0.474
Department FE	No	Yes	Yes	Yes	Yes
Geographic controls	No	No	Yes	No	Yes

Notes. 1981 census cross-section (1,371 districts). Outcomes from the census's retrospective questions, defined over residents with a reported 1976 residence. They are the share who lived in the same province in 1976, whose complement is the share who arrived from another province during 1976–1981, and the share born outside the current district (lifetime migrants). "+Geo" adds altitude, distance to Lima, and terrain ruggedness (available for 1,371 districts). Standard errors clustered at the department level (23 clusters) in parentheses. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A6: Dynamic Difference-in-Differences of Population with the 1972 Pre-Period

Panel A: Very High × year, log population (reference 1981)					
	1972	1981	1993	2007	2017
All units	−0.100	ref.	−0.138	0.031	0.051
	(0.092)		(0.090)	(0.153)	(0.215)
	[−0.283, 0.082]		[−0.316, 0.040]	[−0.272, 0.335]	[−0.374, 0.475]
Observations / units / very-high units	6,550	1,310	35		
Highland units	0.013	ref.	−0.236	−0.125	−0.123
	(0.038)		(0.069)	(0.097)	(0.146)
	[−0.062, 0.087]		[−0.372, −0.100]	[−0.318, 0.068]	[−0.413, 0.167]
Observations / units / very-high units	4,730	946	24		

Panel B: Highland units, p-values under alternative inference					
	VH × 1972	VH × 1993	Change 1993–2007	Change 1993–2017	95% CI, 2017
Unit clusters (946)	0.663	0.002	0.014	0.084	[−0.337, 0.090]
1981-province clusters (104)	0.736	<0.001	0.046	0.233	[−0.413, 0.167]
Department clusters (18)	0.678	<0.001	0.051	0.253	[−0.388, 0.141]
Wild bootstrap, 1981 provinces (104)	0.759	0.076	0.023	0.253	
Wild bootstrap, departments (18)	0.836	0.084	0.138	0.176	
Conley, 50 km	0.922	<0.001	0.078	0.364	[−0.399, 0.152]
Conley, 100 km	0.911	<0.001	0.014	0.307	[−0.372, 0.125]
Conley, 200 km	0.850	<0.001	0.001	0.067	[−0.256, 0.009]
Point estimate	0.013	−0.236	0.111	0.113	

Panel C: All units, alternative clustering			
	SE, VH × 1972	p, VH × 1972	SE, VH × 1993
Unit clusters	(0.052)	0.056	(0.072)
1981-province clusters	(0.092)	0.279	(0.090)
Department clusters	(0.125)	0.431	(0.106)

Notes. Dynamic difference-in-differences of log population on Very High × census-year indicators with district and year fixed effects, reference year 1981, on the districts observed in 1972, 1981, 1993, 2007 and 2017. Panel A reports standard errors clustered by 1981 province in parentheses and 95% confidence intervals in brackets. The 1972 population comes from the 1972 census, and highland districts are those classified as Sierra. Panel B reports p-values for the highland specification, with the number of clusters in parentheses. The change columns test the 2007 (2017) coefficient minus the 1993 coefficient, and the confidence interval refers to the 2017 coefficient. Wild bootstrap rows report p-values only. Panel C reports standard errors and p-values for the all-districts specification under alternative clustering. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A7: Cohort Aged 15–29 in 1981: Share of the Population, by Census Year

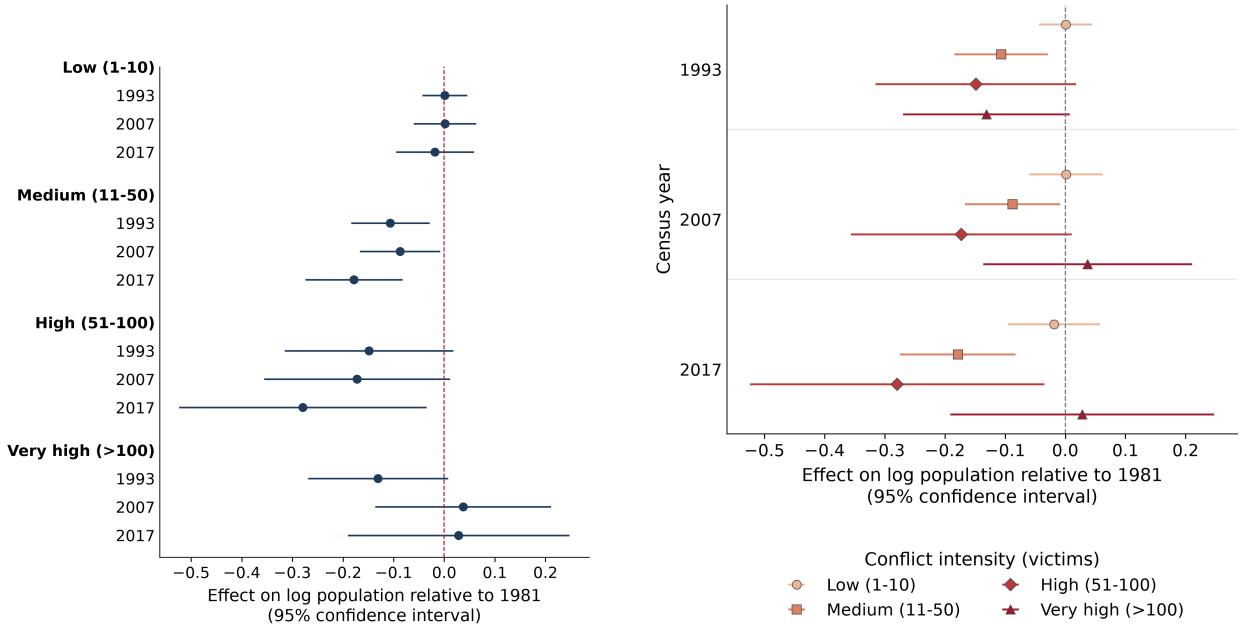
	(a) Baseline			(b)	(c)	(d)	(e)
	Coefficient	p, 1981 provinces	p, wild bootstrap	Initial cond. × year	Department × year FE	Highland districts	Share of ages 15+
Cohort aged 15–29 in 1981, % of population							
Very high × 1993	−2.066***	0.055	0.088	−1.225***	−3.772***	−1.224**	−3.761***
	(0.569)			(0.419)	(0.422)	(0.497)	(0.855)
Very high × 2007	−3.437***	0.064	0.136	−1.424**	−6.284***	−1.463*	−5.594***
	(0.915)			(0.621)	(0.652)	(0.813)	(1.496)
Very high × 2017	−3.729***	0.124	0.299	−1.229	−7.632***	−1.419	−6.260***
	(1.182)			(0.759)	(0.835)	(1.106)	(1.800)
Cohort aged 0–14 in 1981, % of population							
Very high × 1993	−0.504	0.741	0.861	−0.215			
	(0.804)			(0.599)			
Very high × 2007	−2.050***	0.054	0.642	−0.199			
	(0.688)			(0.561)			
Very high × 2017	−2.336***	<0.001	0.072	−0.282			
	(0.547)			(0.514)			
Observations	5,991			5,484	5,991	4,089	5,991
Districts	1,540			1,371	1,540	1,039	1,540
Very-high districts	42			39	42	26	42

Notes. Each cell is the very-high × year coefficient from a regression of the cohort share on district and year fixed effects, with 1981 as the reference year. Standard errors clustered by district in parentheses, with stars from district-clustered p-values. Column (a) also reports the p-values with 1981-province clusters and with the department wild cluster bootstrap (Webb weights). Column (b) adds 1981 initial conditions interacted with year and uses the districts with a 1981 record, Column (c) adds department-by-year fixed effects, and Column (d) restricts the sample to highland districts. Column (e) measures the cohort as a share of the population aged 15 and over. The cohort aged 15–29 in 1981 is aged 27–41 in 1993, 41–55 in 2007 and 51–65 in 2017. Cohorts are followed in the five-year age groups of the census literacy tables. The observation counts refer to the cohort aged 15–29 in 1981 and are the same for the cohort aged 0–14 in 1981. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Figure A3 presents the coefficients of the specification that includes all intensity categories, with zero-victim districts as the omitted group (Appendix Table A4, Column (3)).

The national pattern is not monotonic. In 1993, very-high, high, and medium-intensity districts all grew less than districts without documented violence (−0.13, −0.15, and −0.11 log points), and the very-high and high coefficients cannot be distinguished ($p = 0.87$). A linear contrast across the ordered categories is significant in 1993 ($p = 0.01$ with district clusters, $p = 0.07$ with province clusters). In 2007 and 2017, high and medium-intensity districts

lie below the very-high group. The national comparison therefore does not support a clear dose-response. Within the highlands, on the other hand, where the 1972 census provides a pre-period, the 1993 loss does increase with the intensity threshold (Section 4.1). Interpreting coefficients across intensity levels as a dose-response function also requires assumptions beyond parallel trends (Callaway et al., 2024), so I read these estimates as descriptive.



(a) By conflict intensity

(b) By census year

Figure A3: Population Effects by Conflict Intensity and Census Year

Notes. Both panels show coefficients from the specification with all intensity categories (Appendix Table A4, Column (3)). Panel (a) groups the coefficients by intensity category and panel (b) by census year. The omitted group is districts with no documented victims. Reference year 1981. 95% confidence intervals with standard errors clustered by district.

A demanding version of the identification concern is that very-high-conflict districts differ from the national pool of comparison districts in unobserved ways correlated with regional development trends. Table A8 addresses this by replacing year fixed effects with province-by-year fixed effects, so that each very-high-conflict district is compared only with districts in its own province (the current province of the district that gives the unit its code) in the same census year.

Within provinces, the 1993 population coefficient is close to zero (0.01), and very-high-conflict districts grew faster than their provincial neighbors afterwards (0.20 log points by 2007 and 0.31 by 2017, both significant). The population loss of 1993 was shared by the whole province, consistent with the regional character of the shock documented in Section 4.1. The faster growth of the most affected districts after 1993 within their provinces must be

read with care, because with 1981-province-by-year fixed effects they were also growing faster than their provincial neighbors before the war (1972 coefficient -0.12, $p = 0.02$). Within provinces the literacy coefficients are negative and not significant (-1.4 to -2.5 points), as are the mean-age coefficients. The literacy gains of very-high-conflict districts were shared by the other districts of their provinces, which also started from low levels.

Table A8: Within-Province Comparisons

	Log Population		Literacy Rate, 15+		Mean Age	
	(1)	(2)	(3)	(4)	(5)	(6)
	Base	Prov.×Yr	Base	Prov.×Yr	Base	Prov.×Yr
Very High × 1993	-0.118*	0.011	1.487	-1.438	0.195	0.237
	(0.070)	(0.073)	(1.157)	(1.269)	(0.227)	(0.247)
Very High × 2007	0.049	0.197**	5.508***	-2.511	-0.559*	-0.046
	(0.088)	(0.083)	(1.702)	(1.622)	(0.290)	(0.366)
Very High × 2017	0.057	0.305***	6.615***	-2.147	-0.506	-0.807
	(0.111)	(0.099)	(1.876)	(1.784)	(0.440)	(0.524)
Observations	5991	5951	5991	5951	5991	5951
Adjusted R^2	0.950	0.963	0.856	0.923	0.870	0.901
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	No	Yes	No	Yes	No
Province × year FE	No	Yes	No	Yes	No	Yes

Notes. "Base" columns include district and census-year fixed effects (as in Table A4), with all districts below the very-high threshold as the omitted comparison group. "Prov.×Yr" columns replace year fixed effects with province-by-year fixed effects, so very-high-conflict districts are compared only with below-threshold districts of the same province in the same year. The province is the current province of the district that gives each unit its code. Standard errors clustered by district in parentheses, and the reference year is 1981. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

A.3 Human Capital and Infrastructure

Table A9: Literacy Gap 1961–2017, Conditional Convergence and Age Groups

Panel A: Literacy gap, very-high minus other units (percentage points)							
	1961†	1972†	1981	1993	2007	2017	
All units	−17.219***	−11.920***	−11.315***	−10.750***	−5.707**	−4.421**	
	(5.302)	(3.365)	(4.269)	(3.811)	(2.448)	(2.049)	
p, department clusters	0.032	0.036	0.099	0.077	0.127	0.156	
Units	1,309						
Highland units	−18.624***	−13.082***	−13.171***	−13.477***	−7.441***	−5.440***	
	(4.094)	(2.721)	(3.522)	(2.809)	(1.797)	(1.668)	
p, department clusters	<0.001	<0.001	0.002	<0.001	0.001	0.008	
Units	945						
Panel B: Dynamic difference-in-differences, literacy, Very High × year (reference 1981)							
	1961†	1972†	1981	1993	2007	2017	$\Delta_{93-81} - \Delta_{81-72}$
All units	−5.904***	−0.605	ref.	0.565	5.608**	6.894***	−0.041
	(1.793)	(1.075)		(1.192)	(2.283)	(2.578)	(2.033)
p, department clusters	<0.001	0.643		0.636	0.098	0.079	0.986
Highland units	−5.453***	0.089	ref.	−0.306	5.729***	7.731***	−0.217
	(1.495)	(1.092)		(1.322)	(2.047)	(2.310)	(2.266)
p, department clusters	0.002	0.856		0.717	0.004	<0.001	0.855
Panel C: Beta convergence, very-high units, actual minus predicted change (percentage points)							
	1981–1993			1981–2017			
	Gap	*p*, prov.	*p*, dept.	Gap	*p*, prov.	*p*, dept.	Units
Literacy, ages 15+	−2.381**	0.044	0.060	0.674	0.467	0.333	1,371
	(1.173)			(0.923)			
Higher education, ages 25+	−0.823	0.151	0.280	−1.915	0.116	0.180	1,371
	(0.569)			(1.209)			
Secondary complete, ages 25+	−2.284***	<0.001	<0.001	−5.812***	<0.001	0.001	1,371
	(0.538)			(1.296)			
Panel D: Age groups 15–24 and 25–44, Very High × year (1981-province clusters)							
	1993	2007	2017	Observations			
Literacy, ages 15–24, baseline	3.357*	6.816***	6.572***	5,990			
	[0.067]	[0.002]	[0.004]				
Literacy, ages 15–24, initial conditions × year	0.405	0.943**	−0.084	5,480			
	[0.733]	[0.023]	[0.750]				
Higher education, ages 25–44, baseline	−1.678*	−4.209**	−2.338	5,991			
	[0.056]	[0.015]	[0.260]				
Higher education, ages 25–44, initial conditions × year	−0.879	−0.670	−0.862	5,484			
	[0.242]	[0.601]	[0.560]				

Notes. Literacy is the share literate among persons aged 15 and over, with the same age universe in 1981, 1993, 2007 and 2017. The 1961 and 1972 values (†) are literacy rates computed as 100 minus the illiteracy rates of Albertus (2020), a complementary historical series whose age universe is not documented, so comparisons with 1961 and 1972 cross sources and are not part of the main estimates. Panels A and B use the same districts in every year (complete in the 1972 source and with literacy 15+ in all four censuses). Panel A reports the difference in mean literacy with standard errors clustered by 1981 province in parentheses and stars from the 1981-province *p*-value. Panel B has district and year fixed effects and standard errors clustered by 1981 province in parentheses, and its last column is the change in the coefficient 1981–1993 minus the change 1972–1981. Panel C regresses the change in the outcome on its 1981 level and its square in districts with no recorded conflict, and the gap is the mean of actual minus predicted change for very-high districts, with standard errors clustered by 1981 province in parentheses. Panel D reports *p*-values in brackets, with initial conditions as in Table A33. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Table A10: Effect of Very High Conflict on Infrastructure and Housing Quality

	(1) Electricity	(2) Water	(3) Sewage	(4) Sanitation	(5) Brick Walls	(6) Dirt Floor
Very High × 1993	-3.723*	0.514	-1.121	-4.661	1.125	2.741
	(2.066)	(1.363)	(2.222)	(3.609)	(1.574)	(2.714)
Very High × 2007	-10.005***	-1.398	-3.023	9.846***	-0.262	3.090
	(2.655)	(3.203)	(2.767)	(3.526)	(2.026)	(2.911)
Very High × 2017	0.132	6.301*	-5.910*	6.718**	-0.138	2.008
	(2.626)	(3.756)	(3.033)	(3.409)	(2.625)	(3.118)
Observations	5987	5987	5987	5987	5987	5987
Adjusted R^2	0.773	0.696	0.779	0.816	0.849	0.915
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. All outcomes are percentages (0–100). Electricity is the share of dwellings with electric lighting, Water with a public water network, Sewage with a public sewage connection, Sanitation with any sanitation facility, Brick Walls with brick or cement block walls, and Dirt Floor with earth floors. The omitted comparison group is all districts below the very-high threshold. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A11: Sanitation versus Electricity: Stacked Contrast

	2007		2017		Joint
	Contrast	p-value	Contrast	p-value	p-value
Panel A: main geography					
Unit clusters	19.851***	<0.001	6.587***	0.010	<0.001
	(4.283)		(2.542)		
1981-province clusters	19.851***	0.003	6.587**	0.029	0.017
	(6.566)		(2.984)		
Department clusters	19.851***	<0.001	6.587***	0.006	<0.001
	(5.190)		(2.167)		
Wild bootstrap, departments	19.851*	0.055	6.587**	0.039	0.149
Panel B: robustness geography					
Unit clusters	17.584***	<0.001	4.604**	0.030	<0.001
	(3.822)		(2.118)		
1981-province clusters	17.584***	0.001	4.604**	0.044	0.008
	(5.396)		(2.263)		
Department clusters	17.584***	0.001	4.604**	0.016	0.004
	(4.717)		(1.769)		
Wild bootstrap, departments	17.584*	0.087	4.604*	0.084	0.297

Notes. Stacked regression of sanitation (any facility, namely public network, septic tank or pit latrine, % of dwellings) and electricity (% of dwellings) on Very High × year, with outcome-specific district and year fixed effects. The contrast is the sanitation coefficient minus the electricity coefficient (percentage points, reference 1981). Standard errors in parentheses under the clustering shown in each row, and the wild bootstrap rows report p-values only. Joint refers to the test that the 1993, 2007 and 2017 contrasts are all zero. Panel A uses the main fixed geography and Panel B the alternative geography. Stars from the p-value of the same row. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

A.4 Spatial Spillovers

Table A13 excludes the districts that lie within 25, 50, 75, or 100 km of a very-high-conflict district, measured between centroids. The population coefficients barely move, and the 1993 coefficient ranges from −0.12 to −0.14 and the 2017 coefficient from 0.06 to −0.01 across the four radii, and the 2017 literacy coefficient (Panel B) ranges from 6.4 to 6.8 points.

Table A14 estimates the spillovers directly by adding indicators for the districts within

each radius, interacted with year. The ring districts include every district below the very-high threshold within the radius, including districts with some documented violence. For districts within 25 or 50 km of the conflict core the coefficients are small, between −0.04 and 0.08 log points, and none is individually significant, although the joint test rejects zero for the 50 km ring ($p = 0.02$, Table A30, Panel B). Larger negative coefficients appear in the wider rings, where districts within 100 km lost 0.06 log points by 1993, 0.11 by 2007, and 0.21 by 2017, and the joint tests reject zero for the 75 and 100 km rings ($p < 0.001$). Figure A4 plots the coefficients by radius.

A pattern of small changes near the conflict core and growing losses at 75 to 100 km does not look like a spillover that decays with distance. The wider rings cover a large part of the central and southern highlands. Among the districts observed in 1981, the highlands held 39 percent of the population in 1981 and 30 percent in 2017, and the average highland district shrank by 0.08 log points while the average coastal district grew by 0.45. No ring shows a significant population gain, so there is no sign that refugee absorption inflated the comparison group, and the declining wider rings are more plausibly part of the long-run depopulation of the highlands than an effect of nearby violence.

Table A12 repeats these designs with distances measured between district boundaries, so that adjacent districts are at distance zero. The donuts leave the very-high coefficients close to those of the main estimates, nationally and in the highlands. In the highlands, the ring design splits each ring into districts with and without documented victims. No ring gains population relative to the highland districts beyond 100 km, adjacent districts without victims were growing faster before the war, and the districts without victims at 50 to 100 km lose population after 1993, consistent with the long-run depopulation of the highlands.

Table A12: Spatial Designs with Border-to-Border Distances, Log Population

	1972	1993	2007	2017	Districts in group
Panel A. Donuts, very-high coefficient after dropping other districts within k km					
All districts					
$k = 25$ km	−0.096	−0.136	0.036	0.042	35
	(0.094)	(0.094)	(0.158)	(0.220)	
$k = 50$ km	−0.086	−0.143	0.027	0.024	35
	(0.094)	(0.095)	(0.159)	(0.222)	
$k = 100$ km	−0.073	−0.163*	−0.007	−0.017	35
	(0.095)	(0.096)	(0.161)	(0.224)	
Highland districts					
$k = 25$ km	0.021	−0.235***	−0.122	−0.129	24
	(0.038)	(0.073)	(0.102)	(0.150)	
$k = 50$ km	0.028	−0.240***	−0.128	−0.139	24
	(0.038)	(0.074)	(0.101)	(0.150)	
$k = 100$ km	0.031	−0.263***	−0.169	−0.182	24
	(0.040)	(0.075)	(0.104)	(0.154)	
Panel B. Highland rings around very-high districts, one regression					
Very high conflict	0.031	−0.263***	−0.169	−0.182	24
	(0.040)	(0.075)	(0.104)	(0.154)	
Within 25 km, no victims	0.104***	0.049	−0.016	−0.055	58
	(0.031)	(0.048)	(0.067)	(0.096)	
Within 25 km, 1–100 victims	0.023	−0.049	−0.039	−0.085	166
	(0.026)	(0.053)	(0.055)	(0.097)	
25–50 km, no victims	0.038	−0.072	−0.049	−0.069	83
	(0.027)	(0.052)	(0.081)	(0.094)	
25–50 km, 1–100 victims	0.032	−0.012	−0.100	−0.118	63
	(0.042)	(0.050)	(0.105)	(0.121)	
50–100 km, no victims	0.034	−0.077*	−0.205***	−0.194**	99
	(0.025)	(0.043)	(0.059)	(0.082)	
50–100 km, 1–100 victims	−0.022	−0.069	−0.031	−0.062	76
	(0.036)	(0.055)	(0.071)	(0.086)	
Highland districts in Panel B					946

Notes. Regressions of log population on group indicators interacted with census-year indicators, with district and year fixed effects and 1981 as the reference year, on the districts observed in 1972 and 1981. Distances are measured between district boundaries, so adjacent districts are at distance zero. Panel A drops the districts below the very-high threshold that lie within k km of a very-high-conflict district and reports the very-high coefficient. Panel B estimates the very-high group and six ring groups jointly on highland districts, splitting each ring by whether the district had any documented victim, with the highland districts beyond 100 km as the reference. Standard errors clustered by 1981 province in parentheses, with stars from the same p-values. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Table A13: Donut Specifications: Excluding Districts Near Conflict Zones

	Full Sample	Donut 25km	Donut 50km	Donut 75km	Donut 100km
Panel A: Log Population					
Very High × 1993	-0.118*	-0.120*	-0.117*	-0.129*	-0.138*
	(0.070)	(0.070)	(0.070)	(0.070)	(0.071)
Very High × 2007	0.049	0.050	0.053	0.037	0.011
	(0.088)	(0.088)	(0.088)	(0.088)	(0.088)
Very High × 2017	0.057	0.062	0.052	0.018	-0.014
	(0.111)	(0.111)	(0.111)	(0.111)	(0.111)
Observations	5991	5624	5038	4493	4045
Panel B: Literacy Rate, Ages 15+					
Very High × 1993	1.487	1.394	1.427	1.224	1.007
	(1.157)	(1.158)	(1.160)	(1.163)	(1.166)
Very High × 2007	5.508***	5.548***	5.695***	5.502***	5.319***
	(1.702)	(1.703)	(1.705)	(1.708)	(1.712)
Very High × 2017	6.615***	6.644***	6.750***	6.531***	6.402***
	(1.876)	(1.877)	(1.879)	(1.883)	(1.886)
Observations	5991	5624	5038	4493	4045
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. Donut X km drops every district below the very-high threshold whose centroid lies within X km of the centroid of a very-high-conflict district (centroid distance in every column). The omitted comparison group is the remaining districts below the very-high threshold beyond the donut radius. All specifications include district and year fixed effects, and the reference year is 1981 (pre-conflict baseline). Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Table A14: Ring Analysis: Direct Conflict and Spillover Effects

	Ring 25km	Ring 50km	Ring 75km	Ring 100km
Very High × 1993	-0.120*	-0.117*	-0.129*	-0.138*
	(0.070)	(0.070)	(0.070)	(0.070)
Very High × 2007	0.050	0.053	0.037	0.011
	(0.088)	(0.088)	(0.088)	(0.088)
Very High × 2017	0.062	0.052	0.018	-0.014
	(0.111)	(0.111)	(0.111)	(0.111)
Spillover (25 km) × 1993	-0.038			
	(0.051)			
Spillover (25 km) × 2007	0.022			
	(0.057)			
Spillover (25 km) × 2017	0.079			
	(0.078)			
Spillover (50 km) × 1993		0.004		
		(0.032)		
Spillover (50 km) × 2007		0.028		
		(0.042)		
Spillover (50 km) × 2017		-0.031		
		(0.054)		
Spillover (75 km) × 1993			-0.039	
			(0.025)	
Spillover (75 km) × 2007			-0.044	
			(0.033)	
Spillover (75 km) × 2017			-0.150***	
			(0.042)	
Spillover (100 km) × 1993				-0.057***
				(0.022)
Spillover (100 km) × 2007				-0.111***
				(0.029)
Spillover (100 km) × 2017				-0.211***
				(0.037)
Observations	5991	5991	5991	5991
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes. The dependent variable is log population. Standard errors clustered by district in parentheses. The spillover ring contains every district below the very-high threshold whose centroid lies within the stated distance of the centroid of a very-high-conflict district (centroid distance in every column). The omitted comparison group is the districts below the very-high threshold outside the ring. All specifications include district and year fixed effects, and the reference year is 1981. Joint F-tests for the spillover coefficients are reported in Table A30, Panel B. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

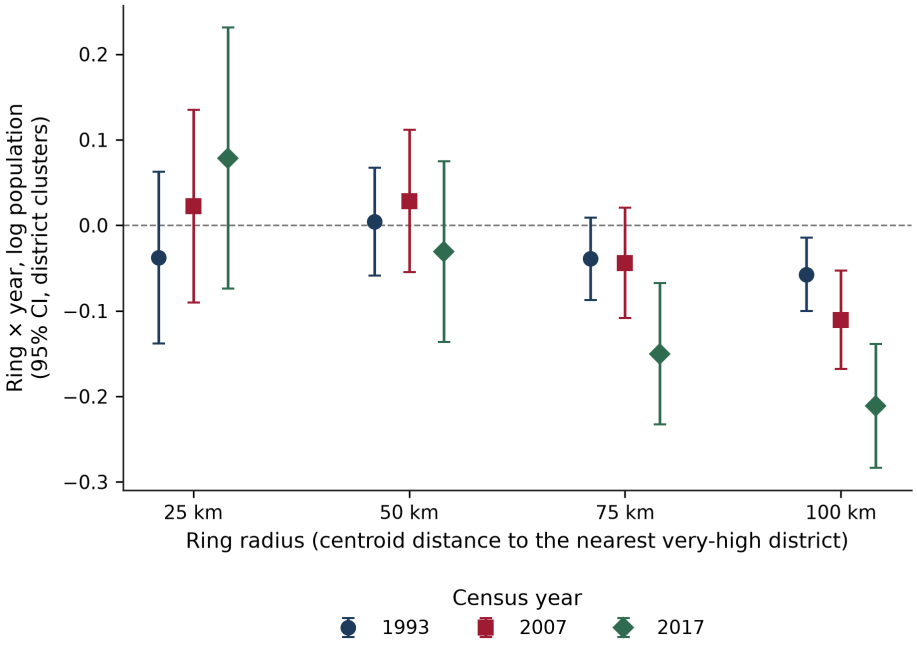


Figure A4: Population of Districts Near the Conflict Core, by Ring Radius

Notes. Coefficients on indicators for the districts below the very-high threshold within 25, 50, 75, and 100 km (centroid distance) of a very-high-conflict district, interacted with census year, from the ring specifications of Columns (1) to (4) of Appendix Table A14. The outcome is log population, with district and year fixed effects and 1981 as the reference year (5,991 observations, 1,540 districts, 42 very high). Bars show 95% confidence intervals with standard errors clustered by district.

Table A15 examines spillovers across outcomes. Districts within 50 km of the conflict core became somewhat less indigenous by 2007 (−2.6 percentage points), their migration rate fell 2.8 points by 2017, and their literacy and mean age show no significant differential.

Table A15: Direct and Spillover Effects on Multiple Outcomes

	(1) Log Pop	(2) Literacy 15+	(3) Migration	(4) Mean Age	(5) Indigenous
Direct Conflict Effects					
× 1993	-0.117*	1.427	1.471	0.215	0.512
	(0.070)	(1.160)	(1.627)	(0.228)	(1.676)
× 2007	0.053	5.695***	4.251	-0.590**	0.610
	(0.088)	(1.705)	(4.963)	(0.292)	(1.838)
× 2017	0.052	6.750***	-1.979	-0.462	-0.702
	(0.111)	(1.879)	(2.561)	(0.443)	(1.732)
Spillover Effects (0–50km)					
× 1993	0.004	-0.394	-1.418**	0.117	-1.455
	(0.032)	(0.497)	(0.696)	(0.155)	(1.132)
× 2007	0.028	1.128	-0.631	-0.193	-2.566**
	(0.042)	(0.755)	(1.964)	(0.189)	(1.294)
× 2017	-0.031	0.806	-2.814***	0.270	-1.133
	(0.054)	(0.796)	(1.086)	(0.214)	(1.127)
Observations	5991	5991	5991	5991	5991
Adjusted R^2	0.950	0.856	0.602	0.870	0.935
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. The spillover ring contains every district below the very-high threshold whose centroid lies within 50 km of the centroid of a very-high-conflict district. The omitted comparison group is the districts below the very-high threshold outside the ring. All specifications include district and year fixed effects, and the reference year is 1981. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

A.5 Demographic Composition and Perpetrators

Table A16: Effect of Very High Conflict on Demographics

	(1) Mean Age	(2) Dep Ratio	(3) Sex Ratio	(4) Children %	(5) Working Age %
Very High × 1993	0.195	0.764	-2.705**	0.435	-0.446
	(0.227)	(1.878)	(1.255)	(0.558)	(0.465)
Very High × 2007	-0.559*	3.862**	-2.173	2.007***	-1.498***
	(0.290)	(1.903)	(1.548)	(0.632)	(0.510)
Very High × 2017	-0.506	0.143	-4.913**	1.058	-0.337
	(0.440)	(1.965)	(1.910)	(0.779)	(0.515)
Observations	5991	5991	5991	5991	5991
Adjusted R^2	0.870	0.763	0.415	0.844	0.765
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. Very high conflict = >100 victims. The omitted comparison group is all districts below the very-high threshold. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A17: Effect of Very High Conflict on Migration and Culture

	(1) Migration	(2) Indigenous	(3) Quechua
Very High × 1993	1.715	0.763	0.676
	(1.621)	(1.663)	(1.621)
Very High × 2007	4.367	1.040	0.990
	(4.955)	(1.823)	(1.866)
Very High × 2017	-1.510	-0.504	-0.584
	(2.556)	(1.718)	(1.753)
Observations	5991	5991	5991
Adjusted R^2	0.602	0.935	0.938
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes. All specifications include district and year fixed effects, with all districts below the very-high threshold as the omitted comparison group and 1981 as the reference year. Standard errors clustered by district in parentheses. The migration rate is the share of the population not born in the current district of residence, and Indigenous is the share reporting an indigenous language as mother tongue. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

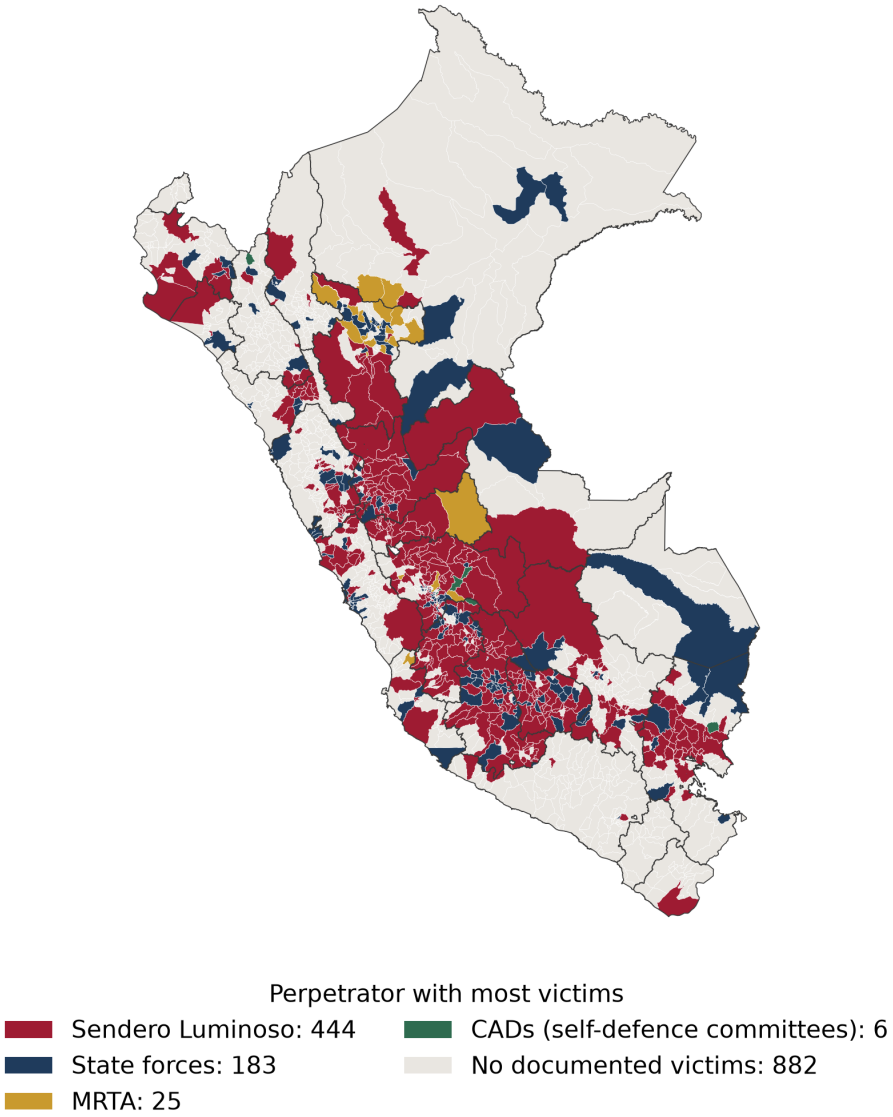


Figure A5: Conflict by Dominant Perpetrator, 1980–2000

Notes. Districts colored by the perpetrator responsible for the largest number of documented victims, which can be Sendero Luminoso, state agents, MRTA, or CADs (Comités de Autodefensa, civilian self-defense committees), with ties assigned in that order. Districts without documented victims in gray. The Sendero-dominant, State-dominant, and Mixed grouping used in the tables applies a majority threshold to the same attribution data (Appendix E.1).

B Additional Historical Background

This appendix expands the historical account in Section 2, covering the origins of the conflict, the state response, and the conflict's end and aftermath.

B.1 Origins of the Conflict

The Shining Path (Sendero Luminoso) emerged from the provincial University of San Cristóbal de Huamanga in Ayacucho, one of Peru's poorest and most neglected departments (Degregori, 2010). Founded by philosophy professor Abimael Guzmán Reynoso in the late 1960s, the organization developed a rigid Maoist ideology that rejected both the Peruvian left's electoral strategies and the Soviet model in favor of protracted people's war modeled on the Chinese revolution (Palmer, 1995). Guzmán, who styled himself "Chairman Gonzalo," cultivated a personality cult and demanded absolute loyalty from followers (Stern, 1998).

The movement drew initial support from university students and teachers frustrated by Peru's persistent inequality and the failure of agrarian reform to transform rural livelihoods (McClintock, 1984). Ayacucho's isolation, both geographic and economic, created fertile ground for revolutionary mobilization. The department had among the highest poverty rates, lowest literacy levels, and weakest state presence in Peru (Degregori, 2010). When Shining Path launched its armed struggle on May 17, 1980, burning ballot boxes in the village of Chuschi on the eve of Peru's return to democracy, it did so in a region where the state had long been experienced primarily as an extractive and repressive force.

B.2 State Response and Human Rights Violations

The Peruvian state's response to Shining Path moved from indiscriminate repression in the emergency zones during the 1980s towards more selective operations later in the conflict (Comisión de la Verdad y Reconciliación, 2003, Conclusiones Generales), within a state whose institutional weakness limited its capacity to respond (Mauceri, 1995). President Fernando Belaúnde Terry (1980–1985) initially dismissed the insurgency as cattle rustlers or delinquents, allowing Shining Path to consolidate control over much of rural Ayacucho. When the military was finally deployed in December 1982, commanders implemented counterinsurgency strategies that treated entire communities as enemy combatants (Comisión de la Verdad y Reconciliación, 2003).

The declaration of emergency zones suspended constitutional rights and granted military commanders authority over civilian governance. In these areas, which eventually covered much of the highlands, arbitrary detention, torture, extrajudicial execution, and forced disappearance became routine (Burt, 2007). The military's failure to distinguish between guerrillas and civilians, compounded by linguistic barriers, cultural prejudice against indigenous populations, and pressure to produce results, generated massive civilian casualties. The CVR attributed approximately 31% of the deaths and disappearances reported to it to state agents, alongside the 54% attributed to Shining Path (Comisión de la Verdad y

Reconciliación, 2003, Anexo 2 and Conclusiones Generales).

The armed forces' abuses proved counterproductive, alienating communities that might otherwise have supported counterinsurgency efforts and providing Shining Path with recruitment opportunities (Comisión de la Verdad y Reconciliación, 2003). Only in the late 1980s did the military begin developing more sophisticated approaches, including support for peasant self-defense patrols (rondas campesinas) that would prove central to eventual victory (Starn, 1998). These community-based organizations, armed and supported by the military but rooted in pre-existing traditions of communal governance, provided an alternative to both Shining Path's authoritarian control and the state's distant and often abusive presence.

B.3 End of Conflict and Persistence

The conflict's turning point came on September 12, 1992, when a specialized police intelligence unit captured Abimael Guzmán in Lima (Comisión de la Verdad y Reconciliación, 2003). The arrest decapitated an organization built around Guzmán's cult of personality and ideological authority. Within months, much of Shining Path's leadership was imprisoned, and the organization fragmented into competing factions. Guzmán's subsequent call for peace negotiations from prison further demoralized remaining fighters (Comisión de la Verdad y Reconciliación, 2003).

The smaller Túpac Amaru Revolutionary Movement (MRTA), which had operated primarily in the central jungle and urban areas with a more conventional leftist ideology, also collapsed in the 1990s. Its last major action, the seizure of the Japanese ambassador's residence in December 1996, ended with a military assault that killed all fourteen hostage-takers. By the late 1990s, organized insurgent activity had effectively ceased outside a few remote areas.

However, the conflict's legacy persists. Remnant Shining Path factions, increasingly tied to drug trafficking, continued low-level operations in the Apurímac, Ene, and Mantaro River Valleys (VRAEM) after 2000. More broadly, the conflict left affected communities with destroyed infrastructure, displaced populations, fractured social trust, and unresolved trauma (Theidon, 2013). The approximately 69,000 estimated deaths and disappearances represent only the most visible cost, because the CVR also documented widespread sexual violence, forced disappearance, and the destruction of entire communities. Three-quarters of victims were rural, Quechua-speaking, and poor, the same populations that had been marginalized before the conflict and would face the greatest challenges in recovery afterwards (Comisión de la Verdad y Reconciliación, 2003).

C Data Construction and Additional Statistics

This appendix documents the construction of the fixed 1981 geography, pre-conflict inequality, and the geography of the conflict. Throughout the appendix, districts are the 1,540 units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

C.1 Construction of the Fixed 1981 Geography

District boundaries changed across censuses. Peru had 1,680 districts at the 1981 census and 1,874 in 2017, and INEI renumbered many districts after 1981 when provinces were created or reorganized, so the same numeric code can designate different districts in 1981 and in later censuses. I therefore link census records by district name and province, and I trace every later district to its district of origin using INEI's register of district creations (Instituto Nacional de Estadística e Informática, 2020). A unit is a 1981 district together with the districts later created from it. When a new district was formed from parts of more than one 1981 district, those districts are merged into a single unit. Five groups of districts whose 1981 or 2007 census records cannot be separated reliably are also merged (the province of Yauyos, Lloque and Ubinas, Eten and Puerto Eten, Lluta and Majes, and Ayacucho and Carmen Alto). Every count is summed over the districts of a unit and every share is recomputed from the summed counts. Of the 1,540 units, 1,483 contain a single 1981 district. The 1981 census database excludes the departments of Apurímac, Loreto, and San Martín, although the six districts of Ucayali province (Contamana), today in Loreto, were enumerated as part of the department of Ucayali and are observed. Therefore 1,371 units are observed in 1981 and all 1,540 in 1993, 2007, and 2017, for 5,991 unit-year observations. A second definition, which assigns each new district only to its main district of origin, yields 1,645 units and is used as a robustness check. Throughout the paper, "district" refers to these units. Table A18 summarizes the sample, and Table A19 shows it by year and conflict status.

I construct the analysis sample in three steps. First, census records of every year are assigned to the 1981 units. Second, CVR victim records are assigned to the same units through the district where the events occurred. Third, geographic covariates are computed on the unit polygons, obtained by dissolving the 2017 district map by unit. The panel comprises 5,991 unit-year observations, and every unit is observed in at least three census years, so all of them enter the fixed-effects estimation. Lima is included in the main specifications, and Column (5) of Table A4 excludes the whole department of Lima.

Table A18: Districts, Census Coverage and Conflict Intensity

	Geography A (main)	Geography B
Panel A. Districts and observations		
Districts	1,540	1,645
Observed in 1981	1,371	1,468
Observed in 1993	1,540	1,645
Observed in 2007	1,540	1,645
Observed in 2017	1,540	1,645
District-year observations	5,991	6,403
Panel B. Districts by CVR-documented victims, 1980–2000		
No conflict (0)	882	938
Low (1–10)	419	448
Medium (11–50)	161	168
High (51–100)	36	40
Very high (>100)	42	51
Panel C. 1981 census coverage		
Departments in the panel	25	25
Departments with 1981 records	23	23
Departments without 1981 records	Apurímac, San Martín	
Departments with 1981 records for under half of the districts	Loreto (6 of 40)	Loreto (6 of 43)
Districts without a 1981 record	169	177
Panel D. Very-high-conflict districts by department		
Ayacucho	21	31
Huánuco	7	5
Junín	7	7
Huancavelica	3	2
San Martín	2	2
Cusco	1	1
Lima	1	1
Ucayali	0	2

Notes. Geography A, the main specification, joins each district created after 1981 with all the 1981 districts it was carved from. Geography B, used as a robustness check, assigns each created district to its first-listed district of origin. A district is observed in a census year when the census records cover its whole 1981 territory. The 1981 census database has no records for two departments and covers few districts of a third (Panel C), so their districts enter from 1993. Departments are assigned by the district that gives each unit its code. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A19: Panel Structure: Districts by Year and Conflict Status

	1981	1993	2007	2017
Very high (>100)	39	42	42	42
High (51–100)	33	36	36	36
Medium (11–50)	121	161	161	161
Low (1–10)	351	419	419	419
No conflict	827	882	882	882
Total	1371	1540	1540	1540

Notes. Conflict intensity based on CVR-documented victims 1980–2000, with Very high (>100), High (51–100), Medium (11–50), Low (1–10). Any conflict = 1+ victims. The panel is unbalanced only in 1981, because districts in departments with no or few 1981 census records (Apurímac and San Martín have none, Loreto has records for 6 of its 40 districts) and districts whose 1981 territory is only partly recorded have no 1981 row. Conflict status is time-invariant (assigned based on total 1980–2000 violence exposure). Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. The panel has 5,991 district-year observations.

C.2 Pre-Conflict Poverty and Inequality

The Gini coefficient is calculated from grouped income data where each observation represents an income value and the count of people earning that amount. Top-coded categories ("1,021 soles o más," etc.) are addressed by estimating a Pareto α parameter from the upper tail of the national distribution via OLS regression of log-survival on log-income, then imputing the conditional mean $E[Y|Y \geq \text{threshold}] = \frac{\alpha}{\alpha-1} \times \text{threshold}$. With the imputed values in place, I compute the Gini from the Lorenz curve, sorting by income, calculating cumulative population shares (p) and cumulative income shares (L), and then applying the trapezoidal rule to estimate the area under the Lorenz curve, where Gini $= 1 - 2 \times$ (Lorenz area). The estimated Pareto α was 2.86, yielding a multiplier of 1.54 for top-coded values. Coverage includes 22 of 25 departments (88%). Loreto, San Martín, and Apurímac are not covered in the 1981 census income tabulations used here, while the six districts of Ucayali province (Contamana), today in Loreto, were enumerated in 1981 as part of the department of Ucayali.

Table A20 reports the department-level poverty and inequality measures from the 1981 census discussed in Section 2, ranking departments separately by mean income and by Gini coefficient.

Table A20: Poorest and Most Unequal Departments, Peru 1981 Census

Top 10 Poorest Departments					**Top 10 Most Unequal Departments**				
Rank	Department	Income	Gini	Population	Rank	Department	Gini	Income	Population
1	Cajamarca	29.6	0.64	205,838	1	Puno	0.67	30.0	179,106
2	Puno	30.0	0.67	179,106	2	Ayacucho	0.66	30.7	90,904
3	Huancavelica	30.5	0.63	64,289	3	Cusco	0.65	34.3	197,866
4	Ayacucho	30.7	0.66	90,904	4	Cajamarca	0.64	29.6	205,838
5	Cusco	34.3	0.65	197,866	5	Huancavelica	0.63	30.5	64,289
6	Chachapoyas	39.1	0.53	52,078	6	Huaraz	0.59	47.0	170,163
7	Huánuco	42.2	0.58	86,912	7	Huánuco	0.58	42.2	86,912
8	Huaraz	47.0	0.59	170,163	8	Huancayo	0.56	49.1	198,572
9	Huancayo	49.1	0.56	198,572	9	Tambopata	0.55	81.8	12,293
10	Chaupimarca	49.7	0.51	44,770	10	Piura	0.55	54.2	233,801

Notes. Income measured as mean income per capita in 1981 soles. Population refers to census-enumerated population in the department capital. Several rows carry the name of the department capital's province, as in the source tabulation, in place of the department name (Chachapoyas for Amazonas, Huaraz for Ancash, Huancayo for Junín, Chaupimarca for Pasco, Tambopata for Madre de Dios). Departments ranked by mean income (left panel) and Gini coefficient (right panel). The five poorest departments are also among the six most unequal.

Table A21 reports the district-level analogue, listing the ten poorest and the ten most unequal districts.

Table A21: Poorest and Most Unequal Districts, Peru 1981 Census

Top 10 Poorest Districts					**Top 10 Most Unequal Districts**				
Rank	District	Department	Income	Gini	Rank	District	Department	Gini	Income
1	Huancaraylla	Ayacucho	3.3	0.48	1	Tupac Amaru	Cusco	0.93	70.7
2	Carapo	Ayacucho	3.3	0.68	2	Layo	Cusco	0.93	38.4
3	Huayllay Grande	Huancavelica	3.4	0.43	3	Parobamba	Huaraz	0.91	128.7
4	Callanmarca	Huancavelica	3.8	0.68	4	Apata	Huancayo	0.90	132.3
5	Huaya	Ayacucho	4.3	0.59	5	Quinuabamba	Huaraz	0.90	253.6
6	Paccaritambo	Cusco	4.6	0.44	6	Macari	Puno	0.90	49.8
7	Capacmarca	Cusco	4.9	0.72	7	Pampamarca	Cusco	0.89	110.9
8	Morcolla	Ayacucho	5.0	0.46	8	Chacabamba	Huánuco	0.89	90.0
9	Ccarhuayo	Cusco	5.4	0.67	9	Coata	Puno	0.89	36.1
10	Pusi	Puno	5.7	0.69	10	Chuschi	Ayacucho	0.89	6.0

Notes. Sample restricted to districts with population >100. Income measured as mean income per capita in 1981 soles. The Department column follows the same capital-province naming convention as Table A20 (for example Huaraz for Ancash, Huancayo for Junín). The poorest districts are concentrated in Ayacucho, Huancavelica, Cusco, and Puno, departments that would experience intense violence. Chuschi (Ayacucho), where Shining Path launched its armed struggle on May 17, 1980, ranks among the ten most unequal districts.

The 1981 census income tabulation gives a Gini coefficient for the 1,322 districts that correspond to exactly one 1981 district, 31 of them very-high-conflict districts. Figure A6 maps these coefficients and outlines the very-high-conflict districts, and Figure A7 shows their distribution by conflict intensity. The median Gini is lowest in districts without documented victims and higher in every conflict category, and the very-high category does not have the highest median.

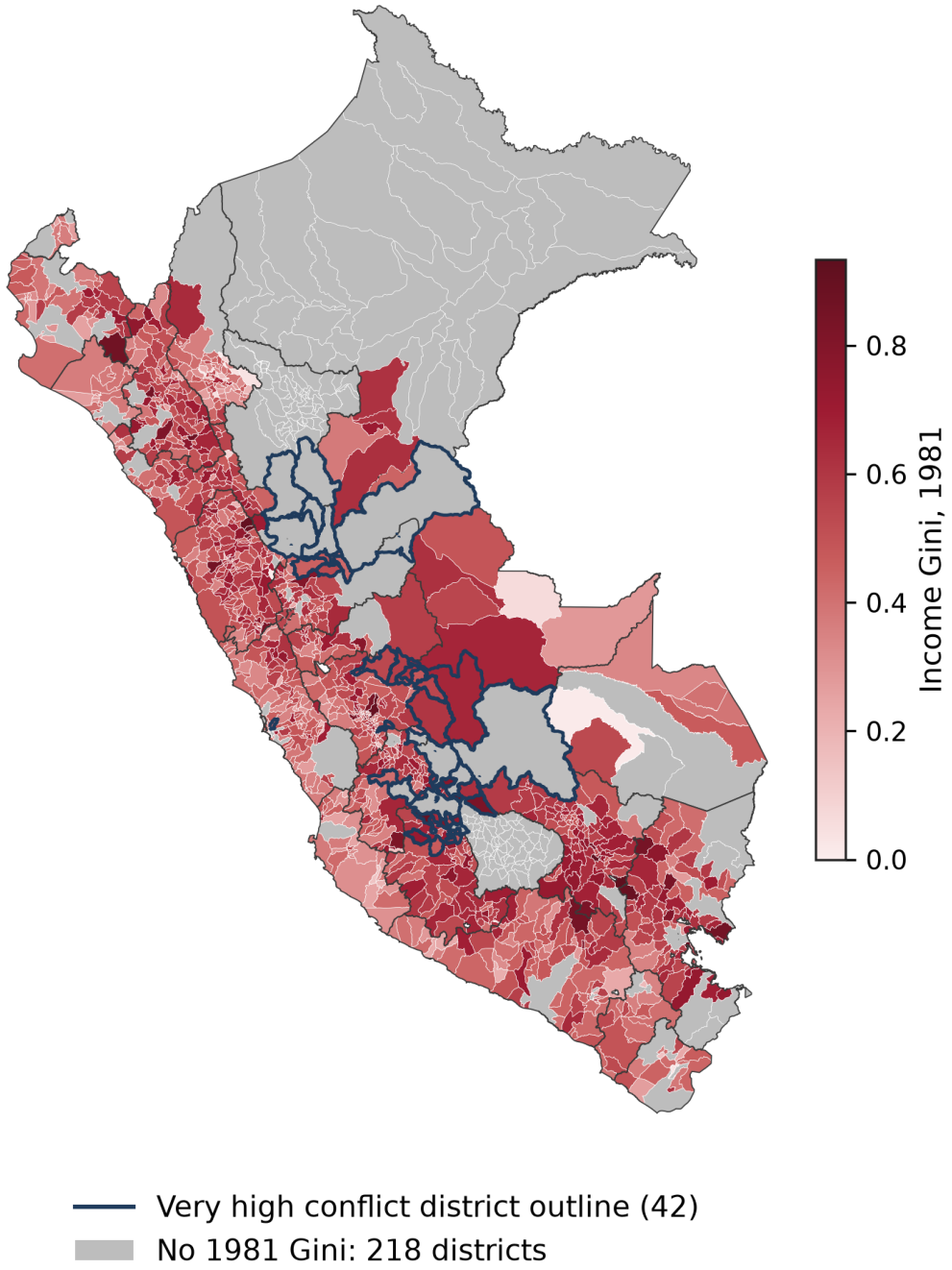


Figure A6: Income Inequality in 1981 by District

Notes. Gini coefficient of declared monetary income in the 1981 census tabulation, on the districts that correspond to exactly one 1981 district. Districts made of several 1981 districts, or without a 1981 tabulation, are shown in gray. Very-high-conflict districts (>100 CVR-documented victims) are outlined.

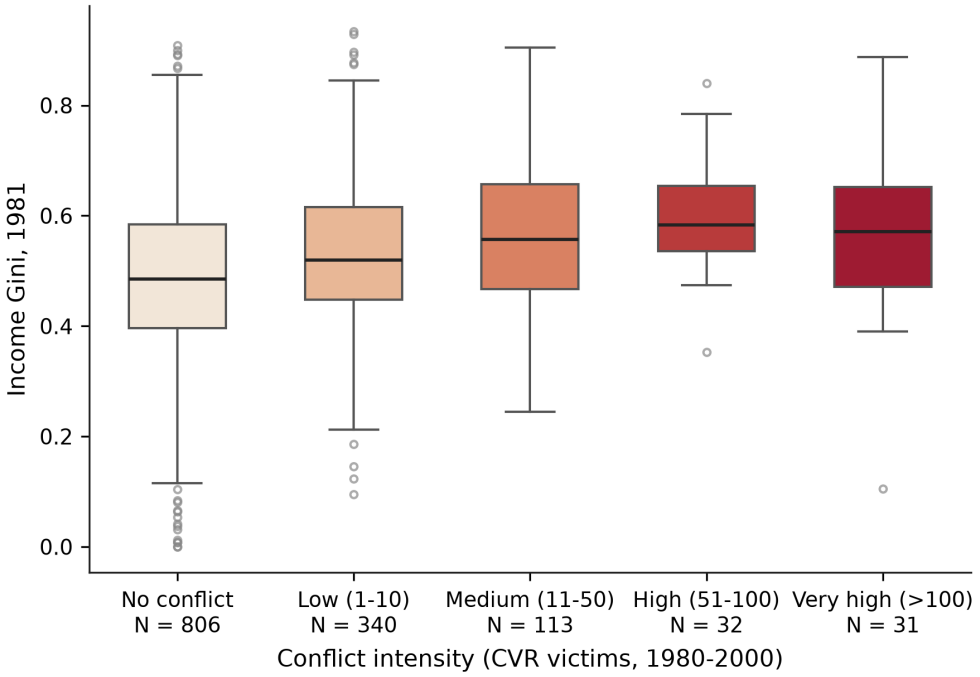


Figure A7: Income Inequality in 1981 by Conflict Intensity

Notes. Distribution of the 1981 district Gini coefficient by conflict intensity (CVR-documented victims 1980–2000). The number of districts with a Gini in each category is shown below each box. Boxes span the interquartile range, the line marks the median, and whiskers extend to 1.5 times the interquartile range.

Table A22 relates the Gini to conflict intensity. Without controls, a one-unit higher Gini goes with 1.861 more log points of victims ($p < 0.01$ with province clusters). Within departments the coefficient falls to 0.762 ($p = 0.004$), and it is 0.717 ($p = 0.003$) with geographic controls and 0.727 ($p = 0.006$) without the flagged small-sample coefficients. The Gini does not predict the very-high indicator once department fixed effects are included (0.022, $p = 0.606$). Inequality in 1981 therefore tracks the spread of violence within departments, and it does not single out the districts that crossed the very-high threshold.

Table A22: Inequality in 1981 and Conflict Intensity

Panel A. Regressions on the 1981 income Gini	Coefficient	SE, province	SE, robust	Observations	Very high	R^2
$\log(1 + victims)$						
No controls	1.861***	(0.385)	(0.251)	1,322	31	0.040
Department fixed effects	0.762***	(0.264)	(0.236)	1,322	31	0.406
Department FE and geography	0.717***	(0.239)	(0.219)	1,322	31	0.486
As above, excluding flagged Gini	0.727***	(0.261)	(0.241)	1,300	31	0.485
Very-high indicator (linear probability)						
No controls	0.074	(0.053)	(0.034)	1,322	31	0.005
Department fixed effects	0.022	(0.043)	(0.035)	1,322	31	0.118
Department FE and geography	0.024	(0.042)	(0.034)	1,322	31	0.213
As above, excluding flagged Gini	0.029	(0.046)	(0.038)	1,300	31	0.214

Panel B. Chuschi (Ayacucho), 1981 income tabulation	
Gini, all income cells	0.887
Gini, excluding cells with fewer than 5 persons	0.801
Persons with a declared income	1,320
Share with zero income	0.406
Persons with income not declared (dropped)	417
Income cells	27
Cells with fewer than 5 persons	12

Notes. Panel A uses the districts made of exactly one 1981 district, for which the 1981 census income tabulation gives a Gini coefficient. Geography adds altitude, distance to Lima and terrain ruggedness. Stars from 1981-province clusters. Flagged Gini values come from a single income level or from fewer than 30 persons in the income tabulation. Panel B rebuilds the Chuschi Gini from the district income cells. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Chuschi, where Sendero Luminoso began its armed struggle, has one of the highest district Gini coefficients in 1981 (Table A21). Panel B of Table A22 rebuilds it from the 27 income cells of the district tabulation. The value of 0.887 reflects a large share of persons with zero declared monetary income (a share of 0.406 among the 1,320 persons with a declared income). Dropping the 12 cells with fewer than 5 persons gives 0.801, so the extreme value does not come from small cells alone. In a subsistence economy, zero monetary income is common and the Gini of declared income overstates inequality in living standards.

C.3 Maps of Conflict Intensity

Figure A8 shows the 42 very-high-conflict districts, those with more than 100 documented victims. They are 2.7 percent of the 1,540 districts and account for 65.0 percent of all documented victims. Twenty-one lie in Ayacucho, seven in Junín, seven in Huánuco, three in Huancavelica, two in San Martín, and one each in Cusco and Lima. Three of them combine

current districts from more than one department.



Figure A8: Very-High-Conflict Districts (>100 Victims)

Notes. The 42 districts with more than 100 CVR-documented victims, 2.7 percent of the 1,540 districts, accounting for 65.0 percent of documented victims. Departments are assigned by the district that gives each unit its code.

Figure A9 enlarges the central and southern highlands. Ayacucho holds 21 of the very-high-conflict districts, Junín seven, and Huancavelica three. Medium- and high-intensity districts surround this core in Apurímac, Huancavelica, and southern Ayacucho.

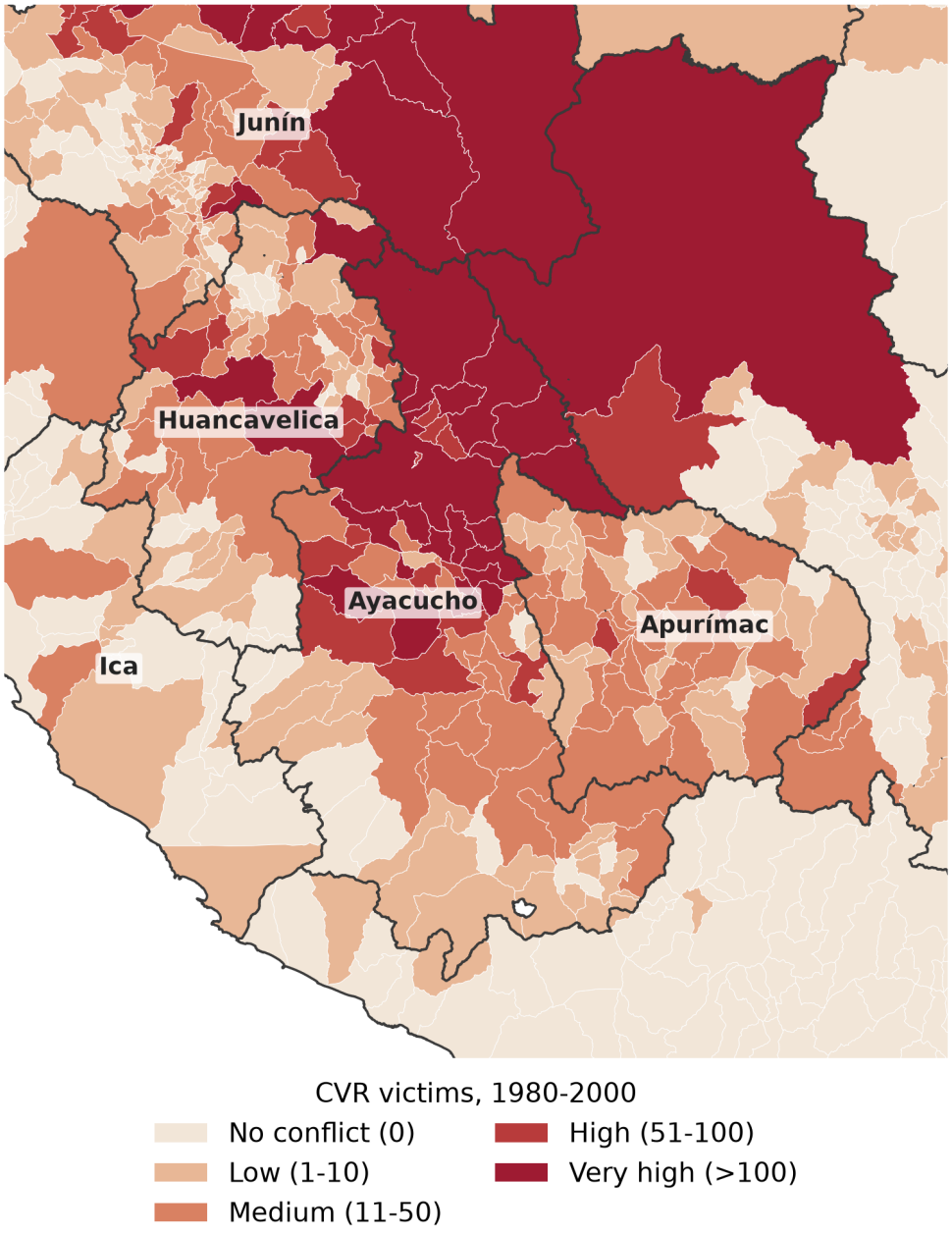
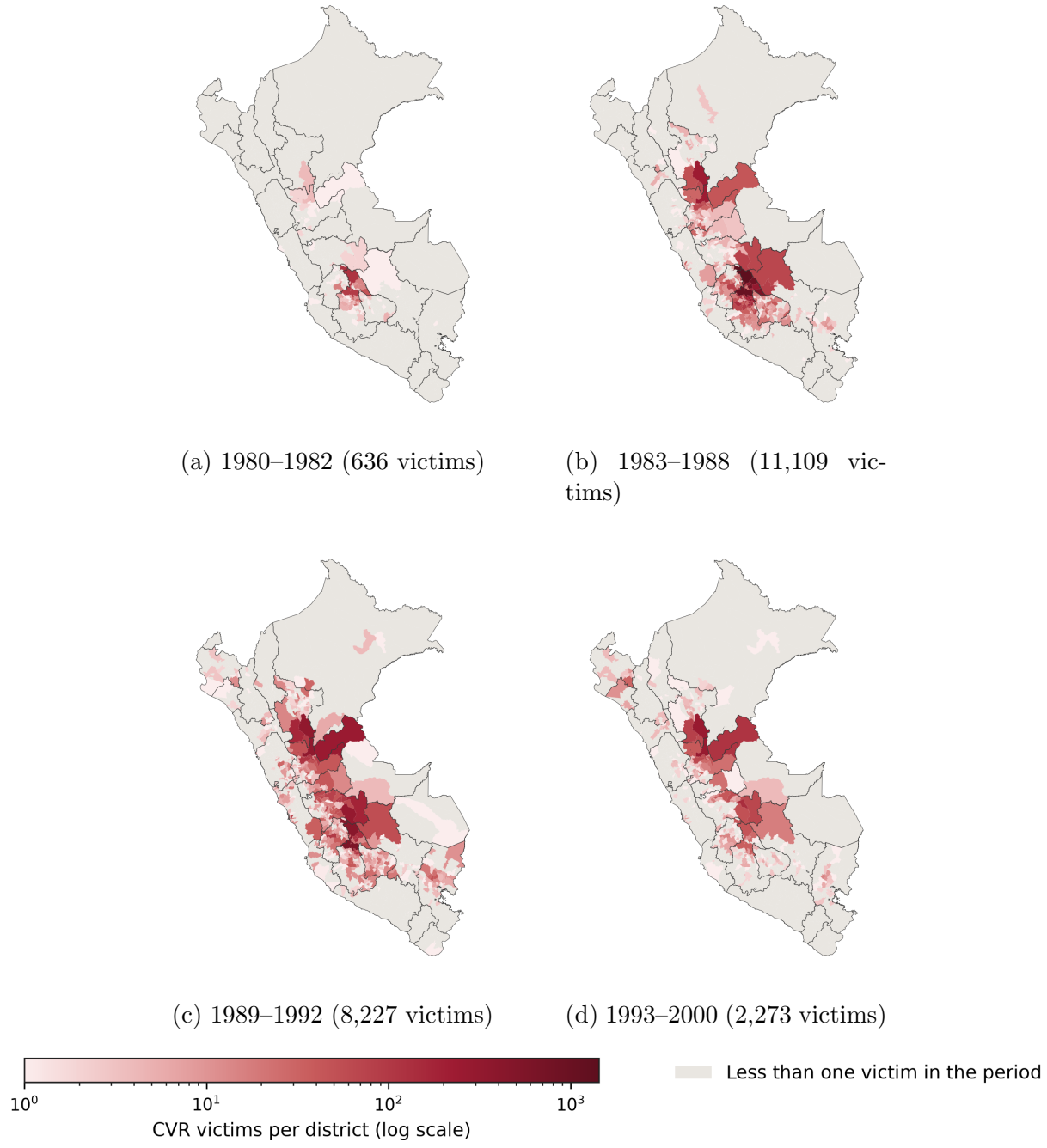


Figure A9: Conflict Epicenter in the Central-Southern Highlands

Notes. Conflict intensity by district in the central and southern highlands, by CVR-documented victims 1980–2000. Nationwide the categories contain 882 districts without victims, 419 Low (1–10), 161 Medium (11–50), 36 High (51–100), and 42 Very high (>100). Department boundaries in dark lines.

Figure A11 shows how victims are distributed across the districts with at least one documented victim, in counts and per 1,000 inhabitants of 1981. Both distributions are skewed, and the thresholds that define the intensity categories cut through continuous distributions. Figure A10 maps victims in the four phases of the conflict, following the periods of the CVR. In 1980–1982, 104 districts recorded at least one victim. Violence then spread to 373 districts in 1983–1988, the period with the most victims, and to 531 districts in 1989–1992, the widest reach of the war. After Guzmán's capture it receded, and 283 districts recorded victims in 1993–2000.



(a) 1980–1982 (636 victims)

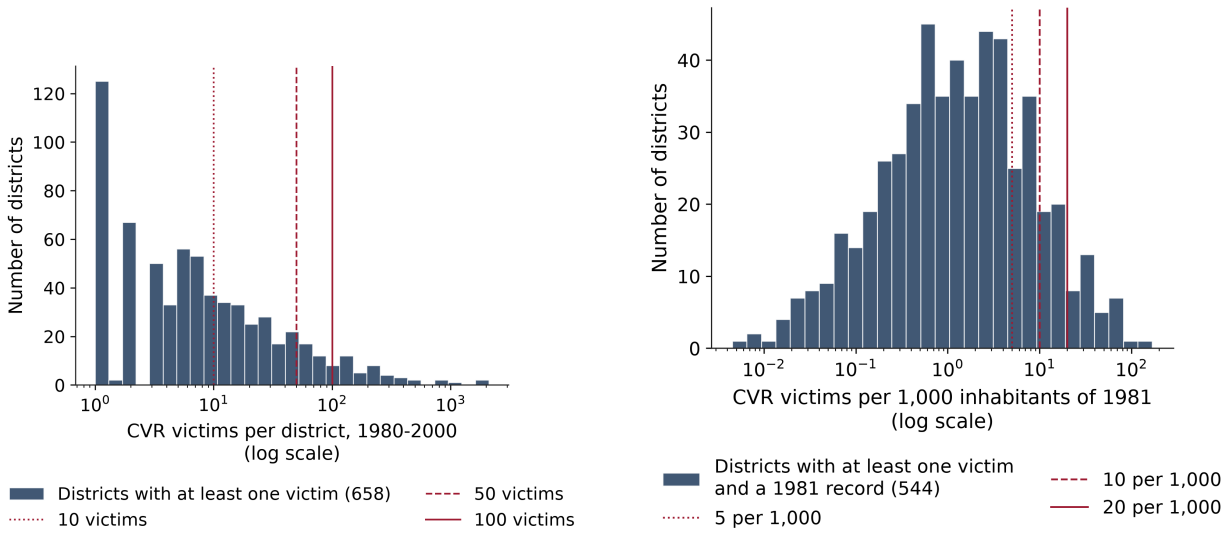
(b) 1983–1988 (11,109 victims)

(c) 1989–1992 (8,227 victims)

(d) 1993–2000 (2,273 victims)

Figure A10: Documented Victims by District and Phase of the Conflict

Notes. CVR-documented victims (deaths and disappearances) per district in four phases of the conflict, on a common logarithmic color scale and on the 1,540 districts of the fixed 1981 geography. The phases follow the periods of the CVR file (1980–1982, 1983–1985 and 1986–1988 combined, 1989–1992, and 1993–2000). Districts with less than one victim in a phase are shown in gray. The numbers of victims and of districts with at least one victim are 636 and 104 in 1980–1982, 11,109 and 373 in 1983–1988, 8,227 and 531 in 1989–1992, and 2,273 and 283 in 1993–2000.



(a) Victims per district

(b) Victims per 1,000 inhabitants of 1981

Figure A11: Distribution of Documented Victims per District

Notes. Number of districts by CVR-documented victims 1980–2000, log scale. Panel (a) uses the 658 districts with at least one victim and marks the thresholds of 10, 50, and 100 victims used for the intensity categories. Panel (b) uses the 544 of them with a 1981 census record and divides victims by the 1981 population, with lines at 5, 10, and 20 victims per 1,000 inhabitants, the thresholds of Table A2. The 1981 census database does not cover Apurímac and San Martín and covers Loreto only through the six districts of Contamana.

D Robustness Checks

This appendix reports alternative conflict measures, heterogeneity by geography, duration, and region, the mediator exercise, literacy trajectories, the auxiliary test statistics, and a set of additional robustness analyses. All specifications use district and year fixed effects with 1981 as the reference year unless stated, and standard errors are clustered by district.

D.1 Alternative Conflict Measures

Table A23 replaces the very-high indicator with an indicator for any documented victim and with the log of victims plus one. Districts with any documented victim grew less than districts without victims, by 0.041 log points in 1993 ($p < 0.05$) and 0.068 in 2017 ($p < 0.05$), with a smaller and insignificant 2007 coefficient (-0.026). The log of victims gives the same pattern, -0.024 per log point in 1993 ($p < 0.01$), -0.010 in 2007, and -0.026 in 2017 ($p < 0.05$). These measures give most weight to the many low- and medium-intensity districts, whose population losses appear in Table A35.

Table A23: Robustness: Alternative Conflict Intensity Measures

	(1) Very High (>100)	(2) Any (1+)	(3) Log(Victims)
× 1993	-0.118*		
	(0.070)		
× 2007	0.049		
	(0.088)		
× 2017	0.057		
	(0.111)		
× 1993		-0.041**	
		(0.021)	
× 2007		-0.026	
		(0.027)	
× 2017		-0.068**	
		(0.034)	
× 1993			-0.024***
			(0.008)
× 2007			-0.010
			(0.010)
× 2017			-0.026**
			(0.012)
Observations	5991	5991	5991
Adjusted R^2	0.950	0.950	0.950
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Comparison group	Below VH	No victims	Continuous

Notes. Standard errors clustered by district in parentheses. Column (1) uses the indicator for more than 100 victims, with all districts below the threshold as the omitted group. Column (2) uses an indicator for any documented victim, with the districts without victims as the omitted group. Column (3) uses log(total victims + 1) as a continuous measure. All columns include district and year fixed effects, and the reference year is 1981. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

D.2 Geographic Heterogeneity

Table A24 splits the sample by terciles of terrain ruggedness, with 14, 15, and 13 very-high-conflict districts. In the most rugged tercile the very-high districts lost 0.326 log points by

1993 ($p < 0.01$) and remained 0.196 below by 2007 ($p < 0.10$). In the least rugged tercile the coefficients are positive, with 0.263 in 2007 ($p < 0.10$), and the middle tercile shows no differential.

Table A24: Population Recovery by Terrain Ruggedness

	(1) Low Rugged	(2) Med Rugged	(3) High Rugged	(4) Full Sample
Very High × 1993	0.130	-0.119	-0.326***	-0.118*
	(0.090)	(0.127)	(0.093)	(0.070)
Very High × 2007	0.263*	0.108	-0.196*	0.049
	(0.156)	(0.134)	(0.107)	(0.088)
Very High × 2017	0.279	0.110	-0.189	0.057
	(0.180)	(0.162)	(0.172)	(0.111)
Observations	1997	1997	1997	5991
Adjusted R^2	0.953	0.942	0.950	0.950
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Comparison group	Below VH, tercile	Below VH, tercile	Below VH, tercile	All below VH

Notes. Standard errors clustered by district in parentheses. Ruggedness terciles based on the Terrain Ruggedness Index of Nunn and Puga (2012). The omitted group is districts below the very-high threshold within the indicated tercile. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A25 splits by distance to Lima. The nine very-high districts in the closest tercile grew faster than their neighbors in every census year (0.302, 0.612, and 0.882 log points, all $p < 0.01$). The 33 very-high districts at intermediate distance lost 0.230 log points by 1993 ($p < 0.01$) and show no differential afterwards. No very-high district lies in the most remote tercile.

Table A25: Population Recovery by Distance to Lima

	(1) Close to Lima	(2) Medium	(3) Far from Lima	(4) Full Sample
Very High × 1993	0.302***	−0.230***	–	−0.118*
	(0.087)	(0.075)		(0.070)
Very High × 2007	0.612***	−0.086	–	0.049
	(0.143)	(0.089)		(0.088)
Very High × 2017	0.882***	−0.142	–	0.057
	(0.155)	(0.109)		(0.111)
Observations	1996	1995	2000	5991
Adjusted R^2	0.950	0.951	0.951	0.950
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Comparison group	Below VH, tercile	Below VH, tercile	Below VH, tercile	All below VH

Notes. Standard errors clustered by district in parentheses. Distance terciles are based on the geodesic distance to central Lima. The Very High × Year interaction in Column (3) is not estimable because no very-high-conflict district falls in the far tercile (all 42 lie in the two closer terciles, 9 close and 33 medium), shown as −. The omitted group is the districts below the very-high threshold within the indicated tercile, and the reference year is 1981. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

D.3 Duration of Violence and Violence Before the 1981 Census

Table A26 restricts the sample to conflict districts with above-median shares of deaths, above-median shares of disappearances, or above-median years of documented violence. None of the very-high coefficients in these subsamples is statistically significant. The long-duration subsample contains all 42 very-high districts, so a short-duration column cannot be estimated, and the high-deaths subsample contains only five.

Table A26: Population Recovery by Conflict Duration

	(1) High Deaths	(2) High Disappearances	(3) Long Duration	(4) Full
Very High × 1993	0.064	-0.104	-0.041	-0.118*
	(0.246)	(0.076)	(0.077)	(0.070)
Very High × 2007	0.304	0.033	0.106	0.049
	(0.337)	(0.091)	(0.093)	(0.088)
Very High × 2017	0.356	0.078	0.170	0.057
	(0.432)	(0.114)	(0.118)	(0.111)
Observations	1258	1256	1054	5991
Adjusted R^2	0.953	0.966	0.963	0.950
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Comparison group	Other conflict	Other conflict	Other conflict	All below VH

Notes. High Deaths/Disappearances = above-median share of deaths (disappearances) among conflict districts. Long Duration = above-median years of documented violence among conflict districts. No short-duration column is shown because it is not estimable, since every very-high-conflict district experienced above-median conflict duration, so no very-high-conflict district appears in the short-duration subsample. Columns (1)–(3) restrict the sample to conflict districts in the indicated category, so Very High coefficients compare very-high-conflict districts with other conflict districts in that category. Column (4) uses the full sample, with all districts below the very-high threshold as the comparison. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Duration rises with intensity, and the median number of years with at least one documented victim is 2 in Low, 6 in Medium, 9 in High, and 14.5 in very-high districts (Table A38, Panel A). Among districts below the very-high threshold, 184 had victims in five or more years. Dropping them leaves the main estimates almost unchanged, with −0.125 for log population in 1993 and −10.414 for electricity in 2007. Prolonged low-intensity violence in the comparison group does not drive the results.

The CVR records give the year of the events but not the month, so violence before the July 1981 census can only be bounded by victims in 1980 and 1981. Twenty-two of the 42 very-high districts have a victim in those years. Excluding every district with a 1980–81 victim leaves 20 very-high districts. The 1993 population coefficient becomes −0.098 and is no longer significant, and the 2007 electricity deficit shrinks to −7.177 ($p = 0.056$ with

district clusters and 0.121 with province clusters) while the 1993 deficit grows to −5.594 ($p < 0.01$). Literacy is unchanged (5.678 in 2007 and 6.450 in 2017, both $p < 0.05$). The excluded districts are the core of the conflict zone, so this check weakens the 1993 population deficit and the precision of the 2007 electricity deficit.

D.4 Regional Heterogeneity

Table A27 estimates the main specification separately for the Sierra and the Selva. Sierra and Selva are coded by district, so a department can contribute districts to both groups. The Costa is excluded, and it contains one of the 42 very-high districts. The 26 very-high districts of the Sierra lost 0.246 log points of population by 1993 ($p < 0.01$) and remained 0.119 below in 2007 ($p < 0.10$) and 0.122 below in 2017 (not significant). The 15 very-high districts of the Selva show positive and insignificant coefficients (0.031, 0.196, and 0.228). Literacy among residents aged 15 and over rose 6.204 points more in very-high Sierra districts by 2007 and 8.126 by 2017 (both $p < 0.01$), while the Selva coefficients are small and insignificant.

Table A27: Regional Heterogeneity: Sierra versus Selva

	Log Population			Literacy Rate, 15+		
	(1) Sierra	(2) Selva	(3) Full	(4) Sierra	(5) Selva	(6) Full
Very High × 1993	-0.246***	0.031	-0.118*	0.559	-0.389	1.487
	(0.073)	(0.118)	(0.070)	(1.350)	(2.373)	(1.157)
Very High × 2007	-0.119*	0.196	0.049	6.204***	0.907	5.508***
	(0.072)	(0.148)	(0.088)	(1.599)	(3.581)	(1.702)
Very High × 2017	-0.122	0.228	0.057	8.126***	1.141	6.615***
	(0.102)	(0.156)	(0.111)	(1.762)	(3.619)	(1.876)
Observations	4089	610	5991	4089	610	5991
Adjusted R^2	0.947	0.944	0.950	0.862	0.802	0.856
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Comparison group	Below VH, Sierra	Below VH, Selva	All below VH	Below VH, Sierra	Below VH, Selva	All below VH

Notes. All specifications include district and year fixed effects, with standard errors clustered by district in parentheses. Sierra and Selva are coded by district, so one department can contribute districts to both groups. Costa districts are excluded (1 of the 42 very-high-conflict districts). Within each sample the omitted comparison group is all districts below the very-high threshold, and 1981 is the reference year. Literacy is the share literate among residents aged 15 and over. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

D.5 Mechanisms and the Decomposition of Population Effects

Table A28 adds potential mediators one at a time to the population regression and then jointly. Among the columns that keep 1981 as the reference year, the 1993 coefficient stays between −0.132 and −0.095, and mean age and the dependency ratio move it the most, to −0.095 (not significant). Columns (2) and (7) drop 1981, so their coefficients are relative to 2017 and cannot be compared with the others, including the −0.108 of the joint specification. Column (6) adds the number of workers employed in the utilities sector, a census employment count that grows with population and is not an index of infrastructure. Urban share is not recorded in 1981, so the columns that include it drop that year and cannot estimate the 2017 coefficient. The 2007 and 2017 coefficients are small and insignificant in every column.

Table A28: Mechanisms: Population Effects Controlling for Potential Mediators

	(1) Baseline	(2) +Urban†	(3) +Migr	(4) +Educ	(5) +Age	(6) +Util.	(7) All†
Very High × 1993	-0.118*	-0.114**	-0.117*	-0.118*	-0.095	-0.132**	-0.108***
	(0.070)	(0.055)	(0.070)	(0.070)	(0.065)	(0.067)	(0.041)
Very High × 2007	0.049	0.058	0.050	0.050	0.013	0.028	0.040
	(0.088)	(0.037)	(0.088)	(0.088)	(0.085)	(0.084)	(0.030)
Very High × 2017	0.057		0.057	0.060	0.008	0.000	
	(0.111)		(0.110)	(0.111)	(0.094)	(0.103)	
Urban (%)		0.004***					0.002***
		(0.000)					(0.000)
Migration Rate (%)			-0.000				-0.000**
			(0.000)				(0.000)
Literacy Rate, 15+ (%)				-0.000			-0.010***
				(0.002)			(0.002)
Mean Age					-0.100***		-0.105***
					(0.004)		(0.004)
Dependency Ratio					-0.005***		-0.006***
					(0.001)		(0.001)
Utilities employment						0.001***	0.000***
						(0.000)	(0.000)
Observations	5991	4506	5991	5991	5991	5991	4506
Adjusted R^2	0.950	0.972	0.950	0.950	0.965	0.952	0.983
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. All specifications include district and year fixed effects, the omitted comparison group is all districts below the very-high threshold, and 1981 is the reference year. Column (2) adds the urban share, Column (3) the migration rate, Column (4) literacy among residents aged 15 and over, Column (5) mean age and the dependency ratio, and Column (6) the number of workers employed in the utilities sector (electricity, gas and water), a count that rises with population and is not an index of infrastructure. Urban share is not recorded in 1981, so Columns (2) and (7) (†) drop that year. In these two columns the 1993 and 2007 coefficients are measured relative to 2017, not 1981, and the 2017 coefficient is not estimated, so they are not comparable with the other columns. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

D.6 Literacy Trajectories

Panel (a) of Figure 4 follows the gap in mean literacy among residents aged 15 and over on the same 1,309 districts in every census. The gap between very-high districts and the rest was 11.3 points in 1981 and 10.8 points in 1993, and it narrowed to 5.7 points in 2007 and 4.4 points in 2017 (Appendix Table A9). The 1961 and 1972 values come from the series of Albertus (2020), whose age universe is not documented, and they are shown for context. Table A29 reports the means for all districts observed in each census, where the gap falls from 11.1 points in 1981 to 3.4 points in 2017.

Table A29: Literacy Rate, Ages 15 and Over: 1981–2017

	1981		1993		2007		2017	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Very high conflict	60.8	17.6	68.9	14.9	81.9	8.8	84.3	7.4
All other units	71.9	16.4	77.4	13.1	86.4	8.8	87.7	7.7
Gap (all other − very high), pp	11.1		8.5		4.5		3.4	
Units, very high	39		42		42		42	
Units, all other	1,332		1,498		1,498		1,498	

Notes. Means and standard deviations across districts of the literacy rate of persons aged 15 and over (same age universe in every census). Very high conflict = more than 100 CVR-documented victims, and all other = every district below that threshold. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. Districts without a 1981 record (departments absent from the 1981 census database) enter from 1993.

D.7 Auxiliary Test Statistics

Table A30 collects test statistics cited in the text. Randomization inference that reassigns the very-high label across districts gives p-values of 0.050 for the 1993 population coefficient and 0.592 and 0.586 for 2007 and 2017. The joint tests of the spillover coefficients reject zero for the 50, 75, and 100 km rings and not for the 25 km ring at the 5 percent level. The PPML estimates for NTL per capita are close to zero. The equality row compares the very-high path of State-dominant districts with that of all other very-high districts and does not reject equality ($p = 0.242$). The Sendero share of victims is jointly related to population, literacy, and mean age ($p < 0.001$ for each).

Table A30: Auxiliary Test Statistics

Panel A: Randomization inference, log population (500 permutations)			
	1993	2007	2017
Estimated coefficient	−0.118	0.049	0.057
Randomization-inference p-value	0.050	0.592	0.586
Panel B: Joint tests that all three spillover coefficients equal zero			
	$F(3, 1539)$		p-value
Ring 25 km	2.18		0.088
Ring 50 km	3.20		0.023
Ring 75 km	10.83		<0.001
Ring 100 km	15.06		<0.001
Panel C: PPML and perpetrator tests			
	Coefficient	SE	p-value
PPML, NTL per capita, Very High × 2007	0.0005	0.226	0.998
PPML, NTL per capita, Very High × 2017	−0.080	0.428	0.852
Equality, State-dominant vs other very high			0.242
Sendero share × year jointly zero, log population			<0.001
Sendero share × year jointly zero, literacy 15+			<0.001
Sendero share × year jointly zero, mean age			<0.001

Notes. Panel A permutes the very-high label across districts, holding the number of very-high districts fixed, and reports the share of 500 placebo estimates at least as large in absolute value as the actual coefficient. Panel B reports joint F-tests from the ring specifications of Table A14. In Panel C the PPML rows estimate the specification of Table A56 by Poisson pseudo-maximum-likelihood on NTL per capita (N = 3,954 after dropping singleton and separated observations, 1,318 district clusters). The equality row tests whether the very-high × year path of State-dominant districts differs from that of all other very-high districts in a pooled model. The joint rows test that the three Sendero-share × year interactions are zero in the specification of Table A44. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district).

D.8 Additional Robustness Analyses

Table A31 applies the same six specifications to every main outcome. The electricity deficit of 2007 (−10.005) holds with initial conditions interacted with year (−10.780), with department-by-year fixed effects (−12.799), with 1981-province clusters ($p = 0.003$), and in the alternative geography (−7.881), and its department wild bootstrap p-value is 0.051. The 2007 sanitation gain (9.846) survives initial conditions and department trends but not the wild bootstrap ($p = 0.147$). Table A32 separates the components of sanitation. The public

sewer deficit of 2017 (−5.910) disappears with initial conditions (0.029), and the gain in pit latrines (9.638 in 2007) holds with initial conditions (8.978) and not with department trends (5.458). The 1993 population deficit is −0.118 at baseline and −0.161 with initial conditions, and it is positive with department-by-year fixed effects (0.052).

Table A31: Uniform Robustness Matrix, Part I: Population, Electricity and Sanitation

	(1) Baseline	(2) Initial cond. × year	(3) Dept. × year FE	(4) p, 1981-prov. clusters	(5) p, wild boot. dept. [95% CI]	(6) Panel B
Log population						
Very High × 1993	−0.118*	−0.161**	0.052	0.203	0.798	−0.203***
	(0.070)	(0.064)	(0.069)		[−0.624, 0.563]	(0.077)
Very High × 2007	0.049	0.014	0.238***	0.740	0.760	−0.017
	(0.088)	(0.077)	(0.083)		[−0.696, 0.782]	(0.082)
Very High × 2017	0.057	0.021	0.374***	0.772	0.803	−0.031
	(0.111)	(0.090)	(0.099)		[−1.497, 0.934]	(0.103)
Observations	5,991	5,484	5,991			6,403
Electricity (%)						
Very High × 1993	−3.723*	−5.536***	−4.503*	0.136	0.291	−3.075*
	(2.066)	(2.056)	(2.473)		[−11.066, 4.988]	(1.779)
Very High × 2007	−10.005***	−10.780***	−12.799***	0.003	0.051	−7.881***
	(2.655)	(2.699)	(3.079)		[−18.516, 0.062]	(2.441)
Very High × 2017	0.132	−2.406	−5.480**	0.972	0.986	2.147
	(2.626)	(1.970)	(2.473)		[−21.808, 31.340]	(2.300)
Observations	5,987	5,480	5,987			6,399
Sanitation, any facility (%)						
Very High × 1993	−4.661	−2.438	2.526	0.396	0.635	−5.141*
	(3.609)	(2.650)	(3.237)		[−32.492, 16.757]	(2.992)
Very High × 2007	9.846***	9.632***	10.139***	0.018	0.147	9.703***
	(3.526)	(2.606)	(3.795)		[−4.123, 20.440]	(3.226)
Very High × 2017	6.719**	5.401***	1.142	0.073	0.465	6.751**
	(3.409)	(1.555)	(3.371)		[−5.062, 30.311]	(2.905)
Observations	5,987	5,480	5,987			6,399
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	Yes
Department × year FE	No	No	Yes	No	No	No
Initial conditions × year	No	Yes	No	No	No	No

Notes. Each row reports the very-high × year coefficient for the outcome in italics, with 1981 as the reference year. Column (1) is the baseline with district and year fixed effects and standard errors clustered by district in parentheses. Column (2) adds 1981 initial conditions interacted with year (own 1981 level where available, literacy, indigenous share, log population and geography) and uses the districts with a 1981 record. Column (3) adds department-by-year fixed effects. Column (4) reports the p-value of the baseline coefficient with 1981-province clusters, and Column (5) the department wild cluster bootstrap p-value with the bootstrap 95% confidence interval in brackets. Column (6) uses the alternative fixed geography (panel B). Stars in Columns (1)–(3) and (6) come from district-clustered p-values. Sanitation, any facility, counts dwellings with a public network, septic tank, or pit latrine, and public sewer counts only the public sewer network. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Table A32: Uniform Robustness Matrix, Part II: Sanitation Components

	(1) Baseline	(2) Initial cond. × year	(3) Dept. × year FE	(4) p, 1981-prov. clusters	(5) p, wild boot. dept. [95% CI]	(6) Panel B
Public sewer (%)						
Very High × 1993	−1.120	−1.012	0.973	0.666	0.846	−1.598
	(2.222)	(1.645)	(2.070)		[−17.816, 5.879]	(1.818)
Very High × 2007	−3.023	−1.328	1.175	0.215	0.645	−4.163*
	(2.767)	(2.351)	(2.843)		[−17.570, 7.628]	(2.362)
Very High × 2017	−5.910*	0.029	−7.316**	0.035	0.146	−6.442**
	(3.033)	(3.040)	(3.230)		[−16.711, 7.488]	(2.855)
Observations	5,987	5,480	5,987			6,399
Pit latrines (%)						
Very High × 1993	−3.371	−1.292	1.601	0.344	0.537	−3.365*
	(2.137)	(2.094)	(2.096)		[−18.323, 11.109]	(1.853)
Very High × 2007	9.638***	8.978***	5.458	0.027	0.481	11.229***
	(3.469)	(2.789)	(3.724)		[−14.412, 49.717]	(3.202)
Very High × 2017	9.501***	4.892	3.747	0.012	0.231	11.006***
	(3.554)	(3.043)	(3.840)		[−8.249, 35.628]	(3.327)
Observations	5,987	5,480	5,987			6,399
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	Yes
Department × year FE	No	No	Yes	No	No	No
Initial conditions × year	No	Yes	No	No	No	No

Notes. Each row reports the very-high × year coefficient for the outcome in italics, with 1981 as the reference year. Column (1) is the baseline with district and year fixed effects and standard errors clustered by district in parentheses. Column (2) adds 1981 initial conditions interacted with year (own 1981 level where available, literacy, indigenous share, log population and geography) and uses the districts with a 1981 record. Column (3) adds department-by-year fixed effects. Column (4) reports the p-value of the baseline coefficient with 1981-province clusters, and Column (5) the department wild cluster bootstrap p-value with the bootstrap 95% confidence interval in brackets. Column (6) uses the alternative fixed geography (panel B). Stars in Columns (1)–(3) and (6) come from district-clustered p-values. Sanitation, any facility, counts dwellings with a public network, septic tank, or pit latrine, and public sewer counts only the public sewer network. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Table A33: Uniform Robustness Matrix, Part III: Education

	(1) Baseline	(2) Initial cond. × year	(3) Dept. × year FE	(4) p, 1981-prov. clusters	(5) p, wild boot. dept. [95% CI]	(6) Panel B
Literacy, ages 15+ (%)						
Very High × 1993	1.487	−1.987**	0.416	0.261	0.378	1.035
	(1.157)	(0.899)	(1.178)		[−2.754, 6.649]	(1.025)
Very High × 2007	5.508***	0.427	0.960	0.023	0.475	6.240***
	(1.702)	(0.726)	(1.634)		[−7.356, 19.792]	(1.561)
Very High × 2017	6.615***	0.630	1.623	0.017	0.487	7.567***
	(1.876)	(0.674)	(1.735)		[−8.023, 23.466]	(1.735)
Observations	5,991	5,484	5,991			6,403
Secondary complete, ages 25+ (%)						
Very High × 1993	−2.354***	−1.341**	−0.443	0.032	0.648	−3.197***
	(0.904)	(0.624)	(0.938)		[−11.934, 4.743]	(0.798)
Very High × 2007	−6.283***	−1.232	−4.500***	<0.001	0.198	−7.536***
	(1.404)	(1.182)	(1.556)		[−17.293, 5.996]	(1.266)
Very High × 2017	−5.495***	−1.615	−4.263**	0.006	0.284	−7.058***
	(1.432)	(1.155)	(1.679)		[−12.744, 6.594]	(1.325)
Observations	5,991	5,484	5,991			6,403
Higher education, ages 25+ (%)						
Very High × 1993	−0.838	−0.735	0.269	0.301	0.848	−1.505**
	(0.703)	(0.488)	(0.731)		[−7.773, 4.465]	(0.587)
Very High × 2007	−3.156**	−0.440	−1.301	0.035	0.501	−4.159***
	(1.271)	(0.927)	(1.395)		[−14.435, 7.908]	(1.105)
Very High × 2017	−1.554	−0.730	−0.498	0.348	0.807	−2.786**
	(1.333)	(0.947)	(1.503)		[−15.762, 9.888]	(1.159)
Observations	5,991	5,484	5,991			6,403
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	No	Yes	Yes	Yes
Department × year FE	No	No	Yes	No	No	No
Initial conditions × year	No	Yes	No	No	No	No

Notes. Each row reports the very-high × year coefficient for the outcome in italics, with 1981 as the reference year. Column (1) is the baseline with district and year fixed effects and standard errors clustered by district in parentheses. Column (2) adds 1981 initial conditions interacted with year (own 1981 level where available, literacy, indigenous share, log population and geography) and uses the districts with a 1981 record. Column (3) adds department-by-year fixed effects. Column (4) reports the p-value of the baseline coefficient with 1981-province clusters, and Column (5) the department wild cluster bootstrap p-value with the bootstrap 95% confidence interval in brackets. Column (6) uses the alternative fixed geography (panel B). Stars in Columns (1)–(3) and (6) come from district-clustered p-values. Literacy and education shares use the same age universe in every census, and education shares divide by the whole age-group population. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

The education rows of Table A33 separate raw and conditional differentials. Literacy among residents aged 15 and over rose 5.508 points more in very-high districts by 2007, and the differential falls to 0.427 once 1981 conditions are interacted with year, where the 1993 coefficient becomes −1.987 ($p < 0.05$). The deficit in completed secondary schooling (−6.283 in 2007) falls to −1.232 with initial conditions and stays at −4.500 with department trends. The higher education deficit (−3.156 in 2007) falls to −0.440 with initial conditions.

Table A34 reports the regional analyses of Section 4. Across highland departments, Ayacucho's population fell 0.143 log points relative to the other departments by 1993 ($p < 0.01$), after no differential change between 1972 and 1981 (−0.023), and the gap was not significant in 2007 (−0.034) or 2017 (0.051). Department victims per inhabitant predict the 1993 loss (−0.049, $p < 0.05$), but not once Ayacucho is excluded (−0.041). Within Ayacucho, district victims do not predict population change (−0.005 in 1993), and the districts of Huamanga province grew 0.353 log points more than the rest of the department by 1993 and 0.893 by 2017.

Table A34: Regional Population Effects: Highland Departments and Ayacucho

Panel A: Highland departments, log population (department and year fixed effects, reference 1981)				
	(1) Intensity	(2) Ayacucho	(3) Intensity, excl. Ayacucho	
ln(1 + victims per 1,000) × 1972	0.029		0.081	
	(0.054)		(0.109)	
ln(1 + victims per 1,000) × 1993	−0.049**		−0.041	
	(0.023)		(0.058)	
ln(1 + victims per 1,000) × 2007	−0.027		−0.046	
	(0.034)		(0.083)	
ln(1 + victims per 1,000) × 2017	0.006		−0.012	
	(0.051)		(0.097)	
Ayacucho × 1972		−0.023		
		(0.059)		
Ayacucho × 1993		−0.143***		
		(0.030)		
Ayacucho × 2007		−0.034		
		(0.038)		
Ayacucho × 2017		0.051		
		(0.058)		
Observations (department-years)	90	90	85	
Panel B: Within Ayacucho, log population (unit and year fixed effects, reference 1981)				
	1972	1993	2007	2017
ln(1 + victims)× year, unit clusters	−0.009	−0.005	0.003	0.023
	(0.010)	(0.019)	(0.020)	(0.027)
ln(1 + victims)× year, 1981-province clusters	−0.009	−0.005	0.003	0.023
	(0.009)	(0.033)	(0.025)	(0.035)
Huamanga province (501) × year	−0.226	0.353	0.585	0.893
	(0.122)	(0.145)	(0.222)	(0.310)
Observations / units	420	84		

Notes. Panel A covers all highland departments, with populations summed over their highland districts in 1972, 1981, 1993, 2007, and 2017 (department-years in the Observations row). Intensity is ln(1 + CVR victims per 1,000 inhabitants of 1981) at the department level, interacted with year. Column (2) uses an Ayacucho indicator interacted with year, and Column (3) repeats Column (1) without Ayacucho. Heteroskedasticity-robust standard errors in parentheses. Panel B covers the districts of Ayacucho. Its first two rows interact the district's log(1+victims) with year, and the last row compares the districts of Huamanga province with the rest of Ayacucho (district clusters). Panel B reports standard errors in parentheses and no stars. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Table A35 estimates all four intensity groups jointly against districts without documented victims. The dose response is not monotonic, since in 1993 the very-high, high, and medium coefficients are similar (−0.131, −0.149, and −0.107), and equality of the very-high and high groups is not rejected ($p = 0.870$). In 2007 and 2017 high- and medium-intensity districts are below the very-high districts. The electricity deficit of 2007 is specific to the very-high group (−11.125), while high-intensity districts are 5.023 points above the comparison group ($p < 0.001$ for the difference).

Table A35: Monotonicity of the Dose Response

	Intensity × year				*p*, equality of adjacent groups			Linear contrast	
	Very high	High	Medium	Low	VH = High	High = Med.	Med. = Low	Estimate	*p*, prov.
Log population									
1993	−0.131	−0.149	−0.107	0.001	0.870	0.648	0.011	−0.411**	0.074
	(0.071)	(0.085)	(0.040)	(0.023)				(0.165)	
2007	0.037	−0.173	−0.088	0.001	0.097	0.390	0.047	−0.099	0.780
	(0.088)	(0.094)	(0.040)	(0.031)				(0.201)	
2017	0.028	−0.279	−0.179	−0.018	0.061	0.438	0.003	−0.204	0.664
	(0.112)	(0.125)	(0.049)	(0.039)				(0.257)	
Electricity (%)									
1993	−4.322	−0.581	−5.102	−0.524	0.269	0.149	0.011	−8.700*	0.185
	(2.116)	(2.813)	(1.673)	(1.164)				(5.122)	
2007	−11.125	5.023	−5.469	−2.726	<0.001	0.002	0.237	−14.500**	0.076
	(2.719)	(2.901)	(2.163)	(1.489)				(6.219)	
2017	−1.039	6.424	−1.289	−4.637	0.019	0.005	0.156	8.983	0.301
	(2.687)	(2.088)	(2.183)	(1.515)				(5.834)	
Districts per group	42	36	161	419					

Notes. One regression per outcome with the four intensity groups interacted with census years, district and year fixed effects, reference year 1981, against districts with no documented victims. Standard errors clustered by district in parentheses, and the four intensity columns are shown without stars. The equality columns test adjacent intensity groups in the same year with district clusters. The linear contrast weights the four groups by their rank (stars from district clusters), and its last column uses 1981-province clusters. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Table A36 assigns the very-high label at random to untreated districts of the same 1981 provinces. The 1993 population deficit lies below 99.2 percent of the placebo estimates, the 2007 electricity deficit below 99.6 percent, and the 2007 and 2017 literacy gains above 97.6 and 99.4 percent. Untreated districts in the same provinces also lag in electricity (placebo mean −3.938 in 2007) and gain in literacy (3.401 in 2007), so part of each differential is shared with the surrounding provinces. The 2007 and 2017 population coefficients and the 2017 electricity coefficient lie inside the placebo distribution.

Table A36: Within-Province Placebo Assignment of the Very-High Label

	Actual	Placebo mean	Placebo SD	Placebo percentiles 5th	Placebo percentiles 95th	Share of placebo ≤ actual	Share with |placebo| ≥ |actual|
Log population							
Very high × 1993	−0.118	0.010	0.057	−0.087	0.103	0.008	0.038
Very high × 2007	0.049	0.051	0.074	−0.069	0.174	0.498	0.586
Very high × 2017	0.057	0.044	0.095	−0.098	0.198	0.566	0.558
Electricity (%)							
Very high × 1993	−3.723	−1.546	1.859	−4.403	1.681	0.118	0.118
Very high × 2007	−10.005	−3.938	2.172	−7.575	−0.313	0.004	0.004
Very high × 2017	0.132	−3.272	2.502	−7.467	0.618	0.932	0.972
Literacy, ages 15+ (%)							
Very high × 1993	1.487	1.019	0.689	−0.087	2.221	0.772	0.228
Very high × 2007	5.508	3.401	1.089	1.643	5.201	0.976	0.024
Very high × 2017	6.615	3.806	1.150	1.988	5.664	0.994	0.006

Notes. In each draw the very-high label is assigned at random to untreated districts of the 1981 provinces that contain a very-high district, keeping the number of labels per province. Where a province has fewer untreated than very-high districts the remaining labels are drawn from the pooled untreated districts of the other treated provinces. The actual very-high districts are dropped from the placebo regressions. 500 draws. Specification as in the baseline (district and year fixed effects, reference year 1981). Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district).

Table A37 reports two sensitivity analyses. With the proportional-selection approach of Oster (2019), controls move the 1993 population coefficient away from zero (−0.131 to −0.161), and the bias-adjusted coefficient is −0.177. For electricity in 2007 the selection on unobservables needed to remove the coefficient is 16.63 times the selection on observables. The relative-magnitude bounds of Rambachan and Roth (2023) for the highland 1993 coefficient exclude zero up to $\bar{M} = 2$ with both clusterings. They include zero from $\bar{M} = 3$ with province clusters and from $\bar{M} = 4$ with district clusters. The bounds rest on a single pre-period.

Table A37: Selection on Unobservables and Violations of Parallel Trends

Panel A. Proportional selection (Oster 2019)

	Baseline (R^2)	Controlled (R^2)	$R_{\max}$	δ for $\beta = 0$	β at $\delta = 1$	Districts / very high
Log population, very high × 1993	−0.131	−0.161	0.285	−20.48	−0.177	1,371 / 39
	(0.069)	(0.219)				
Electricity, very high × 2007	−10.208	−10.780	1.000	16.63	−11.900	1,371 / 39
	(0.739)	(0.843)				

Panel B. Relative-magnitude bounds (Rambachan and Roth 2023), highland districts, very high × 1993

	95% CI, district clusters		95% CI, province clusters	
$\bar{M}$	Lower	Upper	Lower	Upper
Conventional	−0.385	−0.087	−0.370	−0.102
0	−0.384	−0.089	−0.369	−0.103
0.5	−0.393	−0.085	−0.389	−0.091
1	−0.403	−0.078	−0.412	−0.080
1.5	−0.416	−0.068	−0.443	−0.065
2	−0.432	−0.058	−0.478	−0.045
3	−0.478	−0.016	−0.551	0.002
4	−0.533	0.032	−0.622	0.047
5	−0.586	0.085	−0.694	0.095
7.5	−0.729	0.227	−0.877	0.258
10	−0.877	0.372	−1.062	0.443
Very high × 1993 (SE, district / province)	−0.236	(0.076)	(0.069)	
Very high × 1972 (SE, district / province)	0.013	(0.029)	(0.038)	
Districts / very high / province clusters	946	24	104	

Notes. Panel A uses the districts with a 1981 row. The baseline model has district and year fixed effects, and the controlled model adds the 1981 initial conditions interacted with year (own 1981 level where available, literacy, indigenous share, log population and geography). Within R^2 in parentheses. $R_{\max}$ is 1.3 times the controlled R^2, capped at 1. Panel B reports robust 95% confidence sets for the 1993 coefficient of the highland dynamic difference-in-differences with the 1972 census as the single pre-period, allowing post-period violations of parallel trends up to $\bar{M}$ times the 1972–1981 change. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district).

Table A38 collects the duration and pre-census checks discussed in Appendix D.3.

Table A38: Violence Before the 1981 Census and Duration of Violence

Panel A. Victims in 1980–1981 and years with documented victims, by intensity group

	Districts	With a victim in 1980–81	Victims 1980–81	Median years with victims	Max. years	Districts with 5+ years
No conflict	882	0	0.0	0.0	0	0
Low (1–10)	419	10	10.0	2.0	7	30
Medium (11–50)	161	7	13.0	6.0	18	118
High (51–100)	36	4	2.1	9.0	17	36
Very high (>100)	42	22	46.9	14.5	21	42

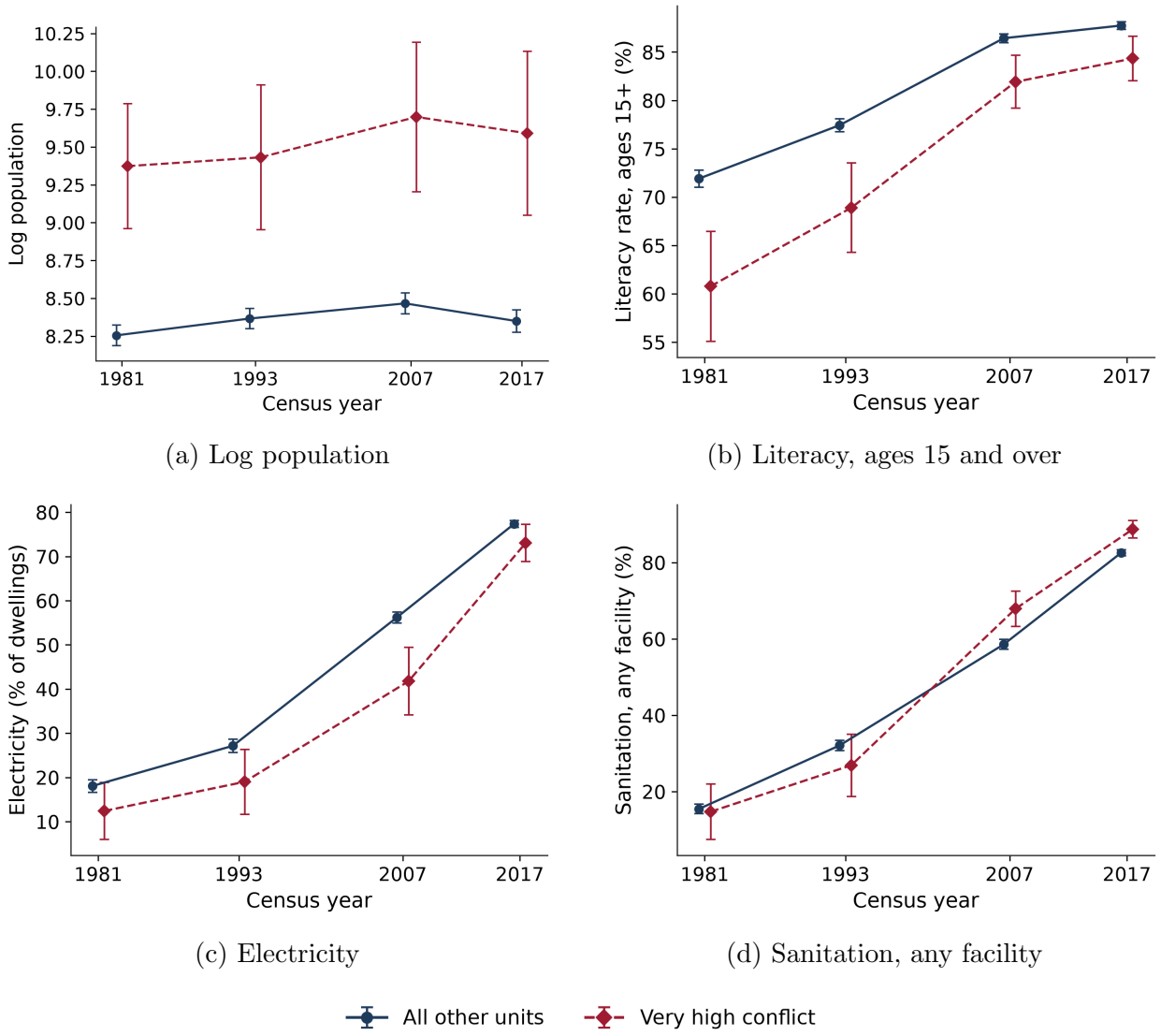
Panel B. Very high × year under sample restrictions

	1993	2007	2017	Observations	Districts	Very high
Log population						
Baseline	−0.118*	0.049	0.057	5,991	1,540	42
	(0.070)	(0.088)	(0.111)			
	[0.203]	[0.740]	[0.772]			
Excluding districts with 1980–81 victims	−0.098	0.153	0.141	5,825	1,497	20
	(0.107)	(0.144)	(0.169)			
	[0.467]	[0.454]	[0.561]			
Excluding other districts with 5+ years of violence	−0.125*	0.044	0.044	5,302	1,356	42
	(0.070)	(0.088)	(0.111)			
	[0.189]	[0.765]	[0.827]			
Electricity (%)						
Baseline	−3.723*	−10.005***	0.132	5,987	1,539	42
	(2.066)	(2.655)	(2.626)			
	[0.136]	[0.003]	[0.972]			
Excluding districts with 1980–81 victims	−5.594***	−7.177*	−1.823	5,821	1,496	20
	(1.766)	(3.757)	(3.742)			
	[0.004]	[0.121]	[0.694]			
Excluding other districts with 5+ years of violence	−4.347**	−10.414***	0.150	5,298	1,355	42
	(2.076)	(2.668)	(2.638)			
	[0.087]	[0.003]	[0.968]			
Literacy, ages 15+ (%)						
Baseline	1.487	5.508***	6.615***	5,991	1,540	42
	(1.157)	(1.702)	(1.876)			
	[0.261]	[0.023]	[0.017]			
Excluding districts with 1980–81 victims	1.439	5.678**	6.450**	5,825	1,497	20
	(1.528)	(2.639)	(2.780)			
	[0.332]	[0.019]	[0.014]			

Notes. Panel A uses the CVR victim records, which give the year of the events but not the month, so 1980–1981 victims are an upper bound for victims before the census of July 1981. Fractional victim counts come from records shared between districts. Panel B reports the very-high × year coefficient with district and year fixed effects, reference year 1981. Standard errors clustered by district in parentheses (stars), p-values with 1981-province clusters in brackets. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Figure A12 shows the raw means behind the estimates. Very-high districts are larger in every census and started with lower literacy (60.8 against 71.9 percent in 1981) and electricity coverage (12.4 against 18.1 percent). Electricity fell further behind by 2007 (41.8 against

56.2 percent) and nearly caught up by 2017 (73.0 against 77.3 percent). Sanitation coverage was similar in 1981 and higher in very-high districts from 2007.



(a) Log population

(b) Literacy, ages 15 and over

(c) Electricity

(d) Sanitation, any facility

Figure A12: Raw Means by Census Year and Conflict Status

Notes. Means and 95% confidence intervals for very-high-conflict districts and all other districts observed in each census year. Sanitation counts dwellings with a public network, septic tank, or pit latrine.

E Perpetrator Heterogeneity

This appendix reports recovery by the main perpetrator of violence.

E.1 Classification of Perpetrator Types

The classification uses the CVR victim records aggregated to districts. A district is Sendero dominant when more than half of its documented victims were attributed to Sendero Luminoso, and State dominant when more than half were attributed to state agents. The remaining conflict districts are Mixed, including the few where the MRTA or the self-defense committees account for most victims. Of the 658 districts with documented victims, 332 are Sendero dominant (24 of them very high), 124 State dominant (7 very high), and 202 Mixed (11 very high).

Table A39 compares these groups in 1981. Sendero-dominant districts were smaller (mean population 7,915 against 29,109 and 33,651), less literate (65.8 percent among residents aged 15 and over against 71.4 and 73.9), and higher (3,048 meters against 2,568 and 2,513). They also record more victims on average (46.0 against 29.7 and 20.9).

Table A39: Baseline Characteristics of Conflict Districts by Dominant Perpetrator

	Sendero Dominant	State Dominant	Mixed Perpetrator	No Conflict (Comparison)
Total Population (1981)	7,915	29,109	33,651	6,464
Literacy Rate, Ages 15+ (%)	65.8	71.4	73.9	73.2
Mean Age (years)	24.2	24.1	24.4	24.6
Altitude (m)	3,048	2,568	2,513	2,273
Total Victims (mean)	46.0	29.7	20.9	0.0

Notes. Mean characteristics of conflict districts (1+ CVR-documented victims) by dominant perpetrator. The final column reports no-conflict districts for comparison. Violence characteristics from CVR data, baseline characteristics from the 1981 census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census.

E.2 Joint Model with a Common Comparison Group

Table 5 interacts the very-high indicator with the three perpetrator groups in one regression, with all districts below the very-high threshold as the comparison group. Sendero-dominant districts show population coefficients of −0.104, 0.141, and 0.130 and State-dominant districts −0.232, −0.224, and −0.256, all insignificant. Literacy rose 8.141 points more by 2017 in Sendero-dominant districts and 12.372 points more in State-dominant districts. Electricity lagged in both groups in 2007 (−8.429 and −14.671). Equality of the Sendero-dominant and State-dominant coefficients is not rejected in any outcome or year with district clusters, and

with province clusters only the 2007 electricity comparison falls below 0.10 ($p = 0.077$). The joint three-year tests reject equality with province clusters ($p = 0.021$, 0.010, and $p < 0.001$) and not with district clusters ($p = 0.143$, 0.227, and 0.227). With seven State-dominant districts concentrated in a few provinces, the province-cluster joint tests rest on very few effective clusters.

Table A40: Long-Run Effects by Dominant Perpetrator (2017)

	Population (log)	Literacy 15+ (%)	Mean Age (years)	Migration Rate (%)
Very High × 2017 Coefficients by Dominant Perpetrator				
Sendero Dominant	0.287**	4.687**	−1.231**	−2.211
	(0.135)	(2.307)	(0.506)	(3.548)
State Dominant	−0.301	12.745***	1.402	0.075
	(0.275)	(4.374)	(1.302)	(6.384)
Mixed Perpetrator	0.020	−0.565	0.364	2.053
	(0.252)	(3.510)	(0.759)	(4.475)
N (Sendero)	1,281	1,281	1,281	1,281
N (State)	472	472	472	472
N (Mixed)	765	765	765	765
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes. Coefficients from separate regressions on subsamples of conflict districts with the indicated dominant perpetrator, so each coefficient compares very-high-conflict districts with lower-intensity conflict districts of the same dominant-perpetrator type. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

E.3 Population and Literacy by Perpetrator

Tables A41 and A42 estimate each perpetrator group separately, so each coefficient compares very-high districts with lower-intensity conflict districts of the same group. Sendero-dominant very-high districts grew 0.203 log points more by 2007 ($p < 0.10$) and 0.287 by 2017 ($p < 0.05$). This within-group gain does not appear in the joint model, where the comparison group includes districts without violence. State-dominant and Mixed coefficients are insignificant. Districts with any MRTA victim show gains of 0.358 in 2007 ($p < 0.05$) and

0.422 in 2017 ($p < 0.01$).

Table A41: Population Recovery by Dominant Perpetrator

	(1) Sendero Dom.	(2) State Dom.	(3) Mixed	(4) MRTA Present	(5) Full Sample
Very High × 1993	-0.027	-0.275	-0.102	0.139	-0.118*
	(0.082)	(0.197)	(0.172)	(0.104)	(0.070)
Very High × 2007	0.203*	-0.263	0.010	0.358**	0.049
	(0.108)	(0.223)	(0.191)	(0.148)	(0.088)
Very High × 2017	0.287**	-0.301	0.020	0.422***	0.057
	(0.135)	(0.275)	(0.252)	(0.154)	(0.111)
Observations	1281	472	765	338	5991
Adjusted R^2	0.938	0.968	0.973	0.970	0.950
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Comparison group	Other Sendero-dom.	Other State-dom.	Other Mixed	Other MRTA	All below VH

Notes. Dependent variable is log population. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. Columns 1–3 restrict the sample to conflict districts where the indicated perpetrator type accounts for the majority of victims, so coefficients compare very-high-conflict districts with lower-intensity conflict districts in the same subsample. These categories are mutually exclusive. Column 4 covers districts with any MRTA victims (not mutually exclusive). The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Literacy rose more in very-high districts of every group except the Mixed one. The 2017 coefficients are 4.687 for Sendero-dominant districts ($p < 0.05$), 12.745 for State-dominant districts ($p < 0.01$), and -0.565 for Mixed districts.

Table A42: Literacy Recovery by Dominant Perpetrator

	(1) Sendero	(2) State	(3) Mixed	(4) Full Sample
Very High × 1993	0.941	5.977**	-1.884	1.487
	(1.594)	(2.306)	(1.448)	(1.157)
Very High × 2007	4.428**	9.402**	-1.233	5.508***
	(2.177)	(3.757)	(2.984)	(1.702)
Very High × 2017	4.687**	12.745***	-0.565	6.615***
	(2.307)	(4.374)	(3.510)	(1.876)
Observations	1281	472	765	5991
Adjusted R^2	0.855	0.878	0.860	0.856
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Comparison group	Other Sendero-dom.	Other State-dom.	Other Mixed	All below VH

Notes. Dependent variable is the literacy rate of persons aged 15 and over (percentage). Columns 1–3 restrict the sample to conflict districts with the indicated dominant perpetrator, so coefficients compare very-high-conflict districts with lower-intensity conflict districts in the same subsample. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

E.4 Demographic Structure by Perpetrator

Table A43 reports mean age and the dependency ratio. Sendero-dominant very-high districts became younger than other Sendero-dominant districts, by 0.727 years in 2007 and 1.231 years in 2017 (both $p < 0.05$), with higher dependency ratios in every census. State-dominant very-high districts were older in 1993 (1.201 years, $p < 0.05$) and had lower dependency ratios in 1993 and 2017. In the full sample the mean age differential is small (-0.506 years in 2017, not significant).

Table A43: Demographic Changes by Dominant Perpetrator

	Mean Age (years)			Dependency Ratio		
	(1) Sendero	(2) State	(3) Full	(4) Sendero	(5) State	(6) Full
Very High × 1993	-0.262	1.201**	0.195	4.723**	-10.561***	0.764
	(0.267)	(0.504)	(0.227)	(2.165)	(2.908)	(1.878)
Very High × 2007	-0.727**	0.664	-0.559*	5.856***	-1.005	3.862**
	(0.291)	(1.026)	(0.290)	(1.988)	(6.208)	(1.903)
Very High × 2017	-1.231**	1.402	-0.506	5.152**	-8.964**	0.143
	(0.506)	(1.302)	(0.440)	(2.355)	(4.253)	(1.965)
Observations	1281	472	5991	1281	472	5991
Adjusted R^2	0.878	0.890	0.870	0.759	0.832	0.763
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Comparison group	Other Sendero-dom.	Other State-dom.	All below VH	Other Sendero-dom.	Other State-dom.	All below VH

Notes. Sendero and State columns restrict the sample to conflict districts with the indicated dominant perpetrator, so coefficients compare very-high-conflict districts with lower-intensity conflict districts in the same subsample. Full columns use the full sample with all below-threshold districts as the comparison. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

E.5 Continuous Perpetrator Effects

Table A44 uses the share of victims attributed to each perpetrator among the conflict districts. A higher Sendero share goes with lower population growth, −0.0034 log points per percentage point by 2017 ($p < 0.01$), and a higher state share with higher growth (0.0016, $p < 0.05$). Literacy moves the other way, because it rises with the Sendero share (0.0530 per point by 2017, $p < 0.01$) and falls with the state share (−0.0319, $p < 0.05$). The continuous estimates use the variation among all conflict districts, most of them below the very-high threshold, and they answer a different question from the joint model of Table 5.

Table A44: Effects of Perpetrator Share (Continuous)

	Log Population		Literacy Rate, 15+	
	(1)	(2)	(3)	(4)
	Sendero	State	Sendero	State
% Sendero × 1993	-0.0016***		0.0045	
	(0.0004)		(0.0074)	
% Sendero × 2007	-0.0014**		0.0456***	
	(0.0006)		(0.0105)	
% Sendero × 2017	-0.0034***		0.0530***	
	(0.0007)		(0.0112)	
% State × 1993		0.0008*		-0.0125
		(0.0005)		(0.0086)
% State × 2007		0.0008		-0.0250*
		(0.0006)		(0.0129)
% State × 2017		0.0016**		-0.0319**
		(0.0008)		(0.0137)
Observations	2518	2518	2518	2518
Adjusted R^2	0.962	0.961	0.864	0.860
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes. Sample restricted to conflict districts (1+ victims). Sendero-share and state-share specifications are estimated separately (one share at a time), not jointly. Coefficients show the effect of a 1 percentage point increase in the perpetrator share. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

F Age Structure

This appendix reports age structure in more detail.

Table A45 reports the differentials for eight age groups. Very-high districts have more children aged 5 to 14 in every census (0.83, 1.77, and 1.42 percentage points, significant at least at the 10 percent level in all three years), and fewer residents aged 15 to 24 in 2007 (-0.85, $p < 0.05$). The other differentials are below one percentage point, and none of the

groups aged 45 and over shows a statistically significant deficit in any census.

Table A45: Detailed Age Distribution Changes in Very-High-Conflict Districts

	(1) 0-4	(2) 5-14	(3) 15-24	(4) 25-34	(5) 35-44	(6) 45-54	(7) 55-64	(8) 65+
Very High × 1993	-0.40	0.83*	-0.70	-0.18	-0.24	0.37*	0.31*	0.01
	(0.27)	(0.46)	(0.46)	(0.28)	(0.24)	(0.21)	(0.18)	(0.24)
Very High × 2007	0.24	1.77***	-0.85**	-0.38	-0.41*	-0.11	0.25	-0.51
	(0.23)	(0.56)	(0.41)	(0.30)	(0.24)	(0.26)	(0.15)	(0.36)
Very High × 2017	-0.36	1.42**	-0.15	-0.40	0.13	0.25	-0.17	-0.72
	(0.25)	(0.64)	(0.51)	(0.28)	(0.22)	(0.22)	(0.22)	(0.60)
Observations	5991	5991	5991	5991	5991	5991	5991	5991
Adjusted R^2	0.738	0.807	0.551	0.586	0.601	0.640	0.709	0.786
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. All variables measured as % of total population. VH = Very High conflict. The reference year is 1981. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A46 follows birth cohorts across censuses. The cohort aged 15 to 29 in 1981 makes up a smaller share of the population of very-high districts than elsewhere, by 2.066 percentage points in 1993, 3.437 in 2007, and 3.729 in 2017 ($p < 0.01$ with district clusters and p between 0.055 and 0.124 with province clusters). The cohort aged 0 to 14 in 1981 falls 2.050 points more by 2007 and 2.336 points more by 2017. The larger child share of very-high districts does not explain the deficit of the cohort aged 15 to 29 in 1981, because measured as a share of the population aged 15 and over it is larger, 3.761 points in 1993, 5.594 in 2007, and 6.260 in 2017, all $p < 0.01$ with district clusters (Table A7, Column (e)). The age group 15 to 29, which holds different people in each census, shows smaller and insignificant differentials as a share of the population aged 15 and over (Table A47). The evidence is consistent with a loss of young people during the conflict, through death or out-migration, and it cannot separate the two. Table A48 reports the mean cohort shares behind these estimates.

Table A46: Age Structure and Birth Cohorts in Very-High-Conflict Districts

Very high × year, percentage points	1993	p, prov.	2007	p, prov.	2017	p, prov.
Cohort aged 0–14 in 1981	−0.504	0.741	−2.050***	0.054	−2.336***	<0.001
	(0.804)		(0.688)		(0.547)	
Cohort aged 15–29 in 1981	−2.066***	0.055	−3.437***	0.064	−3.729***	0.124
	(0.569)		(0.915)		(1.182)	
Cohort aged 30–44 in 1981	−0.533	0.519	−1.100**	0.295	−1.029	0.442
	(0.426)		(0.544)		(0.697)	
Ages 0–14, % of population	0.446	0.594	2.007***	0.039	1.058	0.460
	(0.559)		(0.632)		(0.779)	
Ages 15–29, % of population	−0.750	0.228	−1.050**	0.179	−0.559	0.555
	(0.508)		(0.516)		(0.606)	
Ages 30–44, % of population	−0.388	0.327	−0.593**	0.062	0.141	0.639
	(0.295)		(0.253)		(0.311)	
Ages 45–64, % of population	0.678**	0.044	0.144	0.757	0.081	0.837
	(0.275)		(0.354)		(0.398)	
Ages 65+, % of population	0.014	0.968	−0.508	0.413	−0.720	0.474
	(0.239)		(0.361)		(0.604)	
Observations / districts / very high	5,991	1,540	42			

Notes. Each row reports the very-high × year coefficient from a regression of the share on district and year fixed effects, reference year 1981, with standard errors clustered by district in parentheses (stars) and the p-value with 1981-province clusters. Table A47 reports the age groups as shares of the population aged 15 and over. Five-year age groups come from the census literacy tables, which cover all ages 15 and over in every census but do not cover ages 0–4 in 1993 and cover them only in part in 1981 and 2007, so the 0–14 share is the residual of total population. Table A45 uses the census age tables, which cover all ages, so its child shares differ slightly. Cohort rows follow the persons aged 0–14, 15–29 and 30–44 in 1981 across censuses with a uniform split of five-year groups, as a share of the population in each census. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Table A47: Age Structure in Very-High-Conflict Districts, Shares of the Population Aged 15 and Over

Very high × year, percentage points	1993	p, prov.	2007	p, prov.	2017	p, prov.
Ages 15–29, % of ages 15+	−1.032	0.150	−1.091	0.362	−1.311	0.323
	(0.643)		(0.792)		(0.897)	
Ages 30–44, % of ages 15+	−0.340	0.709	0.100	0.858	0.681	0.311
	(0.610)		(0.462)		(0.579)	
Ages 45–64, % of ages 15+	1.293***	<0.001	1.222*	0.293	1.139	0.353
	(0.402)		(0.662)		(0.756)	
Ages 65+, % of ages 15+	0.079	0.852	−0.232	0.746	−0.510	0.597
	(0.341)		(0.457)		(0.649)	
Observations / districts / very high	5,991	1,540	42			

Notes. Each row reports the very-high × year coefficient from a regression of the share on district and year fixed effects, reference year 1981, with standard errors clustered by district in parentheses (stars) and the p-value with 1981-province clusters. Five-year age groups come from the census literacy tables, which cover all ages 15 and over in every census, and each share is a percent of the population aged 15 and over. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district). *** p<0.01, ** p<0.05, * p<0.10.

Table A48: Mean Cohort Shares by Census Year

	Districts	Aged 0–14 in 1981	Aged 15–29 in 1981	Aged 30–44 in 1981
Very high, 1981	39	45.22	24.79	14.29
Very high, 1993	42	28.68	16.87	10.47
Very high, 2007	42	19.60	12.65	7.42
Very high, 2017	42	17.89	11.97	6.95
All other, 1981	1,332	43.47	23.02	14.44
All other, 1993	1,498	27.20	16.89	11.21
All other, 2007	1,498	19.67	14.03	8.73
All other, 2017	1,498	18.25	13.65	8.19

Notes. Mean share of each 1981 cohort in the population of very-high-conflict districts and of all other districts, percent. Five-year age groups come from the census literacy tables, which cover all ages 15 and over in every census but do not cover ages 0–4 in 1993 and cover them only in part in 1981 and 2007, so the 0–14 share is the residual of total population. Table A45 uses the census age tables, which cover all ages, so its child shares differ slightly. Cohort rows follow the persons aged 0–14, 15–29 and 30–44 in 1981 across censuses with a uniform split of five-year groups, as a share of the population in each census. Districts are the units of the fixed 1981 geography (groups of current districts that map to one 1981 district).

Table A49 compares 1981 and 2017 means. Very-high districts were younger in 1981 (22.79 against 24.51 years) and remained younger in 2017 (30.94 against 33.05). They are larger on average (28,351 against 11,079 residents in 1981), and their literacy among residents aged 15 and over rose from 60.8 to 84.3 percent while the rest rose from 71.9 to 87.7 percent. The 1981 means cover the 39 very-high districts with a 1981 record.

Table A49: Demographic Summary Statistics: Baseline and Long-Run

	1981 (Pre-Conflict)		2017 (Post-Conflict)	
	Very High	All Other	Very High	All Other
Mean Age (years)	22.79	24.51	30.94	33.05
Total Population	28,351	11,079	65,098	17,789
Literacy Rate, Ages 15+ (%)	60.8	71.9	84.3	87.7

Notes. Values are means across districts in each category. Very High Conflict = >100 documented victims. All Other = every district below that threshold (including low-, medium-, and high-conflict districts). Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census.

G Additional Night-Time Lights Results

G.1 Main Window Estimates

Table A50 reports the window contrasts behind Section 8, and Table A51 the estimates within Ayacucho. Estimated by Poisson pseudo-maximum-likelihood, the coefficients are 0.08 for 1993–2007 and 0.01 for 2007–2017, far from significant ($p = 0.69$ and 0.97). The probability of detected light shows no differential either (-0.02 in both periods), and light per capita, which divides by census population, is too imprecise to settle the question. The checks with the other two series agree for the periods in which each series uses a single sensor, and the apparent gains in total light that appear in those series after 2007 coincide with their change of sensor (Section G.2).

Table A50: Night-Time Lights: Total Light, Detected Light, and Lit Area

Dependent variable (transformation, estimator)	Coefficient (SE)	p, district	p, 1981 province	p, bootstrap	Districts (treated)
Product	Chiovelli et al. (2026), DMSP-like series, 1992–2025				
Fixed effects	District and year, all years of the window				
Clustering	Standard errors by district, p-values as in the column heads				
Panel A. Very high × last year of the window					
Period 1993–2007, coefficient of 2007 relative to 1993					
Total light (level, PPML)	0.076 (0.190)	0.689	0.432	0.518	941 (26)
Any light (0/1, LPM)	−0.023 (0.065)	0.725	0.798	0.783	1,540 (42)
Lit-area share (0–1, linear FE)	−0.016 (0.003)	<0.001	<0.001	0.035	1,540 (42)
Light per capita (level, PPML)	−0.020 (0.189)	0.918	0.918	0.936	776 (23)
Period 2007–2017, coefficient of 2017 relative to 2007					
Total light (level, PPML)	0.006 (0.147)	0.969	0.969	0.971	1,115 (29)
Any light (0/1, LPM)	−0.019 (0.055)	0.725	0.755	0.845	1,540 (42)
Lit-area share (0–1, linear FE)	−0.045 (0.006)	<0.001	<0.001	0.043	1,540 (42)
Light per capita (level, PPML)	−0.183 (0.210)	0.384	0.420	0.470	1,028 (28)

Notes. Night-time lights from the harmonized DMSP-like series of Chiovelli, Michalopoulos, Papaioannou and Regan (2026), 1992–2025, with the blooming and top-coding corrections, in digital-number units. The lit-area share is the share of the district's area with a pixel value above zero. The treatment is the very-high indicator. The bootstrap is the wild cluster bootstrap by department (25 departments), Webb weights, 9,999 draws. The last column gives the districts in the estimation sample, with the very-high districts in parentheses. Table A51 reports the within-Ayacucho design and the mean lit-area shares. Light is summed over the fixed district polygons with exact pixel coverage and pixel-area weights, so the polygons are the same in every year. Each coefficient comes from a regression on all years of the window with district and year fixed effects and a full set of treatment × year terms, relative to the first year of the window. Light per capita uses the census years at the two ends of the window. LPM is a linear probability model for the any-light indicator, linear FE a linear regression with district and year fixed effects for the lit-area share, and PPML Poisson pseudo-maximum-likelihood. Standard errors clustered by district in parentheses. For PPML the bootstrap is the score bootstrap of Kline and Santos (2012), and the PPML samples keep the districts with some light in the window. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A51: Night-Time Lights within Ayacucho and Mean Lit-Area Shares

Dependent variable (transformation, estimator)	Coefficient (SE)	p, district	p, 1981 province	p, bootstrap	Districts (with victims)
Panel A. Within Ayacucho, log(1 + victims) × last year of the window					
Period 1993–2007, coefficient of 2007 relative to 1993					
Total light (level, PPML)	−0.111 (0.096)	0.246	0.217	0.217	28 (27)
Light per capita (level, PPML)	−0.203 (0.088)	0.021	<0.001	0.086	15 (15)
Period 2007–2017, coefficient of 2017 relative to 2007					
Total light (level, PPML)	−0.212 (0.108)	0.049	0.005	0.030	46 (42)
Light per capita (level, PPML)	−0.620 (0.168)	<0.001	<0.001	0.019	41 (39)

Panel B. Mean lit-area share (pixel value > 0), share of district area	1993	2007	2017
Very high (42 districts)	0.032	0.041	0.064
Other districts (1,498)	0.080	0.105	0.173

Notes. Same series and definitions as Table A50. Panel A uses the districts of Ayacucho, the treatment is log(1 + victims) of the district, and the bootstrap is the wild cluster bootstrap by 1981 province (7 provinces), Webb weights, 9,999 draws. The last column gives the districts in the estimation sample, with the districts with victims in parentheses. Panel B reports mean shares of district area with a pixel value above zero. Light is summed over the fixed district polygons with exact pixel coverage and pixel-area weights, so the polygons are the same in every year. Each coefficient comes from a regression on all years of the window with district and year fixed effects and a full set of treatment × year terms, relative to the first year of the window. Light per capita uses the census years at the two ends of the window. LPM is a linear probability model for the any-light indicator, linear FE a linear regression with district and year fixed effects for the lit-area share, and PPML Poisson pseudo-maximum-likelihood. Standard errors clustered by district in parentheses. For PPML the bootstrap is the score bootstrap of Kline and Santos (2012), and the PPML samples keep the districts with some light in the window. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

G.2 Series, Sensors, and Census-Year Specifications

Appendix Tables A52 and A53 report the window contrasts for total light, detected light, and lit area with the three series and the four correction variants of Chiovelli et al. (2026). The series of Chen et al. (2024) changes from DMSP-based to VIIRS values in 2012 and that of Li et al. (2020) at its 2013/2014 splice, so their contrasts are reported within each sensor period.

Table A52: Night-Time Lights across Series: Chen et al. (2024) and Li et al. (2020)

Window	Total light (PPML)	Any light (LPM)	Lit share, threshold 1	Lit share, threshold 2	Lit share, threshold 3
Chen et al. (2024), SVNL, thresholds > 0, > 0.5, > 1					
2007 vs 1993	0.147	0.015	−0.003***	−0.006	−0.006
	(0.152)	(0.055)	(0.001)	(0.004)	(0.004)
	[0.285; 0.456]	[0.826; 0.797]	[<0.001; 0.028]	[0.134; 0.353]	[0.135; 0.360]
2017 vs 2007	0.362*	0.020	−0.012***	−0.018***	−0.021***
	(0.204)	(0.077)	(0.003)	(0.004)	(0.005)
	[0.089; 0.371]	[0.863; 0.915]	[<0.001; 0.132]	[<0.001; 0.115]	[<0.001; 0.141]
2011 vs 1992 (DMSP-based)	0.035	−0.044	−0.005***	−0.009*	−0.006
	(0.211)	(0.064)	(0.001)	(0.005)	(0.007)
	[0.758; 0.786]	[0.609; 0.543]	[<0.001; 0.020]	[0.041; 0.317]	[0.274; 0.527]
2023 vs 2012 (VIIRS)	−0.038	−0.031	−0.026***	−0.026***	−0.026***
	(0.035)	(0.055)	(0.004)	(0.004)	(0.004)
	[0.433; 0.634]	[0.596; 0.473]	[<0.001; 0.069]	[<0.001; 0.069]	[<0.001; 0.069]
Li et al. (2020), harmonized digital numbers, thresholds > 0, > 5, > 10					
2007 vs 1993	0.162	−0.095*	−0.025***	−0.018***	−0.010**
	(0.185)	(0.051)	(0.006)	(0.004)	(0.005)
	[0.302; 0.383]	[0.057; 0.118]	[0.002; 0.129]	[<0.001; 0.027]	[0.039; 0.246]

Notes. Very high × last year of each window, relative to its first year, from a regression on all years of the window with district and year fixed effects and a full set of very-high × year terms, all districts. Standard errors clustered by district in parentheses, stars from district-clustered p-values, and in brackets the p-values with 1981-province clusters and from the wild cluster bootstrap by department (Webb weights, 9,999 draws, score bootstrap for PPML). PPML is Poisson pseudo-maximum-likelihood and LPM a linear probability model for the any-light indicator, and the lit shares are estimated by linear regression with district and year fixed effects. Thresholds apply to the pixel values of each series, and the lit share is the share of district area in pixels above the threshold. Light is summed over fixed district polygons with exact pixel coverage and pixel-area weights. Series Chen et al. (2024) is re-extracted on the fixed district polygons, in nW/cm^2/sr, and converts DMSP images to VIIRS-like values for 1992–2011 and uses VIIRS from 2012, so its contrasts are also shown within each sensor period. Series Li et al. (2020) is in DMSP digital numbers, calibrated DMSP for 1992–2013 and simulated from VIIRS for 2014–2020. Its 2007–2017 contrast crosses the 2013/2014 splice, where the light of the overlap year more than doubles between the two files, and is not shown. Table A53 reports the four variants of Chiovelli et al. (2026). *** p<0.01, ** p<0.05, * p<0.10.

Table A53: Night-Time Lights across Variants of Chiovelli et al. (2026)

Window	Total light (PPML)	Any light (LPM)	Lit share, threshold 1	Lit share, threshold 2	Lit share, threshold 3
Chiovelli et al. (2026), blooming and top-coding corrected, thresholds > 0, > 0.5, > 1					
2007 vs 1993	0.076	−0.023	−0.016***	−0.016***	−0.017***
	(0.190)	(0.065)	(0.003)	(0.003)	(0.003)
	[0.432; 0.518]	[0.798; 0.783]	[<0.001; 0.035]	[<0.001; 0.035]	[<0.001; 0.033]
2017 vs 2007	0.006	−0.019	−0.045***	−0.045***	−0.029***
	(0.147)	(0.055)	(0.006)	(0.006)	(0.003)
	[0.969; 0.971]	[0.755; 0.845]	[<0.001; 0.043]	[<0.001; 0.043]	[<0.001; 0.048]
Chiovelli et al. (2026), blooming corrected, thresholds > 0, > 0.5, > 1					
2007 vs 1993	0.022	−0.023	−0.016***	−0.016***	−0.017***
	(0.184)	(0.065)	(0.003)	(0.003)	(0.003)
	[0.844; 0.861]	[0.798; 0.783]	[<0.001; 0.035]	[<0.001; 0.035]	[<0.001; 0.033]
2017 vs 2007	−0.004	−0.030	−0.047***	−0.047***	−0.028***
	(0.166)	(0.055)	(0.006)	(0.006)	(0.003)
	[0.981; 0.986]	[0.628; 0.751]	[<0.001; 0.039]	[<0.001; 0.039]	[<0.001; 0.049]
Chiovelli et al. (2026), top-coding corrected, thresholds > 0, > 0.5, > 1					
2007 vs 1993	0.193	−0.049	−0.026***	−0.026***	−0.026***
	(0.186)	(0.059)	(0.006)	(0.006)	(0.006)
	[0.117; 0.273]	[0.305; 0.368]	[0.002; 0.135]	[0.002; 0.135]	[0.002; 0.135]
2017 vs 2007	0.009	0.039	−0.079***	−0.079***	−0.048***
	(0.158)	(0.071)	(0.018)	(0.018)	(0.009)
	[0.950; 0.957]	[0.710; 0.900]	[0.002; 0.049]	[0.002; 0.049]	[<0.001; 0.046]
Chiovelli et al. (2026), clean DMSP, uncorrected, thresholds > 0, > 0.5, > 1					
2007 vs 1993	0.113	−0.049	−0.026***	−0.026***	−0.026***
	(0.183)	(0.059)	(0.006)	(0.006)	(0.006)
	[0.448; 0.524]	[0.305; 0.368]	[0.002; 0.135]	[0.002; 0.135]	[0.002; 0.135]
2017 vs 2007	−0.032	0.014	−0.081***	−0.081***	−0.044***
	(0.147)	(0.069)	(0.017)	(0.017)	(0.009)
	[0.827; 0.869]	[0.898; 0.943]	[<0.001; 0.047]	[<0.001; 0.047]	[<0.001; 0.046]

Notes. Same specification, inference and definitions as Table A52. Chiovelli, Michalopoulos, Papaioannou and Regan (2026) is in integer DMSP-like digital numbers, so thresholds of 0 and 0.5 select the same pixels. The four blocks differ only in the corrections applied to the DMSP images. *** p<0.01, ** p<0.05, * p<0.10.

G.3 Census-Year Specifications

This subsection reports the census-year specifications with the series of Chen et al. (2024).

G.3.1 Data and Measurement

The data are the annual night-time lights series of Chen et al. (2024), aggregated to the 1981 units. The series converts DMSP-OLS imagery into VIIRS-like values for 1992–2011 and uses VIIRS observations from 2012 onwards. The 1981 census predates the satellite record, so the census-year specifications use 1993 as the reference year, and 1993 itself falls within the conflict. The outcomes are total NTL (sum of pixel values within district boundaries), NTL per capita (total NTL divided by census population), and an indicator for any detected luminosity. Across the three census years, 44.8% of district-year observations record no detectable light.

Table A54: Sample Selection in NTL Per Capita Analysis

	Dropped (Zero NTL)	Retained (Positive NTL)	p-value (difference)
Observations	2,069	2,551	
Share of NTL sample	44.8%	55.2%	
Mean altitude (m)	2,631	2,176	<0.001
Mean population	4,290	27,418	<0.001
Mean distance to Lima (km)	483	468	0.038
Urban share (%)	33.8	48.1	<0.001
Terrain ruggedness	286	231	<0.001
Literacy rate, ages 15+ (%)	78.4	88.0	<0.001
Very high conflict (%)	2.2	3.1	0.054
Any conflict (%)	42.1	43.2	0.436

Notes. Characteristics of district-year observations with zero and with positive NTL, with Welch t-test p-values for the difference. Observations with zero NTL are excluded when using log(NTL per capita) and retained with the IHS and log(x+c) transformations. The zero-preserving specifications use all 4,620 observations, and the positive-only log specification uses 1,964 because the fixed-effects estimator drops districts with positive light in a single census year. Observations with zero NTL come from districts that are higher, smaller, more rural and less literate. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A54 shows that districts without detected light are higher, much smaller, more rural, and less literate than districts with light. Very-high-conflict districts make up 2.2% of the zero-light observations and 3.1% of the positive ones ($p = 0.05$). Dropping zero observations, as a logarithmic specification does, therefore changes the sample in ways related to both development and conflict.

G.3.2 Extensive Margin

Table A55: Extensive Margin: Probability of Any Detected Night-Time Lights

	(1) Basic	(2) +Geo×
Very High × 2007	0.019	0.019
	(0.055)	(0.055)
Very High × 2017	0.034	0.047
	(0.078)	(0.071)
Observations	4620	4620
Adjusted R^2	0.577	0.601
Geo × Year	No	Yes
District FE	Yes	Yes
Year FE	Yes	Yes

Notes. Linear probability model. Standard errors clustered by district in parentheses. Dependent variable equals 1 if the district has any detected night-time luminosity, 0 otherwise. The reference year is 1993. Column (2) includes altitude and terrain ruggedness interacted with year fixed effects to control for differential infrastructure expansion by geography. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A55 presents linear probability models for the indicator of positive NTL. Relative to 1993, very-high-conflict districts were 1.9 percentage points more likely to show detectable light by 2007 and 3.4 points more likely by 2017, and neither estimate is statistically significant. Controlling for geography interacted with year (Column (2)) leaves the estimates unchanged, and so does controlling for 1993 population density interacted with year.

G.3.3 Total and Per-Capita Luminosity

Table A56 presents NTL per capita under alternative transformations. Column (1) drops zero-luminosity observations (1,964 remain), while Columns (2)–(5) retain all 4,620. The 2017 coefficient is -1.61 in logs of positive values, -0.008 in inverse hyperbolic sine units, -0.20 with $\log(x + 0.01)$, and -11.4 percentile points in the within-year ranking.

Table A56: NTL Per Capita: Robustness to Transformation Choice

	NTL Per Capita				
	(1) Log(x)	(2) IHS(x)	(3) Log(x+.01)	(4) Log(x+1)	(5) Percentile
Very High × 2007	-0.590*	-0.001**	-0.010	-0.001**	0.313
	(0.316)	(0.001)	(0.022)	(0.001)	(2.582)
Very High × 2017	-1.606***	-0.008***	-0.198***	-0.007***	-11.368***
	(0.415)	(0.003)	(0.061)	(0.002)	(3.462)
Observations	1964	4620	4620	4620	4620
Adjusted R^2	0.621	0.491	0.627	0.497	0.708
Zero-NTL Districts	Dropped	Preserved	Preserved	Preserved	Preserved
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. All specifications include district and year fixed effects. The reference year is 1993. Column (1) retains only positive-NTL observations. Columns (2)–(5) preserve the full sample using transformations that accommodate zeros. IHS is the inverse hyperbolic sine, $\text{asinh}(x) = \ln(x + \sqrt{x^2+1}) \approx \ln(2x)$ for large x. Percentile = within-year percentile rank (0–100). *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

For outcomes with zeros, the treatment effect estimated with any log-like transformation depends on the units of the outcome and is not a percentage effect when treatment shifts the extensive margin (Chen and Roth, 2024), which explains the spread. I therefore also estimate the per-capita effect with Poisson pseudo-maximum-likelihood (PPML), which is scale-invariant and accommodates zeros (Santos Silva and Tenreyro, 2006; Chen and Roth, 2024). The PPML estimates are 0.001 (standard error 0.226) in 2007 and −0.080 (0.428) in 2017 (Table A30, Panel C). They are not distinguishable from zero, and the standard errors are too wide to rule out moderate gaps in either direction, so the per-capita evidence is inconclusive.

Total luminosity estimated by PPML on the annual series rises in very-high-conflict districts relative to 1993, to 0.52 in 2017 ($p = 0.08$), but most of the increase coincides with the change of sensor, from 0.11 in 2011 to 0.43 in 2012.

Table A57: Demographic versus Economic Recovery: Comparison Across Samples

	Population	Economic Activity			Human Capital
	(1)	(2)	(3)	(4)	(5)
	Log Pop	IHS NTL	IHS NTL/cap	Log NTL/cap	Literacy 15+
Very High × 2007	0.167***	0.351	-0.001**	-0.590*	4.022***
	(0.047)	(0.272)	(0.001)	(0.316)	(1.117)
Very High × 2017	0.175***	0.035	-0.008***	-1.606***	5.129***
	(0.061)	(0.303)	(0.003)	(0.415)	(1.263)
Observations	4620	4620	4620	1964	4620
Adjusted R^2	0.968	0.836	0.491	0.621	0.873
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. All regressions include district and year fixed effects. Columns (1)–(3) and (5) use the full estimation sample of 4,620 district-year observations with NTL data. Column (4) is the positive-only log specification with 1,964 observations. Standard errors clustered by district in parentheses. The reference year is 1993. IHS = inverse hyperbolic sine transformation. *** p<0.01, ** p<0.05, * p<0.10. Literacy is the share literate among persons aged 15 and over, with the same age universe in every census. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

On the same 1993-based sample (Table A57), population grew 0.17 and 0.18 log points more in very-high-conflict districts by 2007 and 2017 and literacy among residents aged 15 and over rose 4.0 and 5.1 points more, while total NTL shows no significant change.

G.3.4 Heterogeneity in Economic Activity

Splitting very-high-conflict districts by their 1981 literacy rate, a characteristic measured before the conflict, shows no difference between initially more and less literate districts in total NTL or in the probability of any light (p between 0.35 and 0.80). Appendix Tables A61 and A62 report the splits by 1993 luminosity and by dominant perpetrator. Because 1993 luminosity is measured with substantial noise and is the reference year of the outcome, the split by initial luminosity is prone to regression to the mean and is reported for completeness.

Table A58 examines lights within periods covered by a single sensor, the DMSP-based years 1992–2011 and the VIIRS years 2012–2023. Detected light spread everywhere, and the share of very-high-conflict districts with any light rose from 40 to 62 percent between 1992 and 2011 and from 83 to 98 percent between 2012 and 2023, against 29 to 55 and 76 to 94 percent in other districts. Within each period, neither the probability of detected

light nor total luminosity changed differentially in very-high-conflict districts, and changing the detection threshold or winsorizing the brightest observations does not produce a positive differential. The VIIRS-period estimate with the top 1 percent winsorized is negative (−0.23).

Table A58: Night-Time Lights within Homogeneous Sensor Periods

Panel A. Districts with any detected light (district light sum above zero), percent						
	1992	2011	2012	2023	Districts	
Very high	40.5	61.9	83.3	97.6	42	
All other	29.4	54.9	76.2	94.4	1,498	
Panel B. Very high × last year of the sensor period, relative to its first year						
	DMSP-based, 2011 vs 1992			VIIRS, 2023 vs 2012		
	Coefficient	p, district	p, province	Coefficient	p, district	p, province
Any detected light (linear probability)	−0.041	0.527	0.636	−0.039	0.482	0.506
	(0.064)			(0.055)		
Districts (very high)	1,540 (42)			1,540 (42)		
Total light (PPML)	0.035	0.869	0.761	−0.038	0.282	0.431
	(0.215)			(0.036)		
Districts (very high)	914 (26)			1,483 (42)		
Total light, top 1% winsorized (PPML)	−0.071	0.845	0.745	−0.229**	0.049	0.022
	(0.363)			(0.116)		
Districts (very high)	914 (26)			1,483 (42)		
Panel C. NTL per capita (PPML), census years, 2007 vs 1993						
Very high × 2007	0.001	0.998	0.998			
	(0.226)					
Districts (very high)	734 (23)					

Notes. The night-time lights series of Chen et al. (2024) converts DMSP-OLS images to VIIRS-like values for 1992–2011 and uses VIIRS VNL V2 from 2012, so comparisons are made within each sensor period. Thresholds apply to the district light sum (sum of pixel values within the district). Panel B reports the very-high coefficient for the last year of each period from a regression on all years of the period with district and year fixed effects, relative to the first year. Standard errors clustered by district in parentheses, with stars from district-clustered p-values, and p-values with 1981-province clusters in the next column. The PPML samples keep the districts with some light in the period. Winsorizing caps the district light sum at the 99th percentile of the pooled district-years of the period. Panel C uses the census years 1993 and 2007, the only pair of census years inside one sensor period. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

In summary, the night-time lights do not show that very-high-conflict districts fell behind or caught up in economic activity after 1993. Neither the extensive margin nor the scale-invariant per-capita estimates are distinguishable from zero, and the apparent gains in total luminosity coincide with the change of sensor.

G.4 Sample Selection Diagnostics

Table A54 compares the district-year observations dropped and retained by the log transformation. The share of very-high observations is 2.2 percent among those dropped and 3.1 percent among those retained ($p = 0.054$), and the share with any conflict is similar (42.1 against 43.2 percent, $p = 0.436$). The dropped observations are higher (2,631 against 2,176 meters), more rugged, much smaller (mean population 4,290 against 27,418), less urban, and less literate (78.4 against 88.0 percent), all with $p < 0.001$. Mean distance to Lima is 483 against 468 km ($p = 0.038$). The log transformation discards the most marginal districts.

G.5 Robustness Checks

Table A59 uses total night-time lights with three transformations that keep the zero observations. No coefficient is statistically significant. Total NTL and NTL per capita are zero for the same observations.

Table A59: Total NTL: Robustness to Transformation Choice

	Total NTL		
	(1)	(2)	(3)
	IHS(x)	Log(x+.01)	Log(x+1)
Very High × 2007	0.351	0.437	0.343
	(0.272)	(0.441)	(0.242)
Very High × 2017	0.035	0.063	0.077
	(0.303)	(0.498)	(0.272)
Observations	4620	4620	4620
Adjusted R^2	0.836	0.799	0.845
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. All specifications include district and year fixed effects, the omitted comparison group is all districts below the very-high threshold, and the reference year is 1993. Total NTL is the sum of pixel values within the district, and IHS is the inverse hyperbolic sine. All columns keep the observations with zero light, which are the same for total NTL and NTL per capita. No coefficient is statistically significant, which is consistent with similar growth in total light in very-high districts and in the comparison group without proving it. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district. *** p<0.01, ** p<0.05, * p<0.10.

Table A60 adds altitude, ruggedness, and distance to Lima interacted with year. The

NTL per capita coefficients stay at −0.001 in 2007 and between −0.008 and −0.007 in 2017, all significant.

Table A60: NTL Per Capita with Geographic Trend Controls (IHS)

	(1) Basic	(2) +Alt×	(3) +Rugg×	(4) +Dist×
Very High × 2007	-0.001**	-0.001**	-0.001**	-0.001**
	(0.001)	(0.001)	(0.001)	(0.000)
Very High × 2017	-0.008***	-0.008***	-0.008***	-0.007***
	(0.003)	(0.003)	(0.003)	(0.003)
Observations	4620	4620	4620	4620
Adjusted R^2	0.491	0.491	0.491	0.490
Altitude × Year	No	Yes	Yes	Yes
Ruggedness × Year	No	No	Yes	Yes
Distance × Year	No	No	No	Yes
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. All specifications include district and year fixed effects. The reference year is 1993. The dependent variable is IHS(NTL per capita). Geographic controls interacted with year indicators allow for differential trends by altitude, ruggedness, and distance to Lima. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

G.6 Heterogeneity

Table A61 splits the districts at the median of 1993 light. The per-capita gap appears among districts above the median (−0.005 in 2007 and −0.019 in 2017, both $p < 0.01$) and not among those below (−0.003 in 2017). These coefficients are in IHS units and are not percentage differences.

Table A61: NTL Per Capita by Initial Development Level (IHS, Full Sample)

	(1) Pooled	(2) Below Median	(3) Above Median
Very High × 2007	-0.001**	0.000	-0.005***
	(0.001)	(0.000)	(0.002)
Very High × 2017	-0.008***	-0.003	-0.019***
	(0.003)	(0.004)	(0.003)
Observations	4620	3054	1566
Adjusted R^2	0.491	0.080	0.555
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes. Districts split at the median 1993 IHS(NTL) level. Standard errors clustered by district in parentheses. All specifications include district and year fixed effects. The reference year is 1993. The dependent variable is IHS(NTL per capita). *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Table A62 splits by dominant perpetrator. The 2017 coefficients are -0.005 for State-dominant districts and -0.006 for Mixed districts (both $p < 0.05$), and -0.002 for Sendero-dominant districts (not significant). The subsamples are small.

Table A62: NTL Per Capita by Dominant Perpetrator (IHS, Full Sample)

	(1) Full Sample	(2) Sendero	(3) State	(4) Mixed
Very High × 2007	-0.001**	0.000	-0.002**	-0.000
	(0.001)	(0.000)	(0.001)	(0.001)
Very High × 2017	-0.008***	-0.002	-0.005**	-0.006**
	(0.003)	(0.004)	(0.002)	(0.003)
Observations	4620	996	372	606
Adjusted R^2	0.491	0.196	0.471	0.099
District FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes. Standard errors clustered by district in parentheses. The reference year is 1993. All specifications include district and year fixed effects. The dependent variable is IHS(NTL per capita). Columns (2)–(4) restrict the sample to conflict districts with the indicated dominant perpetrator, using the CVR attribution described in Appendix E.1, so coefficients compare very-high-conflict districts with lower-intensity conflict districts in the same subsample. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

G.7 NTL Summary Statistics

Table A63 reports 2017 means. Very-high districts have lower light per capita (0.0073 against 0.0169) and higher total light (1,025.4 against 313.5), and more of them have any detected light (95.2 against 85.1 percent).

Table A63: NTL Summary Statistics by Conflict Status (2017)

	All Other Districts		Very High Conflict	
	Mean	SD	Mean	SD
NTL per capita	0.0169	0.0668	0.0073	0.0146
NTL (total)	313.5	1257.0	1025.4	2991.3
Any detected NTL (%)	85.1	35.6	95.2	21.6

Notes. Cross-sectional summary statistics for 2017. Very high conflict = districts with >100 conflict victims. All other districts = every district below that threshold. NTL per capita = total NTL / population. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.

Figure A13 shows the distribution of IHS(NTL per capita) in 2017 for the 42 very-high districts and the 1,498 other districts. Both distributions are concentrated near zero and

overlap widely.

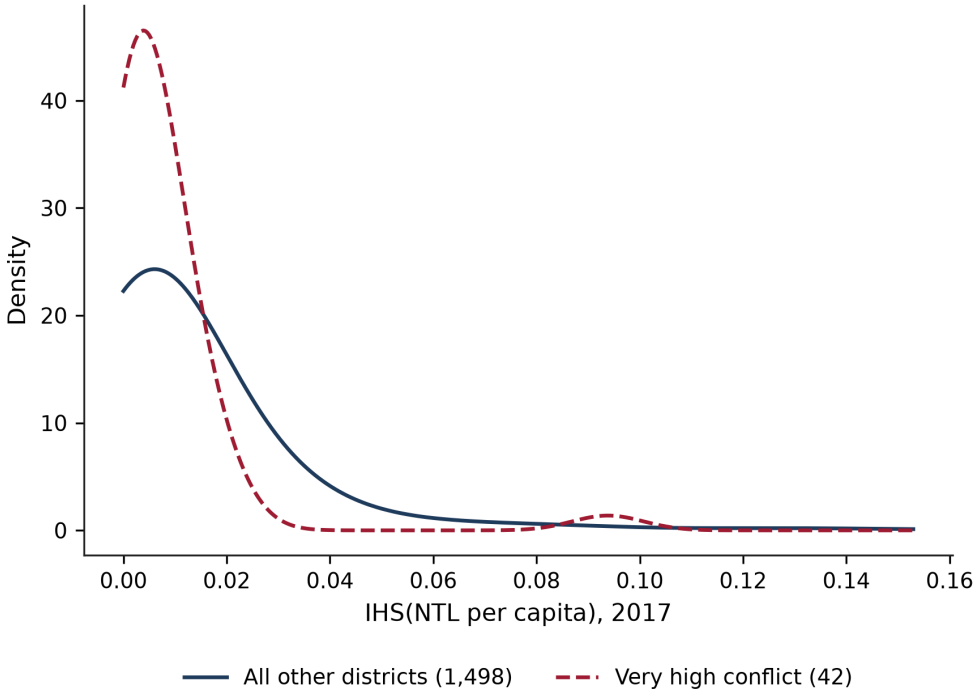
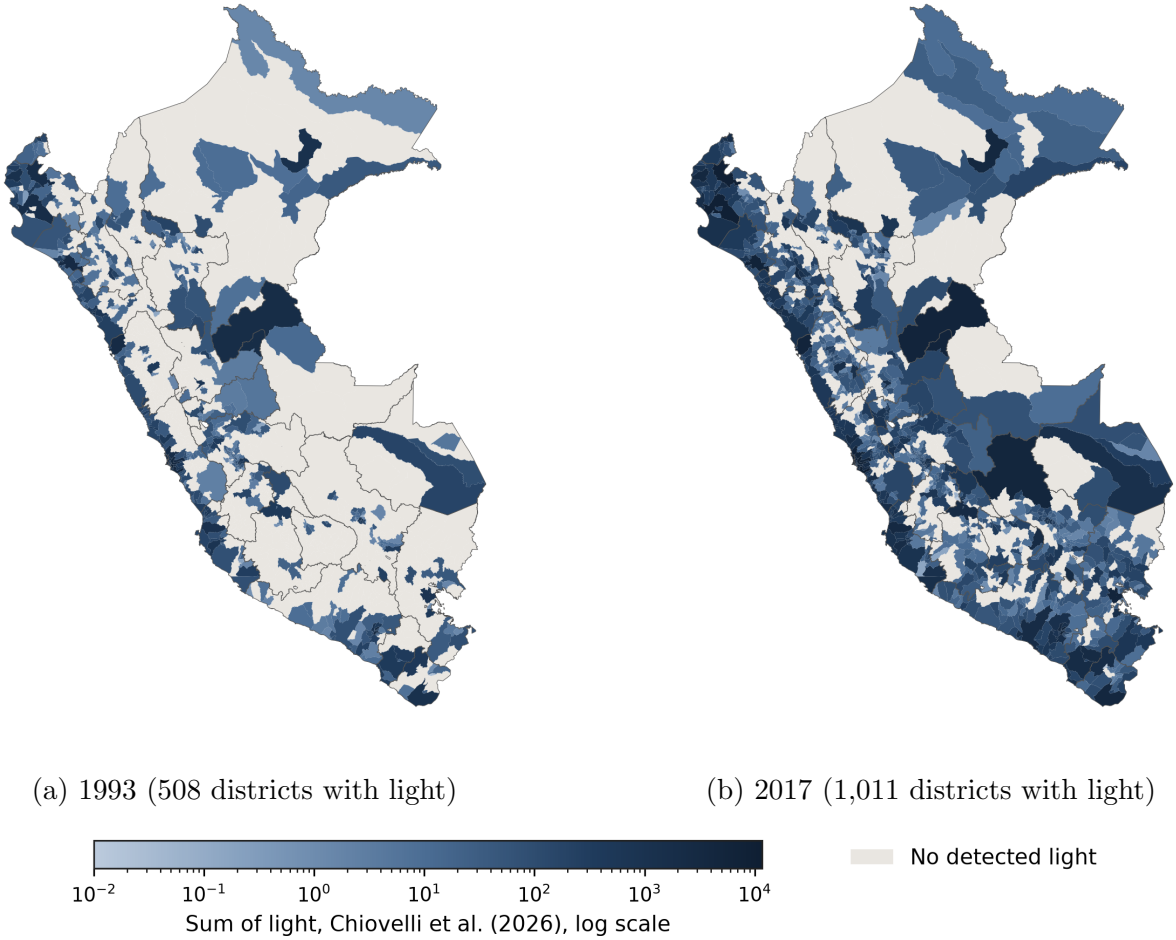


Figure A13: Distribution of Economic Activity by Conflict Status (2017)

Notes. Kernel density of IHS(NTL per capita) in 2017 by conflict status, on the 1981 district units. The IHS transformation keeps districts without detected light.

G.8 Spatial Evolution of Night-Time Lights

Figure A14 maps total light in the main series of Chiovelli et al. (2026) on the 1,540 districts in 1993 and 2017, on a common scale. The number of districts with detected light rose from 508 in 1993 to 1,011 in 2017.



(a) 1993 (508 districts with light)

(b) 2017 (1,011 districts with light)

Figure A14: Night-Time Lights by District, 1993 and 2017

Notes. Total light (sum of digital numbers over the district) in the harmonized DMSP-like series of Chiovelli et al. (2026), main correction variant, on the 1,540 districts of the fixed 1981 geography, with a common logarithmic color scale for both years. Districts without detected light are shown in gray. The number of districts with detected light is 508 in 1993 and 1,011 in 2017.

G.9 Lit Area within Ayacucho

Section 8 reports that within Ayacucho total light and light per capita grew less between 2007 and 2017 in districts with more victims. Figure A15 shows that lit area grew slightly more in those districts, although the slope is not significant with the province wild bootstrap. The lower growth of total light in the districts with more victims therefore came from the intensity of light in places that were already lit.

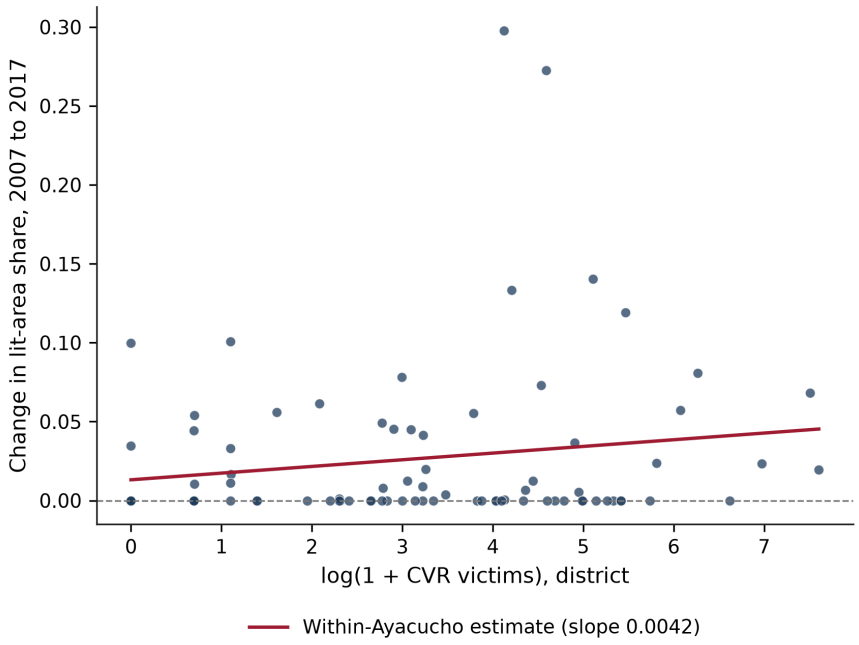


Figure A15: Lit-Area Growth and Victims within Ayacucho, 2007–2017

Notes. Each point is one of the 86 districts of Ayacucho, with the change in the share of district area with positive light between 2007 and 2017 in the main series of Chiovelli et al. (2026) against the logarithm of one plus CVR-documented victims. The line passes through the means with the slope of the window regression of the lit-area share on log(1 + victims) interacted with year, with district and year fixed effects on all years from 2007 to 2017. The slope is 0.0042 (standard error 0.0023 clustered by district, $p = 0.068$), and its p-value with the wild cluster bootstrap over the 7 provinces of 1981 is 0.169. Total light and light per capita grew less in the same districts (Table A51).

H Skills and Sectoral Structure

This appendix reports sectoral employment.

Table A64 shows small sectoral differentials. In 1993 very-high districts had 1.257 percentage points less manufacturing employment and 2.000 points more tertiary employment (both $p < 0.10$). No coefficient is statistically significant in 2007 or 2017. Agricultural and primary employment coincide in these data, so Columns (1) and (4) are identical.

Table A64: Effect of Very High Conflict on Economic Structure

	(1) Agriculture	(2) Manuf.	(3) Commerce	(4) Primary	(5) Tertiary
Very High × 1993	-1.427	-1.257*	0.296	-1.427	2.000*
	(1.647)	(0.716)	(0.370)	(1.647)	(1.066)
Very High × 2007	0.033	-0.341	-0.280	0.033	-0.150
	(2.303)	(0.638)	(0.409)	(2.303)	(1.433)
Very High × 2017	-0.413	-0.171	0.098	-0.413	0.301
	(1.714)	(0.660)	(0.467)	(1.714)	(1.330)
Observations	5991	5991	5991	5991	5991
Adjusted R^2	0.854	0.575	0.827	0.854	0.885
District FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes

Notes. All specifications include district and year fixed effects. Standard errors clustered by district in parentheses. Outcomes in percentage points. Columns (1) and (4) are identical because the primary-sector measure in these data equals agricultural employment. *** p<0.01, ** p<0.05, * p<0.10. Districts are the units of the fixed 1981 geography, each a group of current districts that maps to one 1981 district.